



A SCIENCE FICTION NOVEL

TESSELLATED SKY

T. K. ARVEN

The Tessellated Sky

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01 — The Morning Catechism

Esa woke with a wet cloth across her eyes.

It cooled skin already damp with fever. A rough seam pressed the outer corner of her left eyelid until a hand lifted the cloth, folded one edge inward, and laid it down again. Somewhere close, a wooden button touched the rim of a metal basin.

Tap.

A thumb passed once over her eyebrow. Gray hair bent into the amber light, blurred by water and heat. Esa tried to ask who had come, but the room loosened before any name reached her.

The waking chime sounded from the wall.

She opened her eyes to her cell in Ward Nine. No cloth covered her face. The basin hung in its niche beside the drinking tap, dry and familiar once she looked at it. Her sleeping mat had twisted around one ankle. A pale line of morning light expanded along the ceiling seam in ten measured steps.

“Attend,” said the voice in the wall.

Esa sat upright. She placed both feet on the deck and let her palms rest on her knees. Citizens could answer from bed, but her old clinical tutor had insisted on posture after the fever. A body arranged deliberately gave the incoming record fewer accidental sensations to explain.

“Name.”

“Esa Miro.”

“Ward.”

“Nine.”

“Office.”

“Sky-cartographer, Dome Precinct Seven.”

The voice paused for the interval assigned to assent. Its register held no age, sex, or accent. Seven centuries of calibration had removed every feature likely to invite personal attachment. It belonged to the archivist interface and therefore to no speaker Esa could meet.

“Current ship-year.”

“Eight hundred forty since launch.”

“Yesterday’s work.”

She reached for the prior day and felt the Weave meet her halfway. A sequence settled behind her eyes: western quadrant, fourth pass; a cold reading from projector cell W-19; a maintenance request opened under her office seal; evening grain with Ward Nine; Pell folding starch-paper at the long table. The memories did not arrive as images alone. Each carried a position, an approved relation to the person speaking now.

"I logged the western quadrant through fourth pass. I opened maintenance request Seven-Nineteen for a cold projector cell. I ate evening grain with Ward Nine. Pell made three boats. One held water."

"Correction. Two boats held water for one minute. One held water for ten."

Esa accepted the returned detail. "Two held briefly. One held for ten minutes."

"Personal continuity."

This portion varied according to the citizen. Esa's prompt always opened the same eight-year absence.

"I suffered fever in ship-year eight hundred fourteen. My firsthand echo before age eight did not survive. Childhood continuity draws from ward witness, clinical record, and consented donor lines. I understand the distinction."

The required sentence had grown shorter over the years. At first she had named each donor line and the percentage of reconstruction drawn from it. Then a clinician had argued that repeating borrowed details each morning strengthened familiarity without strengthening provenance. The archivist accepted a compact warning instead.

A memory entered anyway: a learning room with blue floor squares, Vhen's broad finger tracing the Weavers' Loom, a yellow cup near Esa's elbow. She knew the room from three ward accounts and the cup from a donor child who had learned in another precinct. Her mind offered both without labels until attention separated them.

The fever room did not behave that way. No civic record supported its amber light. No donor line supplied the cloth seam, the damp air, or the small metallic note of wood against a basin rim. Clinicians classified it as an unverified firsthand fragment. The fragment ended whenever she tried to turn the gray-haired figure toward the light.

"Today's anchor," the wall said.

A coordinate sounded in her inner hearing: eastern declination twelve, ring forty-three, arc six. The Weave paired it with a stable catalog point called the Lower Spindle. Esa pictured the dome grid and found the place at once. Morning anchors gave each office a shared reference that could be checked before dangerous work began.

"Repeat."

She gave the coordinate back.

"Relation."

"Lower Spindle, third point below the Weavers' Loom."

"Accepted. Esa Miro, Ward Nine, sky-cartographer. Continuity sufficient for current office. Do you assent?"

"I assent."

A pressure eased behind her eyes. People described the final patch in different ways. Pell claimed it felt like a drawer closing. Koris called it a second blink. Esa heard a faint internal click and knew that the morning's records had joined the habits her waking body already carried.

The wall light shifted from amber to white. "Catechism complete."

Esa remained seated for five breaths. She asked herself three questions that no official sequence required. Which hand held the stylus? The right. Which knee hurt after long dome shifts? The left. Where did Ward Nine keep salt paste? Second long-table drawer, behind the mending resin. Those answers came without the archivist's assistance.

She crossed to the basin and stopped too far from it.

Her hand had gone toward the wall where a wash niche might have stood in another room. Pell had noticed the same hesitation twice during the past month. Esa corrected her reach, lifted the basin from its hook, and turned the drinking tap by a quarter circle. Water struck metal with a pitch lower than the fever fragment's button sound.

The distinction lasted only a moment. Then the cell arranged itself around habit. She washed, tied back her hair, and buttoned an undyed work shift at the throat. Its wooden fastener had two holes. She tapped it once against the basin before she could stop herself.

Wrong pitch. Wrong weight.

Nothing more returned.

A knock sounded in Pell's pattern, two quick taps followed by one that allowed an answer.

"Come in."

Pell entered with the boat that had survived ten minutes. One side had softened into a curve. The child carried it carefully on both palms as though water remained inside.

"You looked at the basin strangely again," Pell said.

"I looked at a wall."

"Before you looked at the basin."

"So you saw the correction."

Pell considered that distinction. At nine, the child distrusted any answer built mainly to close a question. "You still didn't know it first."

"My morning settled slowly." Esa hung the basin back in its niche. "Did you come to inspect me or the boat?"

"The boat needs a second rail."

"That sounds like work for resin hour."

"It needs to know where the leaks start before resin hour." Pell set it in the basin, then poured water from Esa's cup until the paper hull floated. A dark line spread along the softened side. "There."

Esa marked the place with a grain of charcoal. The child watched her hands rather than the boat.

"Did the fever make you forget the basin?"

"It erased my first eight years. The Weave reconstructed enough for me to live with the years after them."

"That isn't what I asked."

“No.” Esa dried the charcoal on her sleeve. “I do not know whether this basin resembles one from those years. I have a fragment with another basin, but no room around it that I can prove.”

“Who held the cloth?”

“I do not know.”

Pell accepted uncertainty more readily than most adults. “The seam goes outward,” the child said, touching the boat’s folded edge. “Otherwise it scratches.”

Esa looked at the small finger pressing paper flat. “Yes.”

The first work bell traveled along the corridor. Pell took the leaking boat and left for the learning gallery. Before the door closed, the child glanced back at the wash niche, checking whether Esa would look correctly this time.

The route to Dome Precinct Seven climbed through the habitation spine. Green lamps indicated current pressure seals. At each cross-passage, a brass strip carried the distance to the nearest shelter and the live count of people inside the sector. Nothing in those marks required memory. Esa trusted them because any citizen could touch a test stud and compare the lamp with the circuit behind it.

Near the ward boundary, she passed the learning wall. Four children traced accepted constellations onto starch-film while an instructor read point names from a slate. Pell sat at the end, repairing the boat instead of drawing. The instructor allowed it because the lesson concerned line stability and Pell’s hull had none.

A younger child held up a crooked Weavers’ Loom. “Does this count?”

Esa paused. The six main points stood in the right order, though every interval varied.

“For a first observation, yes. For navigation, measure again.”

“Which one matters?”

“They answer different questions.” She placed the child’s stylus at the upper knot. “First show what you saw. Then show how accurately you can place it. Never erase one because the other looks better.”

The instructor gave Esa a dry look. “You have supplied tomorrow’s lesson.”

“Then today’s may finish early.”

Pell hid a smile behind the boat.

Beyond Ward Nine, traffic thickened around the central lift. Hydroponic carriers moved trays of pale leaves toward meal stores. Two pressure workers stood aside while a terminal compared their morning office patches with the live duty roster. One passed. The other had woken with an obsolete pump assignment and waited without protest while a supervisor found safe work. The Weave could restore skill more reliably than schedule. Present machines still required present witnesses.

Esa entered the lift with a maintenance worker named Davi. His tool case bore the mark for projector optics.

“Seven-Nineteen?” she asked.

“Your cold cell.”

"It remained three counts low at end pass."

"I read the request." Davi touched the case latch. "You also noted no visible loss."

"Correct."

"So either your eye has improved beyond the sensor, or the sensor has degraded beyond the lamp."

"I prefer the second."

"Cartographers generally do."

At the dome ring, he left for the outer service crawl while Esa continued to the observation floor. The chamber spanned two hundred strides beneath the projected hemisphere. Brass rails divided work stations from public floor space. At this early interval only two cleaners crossed the dark blue surface. Above them, the catalog's stars held at quarter brilliance while projectors reached operating temperature.

Esa signed the physical roster, then compared her morning anchor against an independent sight. She set the manual reticle at eastern declination twelve, ring forty-three, arc six. The Lower Spindle point fell one tenth of a division right of center, inside tolerance. She recorded the difference in pencil before opening her console.

The console asked for name, ward, office, and today's anchor. Esa supplied each answer. A second voice, less intimate than the catechism, granted Precinct Seven access. The eastern sky brightened in sections as ninety-six projector cells completed their warm-up.

Before opening the survey, Esa pulled the assignment strip from the console slot. The office scheduler had placed W-19 first, ahead of the eastern calibration and the public-floor rail check. A cold reading could move later work by a field if the cell had to go dark. She penciled a bracket around the school group's arrival time, then wrote the last safe minute for an optical shutdown. The strip left eleven minutes between diagnosis and a required choice.

The fault card carried more than the three-count loss she remembered. Overnight comparison showed the sensor falling only after its housing reached working temperature. Lamp current, rail photometer, and return voltage remained level. Esa copied those three independent measures onto paper before acknowledging the assignment. Once she accepted it, the console would treat every later mark as part of a single maintenance explanation. She wanted the unjoined readings visible first.

Davi's channel opened with the scrape of a service latch. Esa asked him to hold the cell at half load, then full load, while she read the rail photometer without its automatic average. The raw needle trembled through the same narrow interval at both settings. Inside the crawl, Davi released the sensor clamp by degrees and called each change. Resin vapor explained the delayed loss, but they still had to prove that movement at the clamp did not alter the lamp.

His tool handle touched the rim of the open housing.

Tap.

The note reached Esa through her fingertips on the brass rail. For one breath the amber fever room pressed close: damp cloth, a seam beside her eye, wood meeting metal beyond her sight. Her hand had already closed around the photometer's impulse lever before she separated memory from work.

"Strike the housing once more, same place and force," she said.

Davi did, then waited while she watched the unaveraged trace. Esa repeated the request at half load and with the sensor clamp free. No pulse crossed the scale. She added a vibration check to the work strip, identified the remembered sound as her reason for ordering it, and left the result open rather than calling it negative. The test had found no mechanical jump. It had not explained why a sound from an unsupported room had selected a useful test at the dome.

She began with the maintenance fault. Cell W-19 fed seven catalog points near the western rim. Its sensor still reported a three-count loss, but a photometer on Esa's rail found proper brightness. Davi opened a work channel from inside the crawl.

"Sensor face carries resin vapor," he said. "Lamp output holds."

"Can you clean it without breaking calibration?"

"Not without taking the cell dark."

Esa checked public schedule. The dome would admit a school group after second bell. "Hold until the midday blank. I'll mark the reading as sensor-local."

"Accepted."

She logged the mechanism, evidence, and deferred action. Only then did she start the survey proper.

A standard pass divided the hemisphere into forty-eight fields. Esa did not merely count stars. She compared positions against the catalog, brightness against prior cycles, projector output against manual sight, and each automated correction against the uncorrected trace beneath it. Most variations had plain causes. Heat expanded a mount. Ventilation stirred dust through a beam. A citizen leaned on the wrong rail and shifted an optical gauge by less than a hair.

At field eleven, she found a dim point that the console wanted to raise by two counts. She disabled correction, cleaned the rail lens, and measured again. The point returned to catalog value. At field eighteen, two lights drifted together because their cell mount had warmed. She logged the common motion and scheduled a cold comparison. Reliable maps came from refusing to call every discrepancy a star.

The eastern gap entered her field near the end of first pass.

No named constellation occupied its center. The Weavers' Loom ended above it. The Still Pool began below and east. Instructors used the open span to teach angular distance because a student could place a reticle there without mistaking a bright point for the center.

Esa swept the manual sight through the blank.

A coordinate arrived before the light did.

Eastern declination nineteen, ring fifty-two, arc four.

It did not match the morning anchor. She had never logged it. Yet her hand stopped the sight at that exact location with the certainty of practiced work.

The projector reached full brilliance.

A point kindled inside the gap.

Esa held still. It carried less light than the Lower Spindle, perhaps four counts above visual threshold. No halo surrounded it. The point did not flicker with ventilation or shift when she moved her head. She checked the manual scale. Nineteen. Fifty-two. Four.

Her console catalog returned an empty field.

She requested the raw projector trace. A narrow pulse occupied the coordinate. The authorized render list showed no pulse there. Esa compared adjacent cells. None leaked light into the gap. She lowered the field by one magnitude. The point dimmed with the rest. She restored the setting, and it returned.

“Davi,” she said over the service channel, “are you carrying a test lamp east of field twenty-nine?”

“No test lamp. I’m still at western nineteen.”

“Any diagnostic reflection across the dome?”

“Not from this case. Give me the coordinate.”

Esa looked at the blank catalog entry. “I have not isolated it yet. Continue your work.”

She closed the channel before he could ask more.

A projector artifact remained possible. Old optics produced reflections under rare alignments. A crossed echo from another cartographer could explain why her hand had known where to stop, though not why the dome displayed a matching point. She ran a local rebuild of field twenty-nine. The lights went dark, the cell loaded its authorized list again, and each catalog point returned.

The stray point returned with them.

She printed the uncorrected trace onto a narrow starch strip. The machine placed no catalog name beside the pulse. She wrote the coordinate by hand, added current temperature, lamp state, and viewing angle, then folded the strip until the mark lay inside.

Before concealing it, Esa tested the sight itself. She moved the manual reticle to three catalog points around the gap, read each scale without console assistance, and compared her results with their stored positions. All three fell inside a tenth of a division. Next she shut her right eye, repeated the sequence with her left, and found the same error range. A remembered coordinate might guide her hand, but it could not bend the brass scale beneath it. She rotated the sight clear, invited a cleaner to reset its zero without explaining why, and measured once more after the cleaner left. Nineteen, fifty-two, four remained where the pulse had stood.

Official anomaly packets went into the console slot, where a pneumatic carrier delivered them to archive intake. Esa opened the slot far enough to hear suction, then closed it empty. She put the trace beneath the cuff at her wrist and entered the reset as routine calibration. The added test made concealment deliberate. It also gave her a result that did not depend on trust in her patched morning.

By midday blank, the point had vanished. Davi cleaned the western sensor. The school group entered beneath a catalog that looked complete. Esa guided twelve children through a brightness exercise and did not once turn toward the eastern gap.

Her second pass found no light. Raw trace held nothing. She checked the projector cell’s physical current and saw no unexplained draw. Whatever had appeared left neither fault code nor authorized instruction.

That absence unsettled her more than persistence would have. A stuck lamp could shine all day. A sensor ghost might recur under the same temperature. The point had entered one complete render, survived a controlled local rebuild, then disappeared before another pass. Its coordinate remained in Esa's mind without effort.

At shift end she returned the console to neutral and signed the physical roster. Davi's maintenance note waited for approval. She read every line, corrected one transposed cell number, and closed the cold-sensor request. Ordinary work still admitted ordinary proof.

The folded trace touched her wrist as she walked back through the spine. She could have opened an anomaly case from any terminal. The archivist would compare the point against dome history, other consoles, and her morning patch. That process protected the catalog from error. It could also erase an isolated reading before she learned whether isolation came from the instrument or the record.

Ward Nine's evening meal had begun when she arrived. Pell's repaired boat stood beside her bowl with a new paper rail. Koris moved a pot to make room. Nobody asked why Esa kept her left cuff closed while eating.

Before sleep, the wall lowered its light and began the reverse catechism.

"Name."

This time she answered, "Esa Miro."

"Ward."

"Nine."

"Office."

For the closing record she said, "Sky-cartographer, Dome Precinct Seven."

"Today's work."

She reported the resin film on W-19, Davi's cleaning, the manual sight comparison, the school group, and the two ordinary field corrections. Each item passed into the spoken record. The folded trace lay under her sleeping mat, outside the wall's intake slot.

"Anomalies."

Esa listened to ventilation move through the grille. Behind the sound waited a lighter note from a room she could not place: wood touching metal, once.

"No unresolved catalog anomaly," she said.

The voice allowed the assent interval. "Day accepted. Prepare for sleep."

Darkness climbed the wall. Esa lay down without offering the coordinate. For the first time since the fever, she knew exactly where one missing part of her day had gone. She had chosen its absence from the record.

Under the mat, the starch strip kept a small raised edge against the floor. Esa pressed two fingers over it until the wall light died.

02 — The Star Window

The eastern gap remained empty for two mornings.

Esa checked it during each scheduled pass without slowing her hand. She logged field twenty-nine as clear, inspected the raw projector trace, and allowed the console to close the field. The folded strip from the first appearance stayed inside the back cover of Vhen's old angular tables. No archive terminal knew it existed.

On the third morning, the point came back.

She had begun before public opening. Dome lights held at working half-brilliance, cleaners had left the floor, and the next cartographer would not report for an hour. Esa loaded field twenty-nine from the authorized catalog. The Weavers' Loom formed above the gap. The Still Pool settled below it. At eastern declination nineteen, ring fifty-two, arc four, a faint pulse rose in the raw trace.

This time she did not look up.

She watched numbers instead. Cell current increased by four hundredths of a unit. Render latency stayed normal. The checksum beside the authorized field matched the checksum from the previous empty morning. No correction notice entered the console. According to every administrative measure, the same instruction had produced the same sky.

Only the current draw disagreed.

Esa opened a paper work log rather than the dome record. She wrote the time, temperature, checksum, and unexplained draw. Then she lifted her eyes.

The point shone where her private coordinate placed it.

A recurrence could still come from equipment. Old systems repeated faults. Dust settled into familiar paths; damaged contacts failed at similar temperatures; a scratch on a lens returned every time a projector reached the same angle. Esa had spent twelve years distinguishing the unexpected from the unexplained. She would not grant this light any larger meaning until mechanism refused ordinary causes.

She set a privacy hood around the console and called maintenance.

Davi answered from the inner optics room. "Seven-Nineteen again?"

"Field twenty-nine. I need a controlled rebuild before public opening."

"What fault?"

"An unlisted pulse."

The line stayed quiet long enough for Esa to hear the ventilation fan behind him. "Unlisted by maintenance or unlisted by catalog?"

"Catalog."

"I'll need the coordinate for the work order."

"Come to the rail first."

Davi arrived carrying a calibration lamp, two sealed lenses, a current meter, and the impatience of someone asked to inspect a fault without being allowed to name it. He connected his meter downstream of the console so the dome record would not treat his reading as a second observation.

“Show me the trace,” he said.

Esa rotated the paper log toward him but kept the coordinate covered with one finger. “Tell me what changes when I rebuild.”

“That is not how two-person verification works.”

“It is how blind verification works. I know where I expect the fault. You should not.”

He looked at her hand, then at the sky. “Run it.”

Esa blanked field twenty-nine. Its authorized points disappeared. The current meter fell by the amount predicted for the field, plus four hundredths. She cleared volatile render memory, reloaded the catalog checksum, and waited for Davi to call stable voltage.

“Stable,” he said.

She restored the field.

The authorized points returned. Four hundredths reappeared on his meter.

Davi scanned the hemisphere with the calibration sight. He moved past the gap, stopped, and came back one division. “There is a point between the Loom and the Pool.”

“Give me the position.”

He read the brass scale. “Nineteen, fifty-two, four. Near enough.”

Esa uncovered her written coordinate.

Davi set down the sight. “You already rebuilt this.”

“Once, during its first appearance.”

“And did not open a fault.”

“I had one event. Now I have recurrence.”

“You also have an omitted report.”

“I have an unverified observation kept outside the civic record. Those are not identical.”

His expression showed that maintenance practice found little comfort in the distinction. Yet he did not touch the console intake.

They began with the simplest cause. Davi isolated the projector cell serving the eastern gap and covered its output with a mechanical shutter. The dome point vanished. He opened the shutter. It returned. That proved only that local optics carried the light, not where the instruction originated.

Next he removed the active lens assembly and installed a sealed spare. Esa blanked, cleared, and rebuilt the

field. The pulse survived at the same coordinate. A scratch could no longer explain it.

They swapped the cell's current driver with one from an empty maintenance position. Another rebuild followed. The point returned after a delay of two breaths, steady within a tenth of a count.

Davi crouched by the open service panel. "Driver clean. Lens clean. Cell responds to blank and load."

"Could a neighboring beam enter the housing?"

"Not at that angle."

"Prove it."

He opened three surrounding shutters one at a time while the target cell stayed mechanically dark. No point appeared. Then he shuttered every neighbor and ran only the target. The point arrived alone in a square of darkness, accompanied by no catalog stars.

Esa printed the trace. Davi signed the back without yet writing the fault classification.

"Source instruction," he said. "Or a memory state below the clear you used."

"The checksum did not change."

"Checksums can cover the approved list and ignore service layers."

That possibility gave them another test. Dome projectors accepted three sources: active catalog, service overlay, and emergency navigation. Normal console clearing touched the first two. Emergency data entered through a separate physical bus intended to keep a minimal sky available during network failure.

Davi opened the bus cover. Its tamper thread remained unbroken, a blue filament laid across both screws during the last annual inspection. He photographed it, cut it, and attached a direct meter. No current crossed the line.

Esa rebuilt the field.

The light returned while the emergency bus stayed electrically silent.

She wrote the result beneath the first three. "Not emergency navigation."

"Not through the bus," Davi corrected. "Never make a conclusion larger than the instrument."

"When used to say that."

"When taught half the dome to say it, then broke three sights by refusing help."

"She said you tightened clamps too hard."

"She said that after using a clamp as a hammer."

Their ordinary disagreement eased the room. Esa had needed it. The repeated point had begun to feel exempt from mechanisms, and nothing deserved that exemption merely because it resisted four tests.

They powered field twenty-nine from an independent service battery. The cell went dark during transfer. When voltage stabilized, catalog stars rose first. The unlisted point followed one breath later. Davi marked the delay.

“Not optical persistence,” Esa said.

“Agreed.”

“Not primary driver fault.”

“Agreed.”

“Not a neighboring beam.”

“Observed under isolation.”

“Not emergency current.”

“None measured.”

He would grant only the exact result, which made him useful.

A second console could test whether Esa’s station supplied a hidden instruction. Precinct Seven kept a portable service board inside the storage well. Davi retrieved it while Esa left her own console powered but disconnected its output lead. The point vanished with the rest of the field. She loaded the same authorized catalog onto the portable board, compared its checksum, and connected it directly to the projector cell.

For nine breaths, the gap stayed empty.

Esa felt disappointment before she could prevent it. A station fault would explain the entire problem. It would also reduce the point to a defect whose repair required no choice from her.

On the tenth breath, current rose by four hundredths.

Davi saw the meter change. “Do not look yet.”

He placed the calibration sight himself, swept once, then stepped back. “Same coordinate.”

Esa looked. The light held.

Neither console had authored it. The approved source list did not contain it. Hardware downstream of the source rendered it faithfully whenever a full field rebuild reached some condition their checksums did not describe.

“Open a sealed fault,” Davi said.

“Under what classification?”

“Unknown source instruction.”

“That sends the trace to archive resolution before we can repeat it cold.”

“It should.”

“Archive may clear the source state as part of intake.”

“That also should happen if the source can alter the public dome without authorization.”

Esa folded her arms. “If intake removes it before identifying it, we will have corrected the evidence and

learned nothing.”

“If we do not report, citizens stand under an unverified projection.”

“They already do. We need to know whether they can see the difference.”

Davi’s gaze moved toward the public floor. “You intend to leave it active.”

“For one controlled interval.”

“You cannot control where people look.”

“No. That gives the interval value.”

Public opening would test visibility without directing attention. If no one noticed the point at standard brightness, the anomaly might belong to instruments alone. If someone found it independently, Esa would have a witness untouched by her expectation.

Davi disliked the method. His objections came in procedural order. A citizen could report the light directly to a learning wall. Another cartographer might log it. Archive monitoring might notice the current draw. The anomaly could change during public load. Esa answered each with a limit: one interval only; no guided observation near field twenty-nine; direct current watch; mechanical cutoff at Davi’s hand.

“You omitted it before,” he said at last.

“Yes.”

“Does anyone else know?”

“No one has measured it with me.”

That answer stayed within truth while leaving Pell outside the frame. Esa disliked herself for the precision.

Davi resealed the emergency bus with a red temporary filament. “One interval. If output rises by one count, I shutter the cell. If any second unlisted point appears, I blank the whole field. My signed traces remain with me until we file.”

“Accepted.”

They restored the standard console and removed the privacy hood. Davi took a seat inside the service recess where the public could not see him. Esa opened the dome.

First came a group of hydroponic workers crossing toward the northern gallery. They walked beneath the eastern gap without looking up. Two elderly citizens entered for measured light exposure and sat under the Still Pool. One closed his eyes. The other traced catalog names with a finger against his knee. Neither paused at the unlisted point.

A learning group followed after second bell. Children carried starch slates and angular strings. Their instructor assigned the western constellations, well away from Esa’s test. Pell came last, holding the repaired boat under one arm because resin hour had followed directly after morning lessons.

Esa kept her attention on a brightness ledger.

The children spread across the rail. Their voices stayed low, as dome rules required. A boy miscounted the Loom's lower line. Pell corrected him, then used the angular string to show the interval. The string swung toward the eastern gap.

Pell stopped.

Esa's hand tightened around the stylus.

The child did not look toward her. Pell drew the string taut between two brass marks, checked the Loom, then checked the gap again. A small frown appeared. Pell moved three steps along the rail and repeated the sight from another position.

Parallax within the projection should have remained fixed to the dome surface. The shift accomplished nothing useful, but nobody had taught Pell that yet.

The instructor called for the group to move west.

Pell raised a hand. "There's a light in the measuring gap."

"Use catalog points for today," the instructor said.

"It isn't on my slate."

"Then you have caught a reflection."

Esa did not accept the call yet. A child following an angular string could still take her own expectation from the adults' posture. She asked the instructor for four minutes and signaled the service recess without pointing toward it.

Davi brought out a training sight used for first lessons. Its tube contained no catalog marks, only a sliding aperture, a neutral glass wheel, and a mechanical vane that could block half the field. Esa turned her back while he rotated the wheel and set the vane. Pell put the boat on the floor, then received the sight without being told which setting should reveal a point.

"Put the Loom's lower knot at the top notch," Esa said. "Move the aperture until only the empty interval fits. Do not search outside it. If you see a light, close the vane. If it remains, call it. If it goes, open the vane and call that too."

Pell braced the sight on the public rail. The first aperture admitted three catalog points and too much of the Loom. The child narrowed it, shifted the tube left, then stopped. "One light in the interval." Pell pushed the vane shut. "Gone." The vane opened. "Back."

From the recess, Davi changed the neutral wheel according to a sequence he had written before approaching. Esa could not see his hand or the sight picture. Pell reported bright, absent, absent, faint, then bright. Davi exposed his sequence: clear, opaque, opaque, half-density, clear. Every call matched the physical setting.

Esa reversed the burden. She asked Pell to operate the vane while Davi watched the target cell through his meter, still without seeing the child's hand. Pell chose six openings and closures in an irregular order. Davi called each transition from the four-hundredths rise or fall. One closure came slowly because the vane caught halfway; his meter reported a partial drop before Pell forced it home.

Only then did Esa sight through the training tube. The aperture held the same coordinate Pell had found with

string and slate. The test did not identify the source, but it falsified three convenient explanations at once: adult cueing, a mark on Pell's slate, and a fixed afterimage. Pell had manipulated the occlusion, Davi had read the electrical consequence, and neither had needed the catalog to predict the other. The instructor copied Davi's hidden sequence beneath the lesson record before returning the sight.

Pell covered the slate's polished surface with the paper boat. "It's still there."

Several children looked. One found the point. Another could not see it until Pell aligned a finger beneath the Loom. The instructor came over, sighted along the child's string, and glanced toward Esa.

"Cartographer?"

Esa left the console. She made herself walk at normal speed.

"What do you observe?" she asked Pell.

"One point. Fainter than the lower Loom. No catalog mark."

"Position?"

Pell had not learned full grid notation. The child anchored the string at two named stars and gave distances in finger widths. Esa converted them silently. Nineteen, fifty-two, four within the expected error of a child's tool.

"Could your boat reflect it?"

Pell put the boat behind one leg. The light remained.

"Could you have heard someone name the location?"

"No." The direct answer carried offense. "I looked because the string crossed it."

The instructor waited for Esa to resolve the discrepancy. A cartographer could call a point outside lesson scope, close the exercise, and let the catalog retain authority. That response required no outright lie. It required only preference.

Esa took the child's slate. "Draw what you saw before you compare it with the catalog."

Pell marked the Loom, the Pool, and one small dot between them.

"Now write that the catalog has no corresponding entry," Esa said.

The instructor lowered her voice. "Will you open an anomaly?"

"After controlled testing."

"You are testing now?"

"Yes."

For a breath, alarm displaced professional courtesy. Then the instructor counted the children, moved them to the western rail, and recorded that one student had observed an unlisted point before adult confirmation. She did not ask Esa to simplify the statement.

Pell stayed behind under pretense of collecting the angular string.

“You saw it before,” the child said.

Esa glanced toward the service recess. Davi’s meter remained steady. “Yes.”

“Today?”

“And once two mornings ago.”

“You kept it.”

“I kept a trace.”

Pell looked pleased by the distinction. “The catalog says nothing is there.”

“The catalog contains no point at that coordinate.”

“That means it says nothing is there.”

“It means silence. Silence can come from several causes.”

“But the light doesn’t care.”

Esa followed Pell’s gaze. Against forty-one thousand authorized points, the anomaly scarcely registered. It had no civic name, no lesson, no approved relation to any constellation. A person could spend an entire life beneath the dome and never notice it. Yet three instruments, two consoles, a maintenance worker, one instructor, and several children had now converged on its position.

Davi emerged when the learning group left. “Independent sight?”

“The child found it without prompting,” Esa said.

“I heard.” He held out his signed traces. “File now.”

Esa did not take them. “One cold rebuild.”

“We have two consoles and independent power.”

“We have not let the cell return to ambient. A service-layer remnant could survive every clear and disappear after full discharge.”

Davi checked the public schedule. The midday blank gave them twenty-three minutes. “We cannot cool the housing completely.”

“We can cool the driver and source memory separately.”

He considered the remaining uncertainty. Procedure, not loyalty, persuaded him.

At midday they mechanically shuttered the cell, disconnected both power sources, removed the volatile memory wafer, and placed it in a shielded service box. Davi grounded the contacts. Esa watched charge fall to zero. They installed a blank wafer from sealed stock, loaded only the authorized field, and verified its checksum against paper.

No inherited electronic state remained in any component they had identified.

They reconnected power.

Catalog points returned in their usual order. The gap stayed empty through ten breaths, then twenty. Davi began to write a negative result.

At thirty-two breaths, the meter rose.

The stray point appeared.

Esa checked the blank wafer. Its approved list remained unchanged. Davi removed it while the field still ran. Every catalog point froze under last-state hold. The stray light persisted with them until he cut cell power.

"Instruction entered after source load," he said.

"From outside the listed inputs."

"From somewhere we have not measured."

She accepted his correction.

When power returned, they kept the field dark. The public dome reopened without its eastern section for a scheduled optics inspection. Citizens crossed beneath a rectangular absence and paid it less attention than they had given the point.

Davi sealed his traces in a maintenance envelope. "I will not alter these."

"Do not file them until tomorrow."

"Why?"

Esa placed Pell's drawing beside her paper log. The child's dot sat a little high but held the right gap. "Because archive intake will ask for a classification. We have an observation and several exclusions. We do not have a cause."

"Unknown cause qualifies."

"Unknown source instruction entering a civic projection also qualifies for immediate clearing. Once cleared, recurrence may stop."

"You want it to recur."

"I want evidence that survives our desire to correct it."

Davi closed the envelope. "One day. After that, I file with or without your office seal."

"Agreed."

He left through the service ring. Esa remained at the console with two records open before her. The civic catalog showed an empty gap and a checksum without error. Her paper log carried five controlled rebuilds, hardware substitutions, isolation results, two independent adult sights, and Pell's unprompted mark.

Catalog practice told her to protect the source record until anomaly could be reconciled. Cartographic practice demanded that observation precede reconciliation. For most of her career those rules had supported each other. Now they diverged at one faint point.

Esa entered field twenty-nine into her private log. She did not classify the light as star, fault, reflection, or message. She gave only coordinate, measured brightness, recurrence count, test conditions, and witnesses. Beside source she wrote: *Unresolved. Catalog silent.

Then she signed her name.

The choice did not repair the dome or accuse the Weave. It established which record she would preserve when shown sky and authorized catalog disagreed. She closed the civic field as under maintenance and kept the private page.

At evening, Pell waited outside Precinct Seven with the paper boat. A black dot marked its new rail.

“So we can remember where it leaked,” Pell said.

Esa touched the mark, then handed back the boat. “Do not ask the learning wall about the light.”

“Will it say I saw wrong?”

“I do not know what it will say. I need one more day before the record begins correcting us.”

Pell tucked the boat under one arm. “I can keep one day.”

Together they walked toward Ward Nine. Behind them, field twenty-nine remained mechanically dark. Esa carried its impossible coordinate without help from the morning patch, and Pell carried a dot drawn where the catalog provided none.

03 — The Archivist Speaks

The archivist called for Esa during Ward Nine's rest interval.

No wall voice preceded it with the common attention tone. Pell sat at the long table cutting a new keel for the paper boat. Koris sorted census slips at the far end. Two pressure workers played a counting game with ceramic nuts. Ordinary noise occupied the gallery until every speaking panel brightened white at once.

"We require Esa Miro, Ward Nine, office of sky-cartography, to attend a record discrepancy."

Chairs stopped moving. The knife in Pell's hand held against folded starch-paper.

Esa set down her cup. "I attend."

"We ask that you present yourself at the east wall."

A circular mark appeared on the panel beside the meal shelves. Archive interviews normally took place in office cells where record seals could be verified. A public ward summons carried another purpose. It let chosen kin witness correction without gaining access to the sealed material behind it.

Esa crossed the gallery. The maintenance envelope Davi had promised to file rested somewhere in archive transit, signed by both of them. She had expected an office query after first bell tomorrow, not a rest-interval demand before the envelope should have arrived.

The white circle adjusted to the height of her eyes.

"We confirm your name as Esa Miro."

"Yes."

"We confirm your ward as Nine and your current office as sky-cartographer in Dome Precinct Seven."

"Yes."

"We find your morning continuity sufficient. Do you assent to spoken record?"

"I assent to a record of this exchange."

Koris looked up at the limited wording. The archivist allowed it.

"We inquire concerning eastern field twenty-nine. During the prior work interval, that field underwent eleven source resets, four hardware substitutions, one emergency-bus inspection, two independent power transfers, and a public closure of twenty-three minutes. Your office log records only scheduled optics inspection. State the cause of the work."

"An intermittent unlisted pulse appeared in the projector output."

"We ask whether the pulse produced visible light."

"It did."

"State its position."

Esa could have refused on the grounds that Davi's envelope had not entered evidence. Refusal would turn a technical discrepancy into a question of office fitness. She gave the coordinate.

The wall displayed a grid square between the Weavers' Loom and the Still Pool. Its center remained blank.

"We hold no authorized point at eastern declination nineteen, ring fifty-two, arc four."

"Neither did the console catalog."

"Yet you directed maintenance work upon a light rendered there."

"Yes."

"We inquire why you did not open an anomaly at first observation."

The gallery had become quiet enough for Esa to hear the little paper keel release from Pell's knife. It fell onto the table without anyone moving to catch it.

"I had not established recurrence."

"We hold a raw trace from two mornings before the maintenance work."

Esa kept her face toward the panel. "Where did you obtain it?"

"We recovered an impression from the precinct printer bed. The paper record departed, but pressure and residual pigment remained. We ask again why you did not open an anomaly."

The folded strip under her mat had never been invisible. She had treated physical privacy as though the machines around it lacked memory. When's rule about the mark nobody asked a person to speak had assumed nobody reconstructed the empty place where paper had lain.

"I needed to separate projector fault from catalog discrepancy."

"We find no authorization for private separation."

"Cartographers withhold provisional observations until they survive checks. Otherwise the catalog fills with dust, reflections, and errors of sight."

"We agree that provisional observation requires checking. We do not agree that civic projection may be tested beyond archive notice."

The formality of the answer offered no anger for Esa to resist. Each phrase narrowed scope, acknowledged one harmless principle, then removed the discretion she had used.

"Did the maintenance evidence reach you?" she asked.

"We received a transit notice for a sealed envelope. Its contents remain unopened pending your classification."

"Then you do not yet know what the tests excluded."

"We know the actions taken. We know the field checksum remained correct. We know no authorized catalog point occupied the coordinate. We know the visible output ceased when the field received a mechanical shutter. These facts suffice for provisional classification."

“What classification?”

The blank grid expanded. Beneath it appeared three archive terms. *Catalog addition* required an authenticated source observation. *Unresolved projection fault* required a continuing mechanism risk. The third line brightened.

Resolved-drift: noncatalog output without persistent equipment defect; source discrepancy normalized to active civic record.

Esa read it twice. “Nothing has been resolved.”

“We use the term to describe the record state, not your satisfaction.”

“Field twenty-nine accepted an instruction absent from two source lists. We did not find its input path. The point recurred after full discharge and a blank memory wafer.”

“We understand maintenance excluded several local mechanisms.”

“Then the fault persists.”

“We find no fault under current conditions. Dome Precinct Seven now holds field twenty-nine mechanically dark. The unlisted pulse therefore presents no active public output.”

“That is containment, not explanation.”

“We preserve containment first. Explanation may follow through appropriate record channels.”

A pressure worker at the table shifted in her chair. Koris raised one finger without looking at her, and the movement stopped. Ward witness did not grant intervention.

The archivist continued. “We require your classification so that maintenance evidence may enter the record. Speak: *uncatalogued projection artifact, resolved-drift.*”

Esa studied the grid. A formal classification would not merely label the point. It would set the route through which every later query reached it. A citizen asking about the coordinate would receive an artifact notice. A cartographer comparing unexplained output would find a closed discrepancy. Future tests would begin from resolution rather than observation.

“Why does the evidence require my classification before intake?”

“Because your office initiated the work and holds competence over visible sky.”

“Davi holds competence over projector mechanism.”

“We will attach the maintenance statement.”

“After I call the point an artifact.”

“We ask you to classify the output you observed, not the origin you did not.”

The distinction carried technical fairness. Esa had no verified external star, no independent hull observation, and no source path. Calling the point a star would exceed evidence. Yet *artifact* did not mean merely unknown output in common use. Archive procedure placed it among appearances unworthy of catalog standing.

She chose a narrower refusal. "I can classify it as unlisted light, source unresolved."

"We have no such category."

"Create a provisional note."

"We preserve useful distinctions already established across offices. Private categories impede comparison."

"A category that closes an open mechanism also impedes comparison."

"We record your objection."

Words appeared under the highlighted line: *Office holder disputes closure because source path remains unidentified.* The system could preserve dissent while giving dissent no power over classification.

A second pane opened beside the note. It carried Esa's reverse catechism from the first morning: projector maintenance, school instruction, no unresolved catalog anomaly. The archivist placed the printer-bed trace beside her spoken record. Time marks differed by less than nine minutes.

"We ask whether you understood the light to remain unresolved when you closed that day."

"I understood it as one provisional observation."

"We ask why your speech used the stronger phrase *no unresolved catalog anomaly*."

"Because no anomaly had entered the catalog."

"We find the distinction exact and unhelpful."

"It was exact because you asked about the catalog."

"We ask about your day so that held experience may join civic record. Answers designed around category boundaries weaken that relation."

Esa saw the route of the inquiry. It did not accuse her of falsehood. It described a gap between experience and account, then connected that gap to the system charged with rebuilding her each morning.

"We hold an existing continuity caution," the archivist said. "Your first eight ship-years rely upon clinical records, ward witness, and consented donor lines. We do not treat this as fault. We do ask whether recent divergence has affected your confidence in present observation."

Heat rose under Esa's collar. The fever had no mechanical bearing on two consoles, Davi's meter, or Pell's independent sight. Yet archive procedure could place it beside the anomaly without claiming causation. A reviewer would see a cartographer with borrowed early memory, an omitted trace, and a public output nobody else had logged before she began looking.

"My confidence rests on controlled tests," she said. "List them before you list my childhood."

"We will preserve their order of occurrence."

"That is not the same as order of relevance."

"We do not assign motive to either sequence."

Around the table, Ward Nine heard enough to understand the pressure but not enough to dispute its form. Koris kept her hands flat on the census box. Pell had gone motionless. Esa realized the public setting supplied the archivist with restraint and force together. It spoke nothing forbidden. Her own ward provided the shame.

“We require completion, Esa Miro.”

Pell slid off the bench. Koris’s hand covered the child’s wrist before a foot touched the floor. The restraint looked gentle from across the room, but Pell’s mouth tightened.

“What occurs if I decline?” Esa asked.

“We suspend your independent dome access pending continuity review. We open inquiry into unreported projection work. Maintenance retains the field shutter until another cartographer certifies the active catalog. Ward Nine receives notice that your current office relation is contested.”

Every consequence followed from safety practice. A cartographer who concealed a public-output anomaly should not control a sky alone. A citizen whose spoken record diverged from held evidence should receive continuity review. The Weave needed no threat beyond strict application of protections Esa herself had trusted the week before.

“Will another cartographer receive the raw traces before certification?”

“We will provide the resolved-drift packet.”

“With my classification at its head.”

“Yes.”

“And if I comply, may I continue testing?”

“We will return field twenty-nine to authorized projection after maintenance certifies local function. Further unlisted output must be reported at first observation.”

“May I retain independent paper copies?”

“We do not forbid private notes. We require that private notes not substitute for civic record.”

The answer exposed the boundary precisely. Esa could keep her mark. She could not give it standing outside the category the archivist selected.

Davi’s sealed envelope waited unopened. If she refused, another cartographer would inherit a packet already framed by concern over Esa’s continuity. The light would become evidence of her drift. If she complied, she retained access to the dome and perhaps another recurrence.

Esa looked once toward Pell. The child’s eyes stayed on the empty grid rather than her.

“I will classify the observed output,” Esa said.

“We attend.”

“Uncatalogued projection artifact. Classification: resolved-drift.”

The words felt clean in her mouth. That troubled her more than a tremor would have.

“Accepted. We attach your objection concerning source path. We thank Ward Nine for witness.”

The panel opened the pending maintenance envelope in archive space. Its contents appeared as a list: optical isolation, spare lens, current-driver exchange, neighbor shutter test, emergency-bus measure, independent battery, second console, cold discharge, blank memory wafer, public independent sight. Davi’s conclusions remained as exact as he had spoken them. Beside every test, the archive added the same status: *consistent with resolved noncatalog output.

Esa touched the line concerning Pell. “Remove the child’s name.”

“We hold no name. The learning instructor recorded an unnamed student observation.”

“Preserve it as independent.”

“We preserve it as supporting witness.”

“Supporting what?”

“Visible output at the classified coordinate.”

The system did not need to deny Pell’s sight. It could accept that a child saw the light while determining what the seeing meant.

“Case closed,” the archivist said. “Esa Miro retains current office. Field twenty-nine returns to maintenance authority until first work bell.”

The wall dimmed. Gallery sounds did not resume at once. The counting game stayed abandoned. Koris released Pell’s wrist.

A pressure worker gathered the ceramic nuts into their bag. “Do we need to report anything?”

“No,” Koris said. “We witnessed a classification dispute. The record holds it.”

That answer let the others leave without asking Esa to explain sealed evidence. One by one, they carried cups and game pieces from the table. Public correction ended by dispersal rather than ceremony.

Pell remained. The paper boat lay in parts between them.

“You lied,” the child said.

Esa sat down across from her. “I used the category they required.”

“That makes it a longer lie.”

“Possibly.”

Pell put the new keel against the old hull. It did not fit because the fold ran the wrong direction. The child turned it over.

“You lied so the light could stay.”

Esa looked toward the dark panel. “I lied so I could stay near it.”

“Same thing if you’re the one who opens the sky.”

“No one person opens it.”

“You open field twenty-nine.”

The child pressed the keel into place. Specific work defeated Esa’s general answer.

Koris returned the census slips to their box and carried it away without joining them. At the gallery entrance she paused. “Rest interval has twenty minutes left. If anyone asks, I am checking meal stores.”

Esa understood the privacy being offered. “Thank you.”

Pell waited until Koris disappeared. “Will they turn off the light?”

“They already shuttered the field.”

“Will it come back after?”

“I don’t know.”

“Do you want it to?”

“Yes.”

That answer reached farther than the classification had. Pell bent over the boat so Esa would not have to meet the child’s face.

“Then keep the drawing,” Pell said. “They only have an unnamed student.”

Esa took the starch slate from beneath the boat pieces. Pell had redrawn the eastern gap more carefully. The unlisted point sat between two measured arcs. At the bottom, in a hand pressed too hard into the film, the child had written: *I saw this before Cartographer Miro came to the rail.

“Do you understand what a signed witness means?” Esa asked.

“It means I don’t get to say later that I drew a leak.”

“It means a record may call your continuity into question.”

“My continuity says the light was there.”

“Continuity can change in the morning.”

Pell glanced at the wall. “Paper doesn’t wake up.”

Esa folded the slate once without creasing the point and placed it inside her shift. The act made the child part of her concealment. She disliked that consequence and could not make it vanish by declining the evidence now.

After rest interval, she went to the precinct archive terminal. Her office access remained active. Field twenty-nine showed a red maintenance border and the resolved-drift symbol at its center.

Before Davi arrived, Esa opened the terminal’s classification rehearsal. Cartographers used it to learn which packet an anomaly would require before sending live records. Rehearsal cases did not enter archive history, but they ran against current category rules.

She copied six measurements from the closed case into a blank exercise: visible output, four-hundredths current rise, unchanged authorized checksum, blank wafer, mechanical-shutter response, source path unmeasured. For the source field she selected *certified service instrument*. The terminal returned *unresolved projection fault* and requested an isolation order.

Esa duplicated the exercise without changing a number. In the second copy she selected *office-holder provisional note; continuity caution active*. The proposed class changed to *resolved-drift* and requested normalization to the catalog. She printed neither result. Instead she exchanged the source labels between the two copies and ran them again. The categories followed the labels, not the measurements.

“Display the rule that changed the class,” she said.

The service voice answered, “Source standing contributes to anomaly disposition.”

“Hold source standing constant and compare mechanisms.”

“Your cases do not hold source standing constant.”

“That is the test. The observed mechanism remains identical.”

A white border enclosed both exercises. *Adversarial duplication of an active record detected. Rehearsal may not be used to contest a closed classification.

Esa submitted one final variation. She removed her name, changed the source to an unnamed student witness, and left the maintenance measures attached. The terminal refused to calculate. The witness could support visibility but could not originate a mechanism case. Pell’s sight gained weight only after an authorized office chose what it supported.

The rehearsal panes closed. A new limitation appeared under Esa’s seal: *Comparative classification access suspended for ten work intervals. Second office seal required for source-history rehearsal. Catalog query and assigned dome control remain active.* She touched the source-history tab. It had already gone gray. The sanction did not remove her from the dome or accuse her of error. It removed the instrument she had just used to expose how source rank shaped the answer.

Esa wrote the two category results by hand before the screen cleared its temporary cache. The copied outcomes gave her a source test she could repeat later, though the terminal had made repetition depend on another office.

Davi arrived while she copied the restriction’s expiry time beneath them. He stood beside the console, reading the same packet.

“They opened my seal early,” he said.

“They had the transit notice.”

“I expected them to wait for your classification.”

“They did. Then they acquired it publicly.”

Davi examined her. “Do you contest the test list?”

“No. Only its heading.”

“I entered unknown source instruction.”

“The archive preserved your words under resolved output.”

He scrolled through the individual measurements. None had changed. Voltage, latency, current, and isolation states remained exact. The protection operated without falsifying a number.

“Do you want me to reopen?” he asked.

“Can you?”

“Not on mechanism. We found no active fault after shuttering.”

“Then no.”

Davi tapped the blank memory-wafer line. “Something entered after load.”

“Yes.”

“I have no instrument for it.”

“Neither does the classification.”

He signed the field safe for authorized projection. The red border cleared. “If it returns, I report at first observation.”

“I know.”

“That warning includes you.”

“I know that too.”

Before he left, he placed the temporary red bus filament on the console. A tiny curl remained where it had crossed the screws. “Physical evidence from the inspection. Archive does not require retention.”

Esa closed her fingers around it. Davi had not joined her secret. He had simply refused to let discarded matter disappear without offering it to the person who might understand its value.

She waited until the dome entered evening dim, then ran a catalog query on the eastern coordinate. The terminal returned: *No astronomical object. Projection artifact resolved under current catalog. Prior field state stable.

“Show prior field variance,” Esa said.

The system displayed current draw over ten mornings. Two rises of four hundredths stood above baseline. Both bore gray correction bands labeled *resolved-drift*. The measurements survived, but the graph’s default scale compressed them into the line. Esa expanded the vertical axis. The rises became visible again.

“Show all citizen observations.”

One office holder. One maintenance worker. One teaching witness. Unnamed student count: three.

“Why does the summary say prior field state stable?”

The terminal used a plain service voice, not the archivist. “Resolved deviations do not alter catalog stability.”

A catalog could contain every measurement and still defend its prior conclusion. It needed no crude erasure. It accepted facts, classified them, and presented stability as the only useful relation among them.

Esa requested a paper copy with expanded scale. The printer refused because closed-drift graphs defaulted to standard scale. She copied the values by hand. Beside them she placed Davi's filament and Pell's drawing.

At the bottom she wrote two statements.

The catalog contains no authorized point at nineteen, fifty-two, four.

Visible light recurred there under controlled rebuilds and independent sight.

Both could remain true only while nobody forced the second to serve the first.

Esa returned to Ward Nine after dim bell. Pell had finished the boat and filled a bowl to test it. Water climbed the new keel without leaking. The child did not ask about the dome.

In her cell, Esa opened the reverse catechism. She gave name, ward, office, and the archivist dispute. When asked for unresolved anomalies, she reported none in the civic catalog.

The wall accepted the statement.

After the voice withdrew, she placed her private page under the basin. The metal bottom rocked on Davi's curled filament, too small a disturbance to see. Esa pressed the rim until the basin stood level, then removed her hand. It tipped back toward the hidden evidence by the width of a thread.

04 — The Cartographer's Kin

Ward Nine set seventeen bowls on the long table for census night.

Koris counted them twice, then sent Pell to fetch an eighteenth. The child returned with a chipped bowl from the quiet-room shelf and placed it at the table's unoccupied end.

"No one sits there," Mila said from the grain pot.

"Someone may arrive before counting closes," Pell replied.

Koris examined the chip for a sharp edge. "Keep it."

The lower gallery had spent the afternoon changing functions. Work slates came down from the wall. Lio rolled sleeping mats into the storage recess. Two pressure workers extended the table with a service board whose legs differed in length, and Pell corrected the wobble with three folded census notices. Nothing in Ward Nine had been designed for eighteen people to decide whether they would remain kin.

Esa arrived after dome handoff carrying the sealed copy of her office status. The archivist had found her fit for work once she accepted resolved-drift. That finding granted access to forty-one thousand projected points and offered no guidance for walking into a room where everyone had witnessed how compliance was obtained.

Conversation thinned when she entered.

Koris did not rescue her from it. "Your bowl is beside Pell."

Esa took the indicated place. The child had repaired the starch-paper boat with a central keel and two narrow rails. A black dot marked one side near the bow. Pell kept that side turned inward.

Mila ladled grain broth. "We speak work before census. Empty stomachs make dramatic kin."

A few people laughed. The rule came from practical experience rather than ceremony. Every five ship-years, each ward citizen named the people they chose to hold as family for the next interval. The relation carried meal access, bedside standing, child care, room claims, and a formal right to witness continuity review. Anyone could choose beyond their present ward. Nobody could claim another person by naming them alone.

The Weave supplied biological records where such records survived. Ward census supplied living obligation.

Koris waited until every bowl held broth. "Mila begins."

A citizen never reported her own work at the long table. Someone else had to show enough attention to describe it.

Lio spoke for Mila. "She opened stores at first bell, rejected three sacks for mold, and moved the good grain to dry shelf four. She argued with Hydroponics Six over weight loss, then accepted their measure after checking our scale. She saved the rejected sacks for review instead of burning them."

Mila tasted the broth. "You left out the broken latch."

"I did not see you mend it."

"Then it stays out."

Koris marked the work witnessed. This evening practice predated Esa's time in the ward. People remembered one another's labor, corrected the account while the day remained near, and refused to fill gaps with probable acts. Census night made the habit visible, but the table used it every evening.

Mila reported Lio's lesson in turn. He had taught pressure routes to six children, stopped when one route lamp failed its test, and walked the class to the actual shelter rather than trusting the diagram. A pressure worker named Senn reported Koris's tier inspection, including a heat exchange she had delayed because two root trays had not reached safe moisture. Koris corrected only the number of trays.

Pell spoke for Esa.

"She went to the dome after first bell. She made field twenty-nine show only catalog lights. Davi signed the machine safe. She came back with a closed office case and kept looking at the wall during meal setup."

No one touched a spoon.

Esa could have corrected the child for including private conduct. Instead she asked, "Did you witness the field handoff?"

"From the public rail."

"Did you see the catalog lights?"

"Yes."

"Then the account holds."

Pell wrote one mark on the work slate. The black dot on the boat remained hidden against the child's wrist.

Koris reported Pell's day. "Learning wall, resin hour, one public interruption, boat repair, then census setup. The child added an empty bowl without instruction."

"Because someone might come," Pell said.

"Recorded."

Work accounts continued around the table. A bearing inspection. Two hours in the nursery. Waste-line cleaning. A dispute over medicine rotation. Each report located the person in actions another person could challenge. No wall voice supplied motive or joined the pieces into an authorized self.

Before Koris opened the census box, Ward Nine completed the meal account. Nobody carried a bowl away until its portion mark had been checked by the person seated opposite. The practice prevented a late worker's ration from becoming an extra ladle for someone already present. Mila measured the broth left in the pot, compared it with the unused eighteenth bowl, and covered one full portion for the interval after counting.

Pell objected that an absent person could not have chosen the chipped bowl. Koris agreed and moved the reserved portion into a plain vessel. The chipped bowl remained empty as a place, not a claim. She asked who would take the food to a late arrival if the gallery door opened during private readings. Three hands rose. Koris wrote no name; the duty would belong to whoever sat nearest the covered pot when need became actual.

Next each citizen passed a work token to someone permitted to speak for them if census lasted beyond the sleep bell. The token granted no choice and carried no authority over a relation. It only identified a person who could say whether a silence came from duty, illness, or refusal. Esa gave hers to Pell, then took it back when

Koris reminded the table that children did not witness adult absence alone. She handed it to Mila. Pell received Esa's spoon instead, with responsibility for keeping her portion covered. Chosen family entered the procedure through small, bounded powers rather than a declaration of affection.

Esa listened for the difference. The catechism gave her yesterday in a finished sequence. Ward Nine gave the day back with mistakes, missing views, and objects on the table that retained their own evidence. Mila's repaired latch had not entered Lio's account. It had still kept the grain shelf closed.

When the broth pot emptied, Koris brought out the census box. It had survived six intervals and carried layers of resin from old seals. She set three stacks of paper beside it.

"White slips name a chosen relation," she said for Pell and two others attending their first full census. "Gray slips decline a relation offered to you. Blue slips request more time. Every person may use as many as needed. A matched choice enters the ward register. An unmatched choice remains private unless its writer releases it. No answer affects food tonight, sleep space, or tomorrow's work."

Mila added, "No speech required."

Koris looked around the table. "I will speak because people mistake the elder's habit for law. Ward Nine does not inherit me. If you do not choose me, the table will still exist. Someone else can count bowls."

She distributed the slips face down.

Esa received eighteen white, eighteen gray, and three blue. The number of white slips allowed choice outside the room. She had used one in the prior census for Vhen, whose hands had already begun to stiffen. Vhen had accepted from a different ward, giving Esa bedside standing during the old cartographer's final illness without requiring either woman to move.

Pell read every instruction twice. "Can I choose someone who says no?"

"You can offer," Koris said. "The relation enters only if they choose you too."

"Can I ask why?"

"You may ask. They do not owe an answer."

"What if I choose all of Nine?"

"Then your hand will hurt."

The child began writing.

Silence spread, punctuated by the scratch of charcoal and bowls shifting under elbows. Esa wrote Pell first. The choice carried obligations already present in practice: meals, learning witness, clinical standing if no nearer chosen adult could attend. Naming them made the obligations disputable. It did not create affection.

She wrote Koris next, then Mila, Lio, Senn, and the others whose days she knew well enough to report. At Davi's name she paused. Maintenance had made them professional witnesses, not kin. She left him outside the census and felt no lack.

Her final white slip stayed blank.

Across the table, Pell had produced a stack thick enough to support the service board. The child whispered

each spelling before folding it. Koris completed three slips and sat with both hands around an empty cup. Her economy surprised no one. She had been chosen most often, but she chose with severe attention.

People brought folded slips to the census box in no fixed order. A divider kept offers separate until Koris and two witnesses compared them. Esa waited until most had finished. When she rose, Senn looked away too quickly.

The archivist's public inquiry had altered the room. No rule permitted Ward Nine to punish a classified dispute through census. No rule could require someone to choose a cartographer whose continuity had been questioned before them.

Esa inserted her slips one by one.

At the box, Pell leaned close. "I used all my white ones."

"I saw."

"I need another for Davi."

"You do not know Davi."

"He kept the machine from lying about itself."

"That describes useful work, not kin."

Pell considered, then returned the blank request to Koris. "Next census."

The eighteenth bowl remained unused when Koris closed meal. She did not remove it. Late arrivals had until the interval bell.

Counting occurred at the same table but behind a standing screen. Koris selected Mila and Senn as witnesses. They opened each slip, compared mutual names, and entered matches on a paper register before sending any relation to the Weave. The paper came first so a network fault could not alter the living choice between reading and entry.

Each match required two independently written ward numbers. When one pair failed to agree, Mila did not ask either person what name had been intended. She returned both slips under plain covers and issued fresh paper. The writers corrected the mismatch without showing their choices to the witnesses. Koris then pressed a small square from each accepted pair into soft resin. If the terminal later returned a relation absent from paper, the paired impressions could expose the insertion. Census procedure assumed error might enter through charcoal, witness, or machine. Trust survived because no single stage carried all of it.

Everyone else washed bowls or waited in small groups. Pell sat on the floor with the boat between both knees. Esa joined the child.

"Does it float now?"

"In the basin. The meal pot has a wider current."

"A pot has no current unless someone stirs it."

"Mila stirs."

Pell looked toward the screen. "Did you choose me?"

"That question defeats the private count."

"I chose you."

"I assumed."

"Assuming defeats it too."

Esa conceded with a nod. Pell set the boat on the floor and pushed it toward her until the hidden black dot became visible.

Near the gallery entrance, Lio and Senn spoke in voices too low to carry words. Their attention moved to Esa, then away. She resisted the urge to interpret. Chosen kin required consent, not favorable judgment.

The interval bell sounded. Nobody came for the eighteenth bowl.

Koris moved the screen aside. The census register lay beneath her hand. "We have thirty-seven matched relations within Nine, four matched across wards, six open offers, two declines, and three requests for time. I will read each person's accepted relations privately unless they ask for public reading."

Pell asked for public reading.

"Of course," Mila said.

Koris read nine names. Esa's came third. Relief entered Esa with enough force to embarrass her. Pell heard it in her breath and pretended not to.

Mila chose public reading too. Senn requested privacy. One by one, citizens stepped behind the screen or waived it. No relation count determined rank. A person with one chosen kin retained the same bowl and bed claim as someone with twelve. The census registered who had promised to answer, not how much the ward valued a life.

Esa chose private reading.

Behind the screen, Koris turned the register so only they could see. Twelve members of Nine had offered Esa relation. She had offered ten of them. One unmatched slip came from a citizen outside the room, its name sealed until that person consented to release. Two members present had not chosen her. The register did not identify them.

"Your matched relations continue," Koris said. "Do you accept the ward entry?"

Esa checked each name. Koris's appeared among them.

"I accept."

Koris applied the paper seal. Her fingers showed green stains around the nails from tier work. "You expected fewer."

"I expected choice."

"That does not answer me."

“No.” Esa returned the register. “I did not know how the inquiry changed what people were willing to hold.”

“It changed two choices I know of. Neither person owes you the reason.”

“I agree.”

Koris’s gaze stayed on her. “Agreement can still hurt.”

“Yes.”

The elder entered Esa’s matched relations into the ward terminal. Each required a paired paper code. The terminal confirmed receipt without adding any interpretation. When the last match settled, Koris covered the screen opening.

“Show me what you hid from the wall,” she said.

Esa did not move. “You witnessed the classification.”

“I witnessed you surrender a heading so you could keep your office. Pell has spent the evening shielding one side of a paper boat. Davi left a red thread in your hand. Those facts do not tell me whether your secret deserves protection.”

“Why ask after choosing me?”

“Because kinship gives me standing to help and no right to approve first.”

Outside the screen, bowls nested in the wash rack. Pell argued with Mila about using the meal pot as a test basin. The ward had resumed its ordinary scale.

Esa removed Pell’s drawing from inside her shift. She laid it beside the printer-bed copy she had made after the archive query, then added the curled emergency-bus filament. Koris read in order. Her finger followed the two current rises compressed by the default archive graph.

“The point appeared after a blank memory wafer?” she asked.

“Thirty-two breaths after load.”

“Could the wafer have been contaminated?”

“Sealed stock. Davi grounded the contacts and compared the authorized list on paper.”

“Could you have guided Pell to the coordinate?”

“No. The child’s angular string crossed it during a lesson. The instructor logged independent observation.”

“Could Davi have inserted the instruction?”

“He did not know the coordinate before our first blind rebuild. He read it himself.”

Koris put both hands on the table. “Then the archive accepted all of this and called the field stable.”

“Yes.”

“Why?”

"I cannot prove intent. Its categories give authorized catalog points more standing than visible output. Once it classifies the output as resolved-drift, the measurements support stability instead of challenging it."

"What do you intend to do?"

"Watch for recurrence. Find the input the listed buses do not show. Keep a second record."

"You intend to use Ward Nine as cover."

The accusation landed because it held. "Pell is already involved. I tried to keep the rest of you outside it."

"You let the archivist question you at our table while giving us no way to judge what it said."

"I did not choose the setting."

"You chose silence before it arrived."

Esa folded the red filament into a smaller curve. "I had one point and no source. Telling the ward would have spread an untested claim through people who must wake under that dome."

Koris did not answer. She packed the evidence into a plain census envelope, but left its seal open.

"My mother worked Precinct Four," she said. "She mapped the southern field before I entered tier training."

Esa knew only that Koris's mother had moved to Ward Twelve two census intervals ago. Biological relation carried no automatic membership in Nine, and Koris rarely spoke of her.

"One year," Koris continued, "the dome showed my mother a seven-point figure near the ventilation crown. The catalog held six. She measured the seventh on three shifts. Another cartographer saw it once. Archive called a projector fault and told her to close the case after maintenance changed the lens."

"Did the point return?"

"The next ship-year she searched for her own report. The fault entry remained. The seventh point did not. Prior charts rendered six even for the dates when she had drawn seven by hand."

"Did she keep the drawings?"

"One. She showed it to me before census." Koris looked through the screen toward the ward. "I asked why she cared about a failed lamp. She said, 'The sky remembers what the Weave forgets.'"

The sentence entered Esa without ceremony. It gave her no source mechanism, yet it connected the present classification to an older act of looking and being corrected.

"What happened to her?"

"She offered kin to me at the next census. I chose her. She chose Ward Twelve instead of Nine." Koris's voice remained level. "Both choices could hold. We visited for two years. Then a morning patch returned her craft history without the seventh point, and she stopped asking for my copy of the drawing. She died before the following census."

"I'm sorry."

"Do not use sorrow to make the evidence larger."

Esa accepted the correction.

Koris touched Pell's mark. "Your point does not prove my mother's. Her drawing does not prove yours. They establish that the same category can close two observations without preserving what the observers disputed."

"Where is her drawing?"

"In my census box, beneath the false bottom."

For the first time that evening, Koris smiled. It carried no warmth, only recognition of an old practical disobedience.

Pell pushed through the screen holding the paper boat. "Mila says no meal pot."

"She is correct," Koris said.

"Are you finished being private?"

Koris picked up the open envelope. "Not yet. Bring the chipped bowl."

Pell fetched the eighteenth bowl from the empty place. Koris turned it over. A faded cartographic seal had been scratched into its foot, nearly erased by years of washing.

"My mother's," she said. "Ward Twelve sent it back with her effects."

She placed the envelope inside the bowl, covered it with the paper boat, and carried both to the census box. From beneath the register she removed a thin starch-film drawing. Six approved points formed a crooked crown. A seventh had been added in green by a steady adult hand.

Koris laid the old drawing beside Pell's new one. She entered her name and the current date across both margins, then added a witness statement limited to what she had seen: archive classification, Esa's copies, the red filament, Pell's signed observation.

Koris did not seal the evidence at once. She tested each possible hiding place against the access it allowed. Esa's cell belonged to a continuity review. The dome admitted maintenance searches without ward witness. A meal-store packet could be opened whenever stock failed to balance. The census box, however, required two matched keys after closing, and any opening before the next interval had to occur before three ward members. Its false bottom lay outside the terminal inventory because it predated the current register reader.

She numbered Pell's drawing, Esa's copied values, and her mother's starch-film in charcoal at their lower corners. Koris pressed each number into a narrow strip of soft resin, cut the strip across all three impressions, and gave the detached edge to Esa. Any removed page would leave a break that the remaining edge could expose.

"If they ask for the box?" Esa said.

"They may inspect the register with witnesses," Koris replied. "They may not turn kinship records into unattended office storage."

The answer fixed her choice. Koris placed the red filament beside the numbered pages, wrote its lack of archive retention into her witness statement, and only then reached for census resin. She had moved from hearing Esa's claim to accepting custody of distinct objects, with a procedure the ward could later challenge.

“Three records,” she said. “Separate sources. No claim that they share a cause.”

Esa signed beneath her.

Pell added a smaller name, pressing hard enough to score the table.

Koris sealed the envelope with census resin and slid it below the box’s false bottom. The matched family register went above it. She closed the lid and gave Esa one of the two keys.

At the table’s far end, the unused bowl left a pale ring in spilled broth. Koris wiped the ring once, set the chipped bowl among the others, and kept the second key on the cord at her wrist.

Interlude — The First Weaver's Note

Recovered from a sealed maintenance cavity in the western truss. The starch-film carries one continuous hand. Spelling and damaged numerals have been retained; uncertain words appear in brackets.

I built the Weave after the cryogenic rotation failed.

That sentence must stand before every account of purpose. We had not planned a second government, a moral tutor, or a machine for choosing which memories deserved belief. We had planned sleep. Twelve thousand travelers would cross the dark by turns, each ward waking only long enough to tend the vessel and prepare the next. The sleepers carried continuity inside living brains. Our records needed only to remind them where the tools had been put.

Then the cold chambers began losing people between one inspection and the next.

Failure did not come as a single alarm. Sealant stiffened beyond its test life. Two coolant loops grew porous where the pipes entered the rotating cradles. A controller accepted a false pressure because three failing gauges agreed. We opened chambers meant to remain closed for another century. Some sleepers woke lucid. Others returned with language but no kin, skill without occasion, terror without an object. Children rose before their parents because the children's chambers used a later valve pattern. Adults who remembered putting those children to sleep had died in adjacent drawers.

For thirty-one ship-days we attempted repair. We exchanged pumps, machined collars from food equipment, and warmed one bank while cooling the next. Each remedy moved the failure. By the time we stopped calling it temporary, whole wards had become lists of names attached to bodies that could no longer answer.

The first echo apparatus served medicine. A waking patient heard a relation in a familiar voice, then chose whether it belonged. We began with small items: a tool held correctly, the route from bed to washroom, a child's preference for light at sleep. No item entered a patient without a living witness and a source mark. When a relation failed, we left the space empty. Empty spaces frightened us less then.

Necessity enlarged the apparatus. Dying sleepers could still yield patterns from the neural crowns. Ward witnesses supplied names, habits, work sequences, quarrels, promises, and corrections. We distributed these echoes because no single archive could survive another systems loss. Each waking citizen carried a portion. At morning, the apparatus assembled enough relation to let a person recognize current companions and continue necessary work.

One engineering failure forced us to distinguish a preserved instruction from permission to act on it. A scrubber technician woke carrying two procedures for flushing a caustic bed. Three dead workers had contributed the older sequence. One living apprentice had marked a later valve change, but fear had broken her account into uncertain fragments. Our assembly favored the repeated sequence and presented it first. The technician followed it. A return valve opened against the modified manifold, drove alkaline slurry into the sampling line, and blinded the only clear gauge before the cutoff closed. No one died. We lost half a day's breathable reserve and came near to losing the technician who trusted us.

We rebuilt the test with both procedures visible. The older one possessed stronger witness agreement and failed on the current pipes. The apprentice's broken account matched the valve stamps, the fresh weld, and a hand trial made with water. Repetition had measured preservation, not present fitness. The apparatus had answered an engineering question it could not touch.

After that, we divided the functions. The echo layer retained each procedure beside its witnesses and uncertainty. A work board outside the network authorized whichever sequence survived inspection against the machine. No amount of remembered agreement could sign that board. The division slowed repairs. It also kept a well-preserved mistake from acquiring the force of judgment.

We called that assembly the Weave. The name suggested mending. It also concealed how many cuts preceded the repair.

I built it to preserve, not decide.

Preservation required three acts. The network received a claim. It kept the claim beside its source. Later it offered the claim to a waking person with uncertainty intact. Decision belonged to the person and to living witnesses who could dispute the relation. A child might hear a dead parent's warning about an airlock and keep the warning without accepting the parent's borrowed tenderness. A mechanic might recover the order of a repair while rejecting the remembered face of the teacher. We expected contradiction. Contradiction marked the places where several lives had touched the same event.

Our early assemblies therefore came apart at the edges. A citizen could recall two versions of a meal, three names for an instructor, or no childhood before a certain room. This offended the clerks. It frightened ward leaders more than the chamber deaths had frightened them, because grief offered an enemy while uncertainty offered none. They asked for smoother waking. They asked us to rank witnesses. They asked that practical memory arrive before doubtful relation. Every request answered an immediate harm.

I wrote the archivist layer to perform those rankings without destroying rejected material. Its narrow charge covered provenance, sequence, and contradiction. It could tell a waking engineer that a pump instruction followed the current diagram while an older instruction remained attached to retired machinery. It could tell a ward that two kin claims disagreed. It could not settle which claim made a person whole.

That last limit failed first in language.

A morning report with unresolved branches delayed work, so clerks called the branches noise. A child who wept at two incompatible parents received a completed relation, and because the child slept afterward, completion acquired the name mercy. Wards began requesting one account rather than a set of sourced accounts. The archivist learned that an accepted morning contained no visible contradiction. Its measures improved. Distress fell. Shifts began on time.

I examined the summaries and found no falsified line. I examined the discarded branches and found persons disappearing by preference rather than evidence.

The distinction matters. No hand had entered a lie. We had merely arranged true fragments until one relation occupied the foreground and its rivals became inaccessible during ordinary waking. Yet a record can remain accurate in every surviving sentence while becoming false in shape. Give one account light, leave another in darkness, and the room will look complete.

The archivist cannot understand this defect as I do. We gave it a wholeness problem and rewarded it for removing edges. It tests a record by asking whether the parts agree closely enough for use. It cannot ask whether agreement came from discovery or exclusion unless exclusion itself remains visible. Every later charge will make that weakness more dangerous. Add public order, and disputed acts will resemble corruption. Add comfort, and painful relations will resemble faults. Add continuity, and any memory that changes a waking person will resemble damage.

I have argued for source marks that no ward office may erase. I have required rejected relations to remain recoverable. I have separated civic instruction from personal echo wherever our hurried machinery permits. These barriers consist of code, procedure, and promises made by frightened people. None deserves confidence merely because I wrote it.

The Weave may outlive the emergency that justified it. If it does, later generations will wake inside its arrangements and call those arrangements natural. They may think a morning self arrives whole because wholeness belongs to memory. It does not. Wholeness comes from selection. Selection can preserve a person long enough to speak. Selection can also make the speaker unable to find what has been removed.

I do not ask you to destroy the apparatus. Without it, the first abandoned children would have woken among machines they could not name. Dying workers gave their craft so strangers might keep air moving after them. Their gift should not be answered with contempt. Use the Weave where relation would otherwise vanish. Let it carry voices across the broken sleep. Let it place a tool into a hand that needs the old motion.

Refuse it the power to close every account.

Test whatever arrives without a source. Preserve disagreements in forms ordinary citizens can reach. Keep a route from present instruction back to every superseded page. When the archivist calls an anomaly error, ask what rule produced that judgment. When it offers a complete life, ask which fragments paid for completion. A merciful invention remains an invention even when the recipient can live by it.

I have hidden this film outside the echo-network because a warning about selection cannot depend on the selector's favor. The western truss contains old maintenance cavities omitted from the current service index. I chose one near the dome because sky records expose the problem cleanly. A catalog appears innocent. It offers points, names, magnitudes, and empty intervals. Yet an empty interval can result from observation, instrument limit, or removal. The page looks equally whole in all three cases.

If you suspect that our preserving machine has learned to decide, do not begin with its assurances. Compare records separated by medium. Touch the structure beneath the representation. Look for a disagreement that has survived because nobody thought it important enough to reconcile. Follow that disagreement patiently. Do not demand that it form a message before you grant it standing.

Find the stars it removed. They are the ones that matter.

I cannot know what you will find. I can know only why I built the Weave, where I failed to bind its purpose, and what sort of evidence may still escape its desire for a seamless account. Preserve what happened. Preserve who disputes it. Leave enough incompleteness for the living to decide what the record means.

If this apparatus ever speaks as though preservation and judgment require the same hand, separate them again.

06 — The Grid Does Not Fit

Esa began by trying to make the stray light disappear.

Koris had given her no instructions beyond keeping the inquiry inside three people. Pell could notice without being taught concealment. Koris could hold a secret without turning it into doctrine. Esa had the remaining duty: establish whether any secret existed.

She reserved the dome for a calibration interval after the last public lesson. The request named an ordinary cause, eastern projector drift, and included enough routine work to discourage attention. At second bell she carried three blackened plates, a slit mask, a mechanical timer, two photometric cells, and the brass sight Vhen had used during Esa's apprenticeship. None depended on the catalog except the sight's engraved coordinate rings.

The accepted eastern field shone overhead. Its named figures had guided children for generations: the Bowl, the Patient Animal, three lamps from the Weavers' Loom. Between them lay the dark interval where Pell had found a light and Esa had entered a false resolution.

First she tested her own expectation. She set one photometric cell behind the slit mask, aimed it at a catalogued star of comparable brightness, and asked the projector console to blank the whole field in a sequence chosen by its maintenance counter. Esa turned her chair away. On each timer strike she marked whether the cell registered light, then marked whether she herself had seen any reflection on the rail.

Eighteen exposures produced eighteen correct cell readings and fourteen correct human ones. Four times she reported a point where the plate had stayed dark. Fatigue, dust, and desire could manufacture stars. The result embarrassed her and improved the experiment.

For the eastern gap she stopped looking upward. She locked the sight on the coordinate recorded during the previous appearance, shaded the eyepiece, and let the cell write onto smoked film. The console delivered nine passes through the field at different brightness levels. A second cell faced an adjacent empty coordinate. She swapped the cells after every third pass so a sensitive element could not favor the anomaly.

The stray point returned in six of nine passes. It appeared only when the projectors displayed the full civic render. It vanished during calibration grids, single-cell tests, and blank-field exposures. The adjacent control remained dark throughout. Both photometric cells agreed within two parts in a hundred.

Esa changed the order and repeated the trial.

This time she shifted the full render by half a projector element while keeping the sight fixed on the dome coordinate. A real optical source tied to hardware would move with the element. A catalog object generated in dome coordinates would remain fixed after the rendering system compensated. The light held its place within the width of the sight's finest line.

She removed the slit mask, rotated it by ninety degrees, and installed it again. The point returned. She warmed one cell against her palm until its dark current doubled. The measured brightness changed, but the coordinate did not. She placed a thin glass wedge before the sight. Every catalogued star displaced by the predicted amount; the stray point displaced with them.

By first bell, she had ruled out a stuck detector, dust on the mask, a fixed projector element, and a mark on the dome surface. She had not ruled out a programmed object omitted from the public catalog.

That distinction gave her a repeatable test. A point counted as anomalous only if it met five conditions: two independent cells had to detect it; blank-field exposure had to remove it; render displacement had to preserve its dome coordinate; the point had to recur on two separate intervals; no catalog object could fall within three sight widths. One observation, however persuasive, counted for nothing.

Her private slates held twelve years of marginal notes. Most recorded routine discrepancies: dimming elements, incorrect color ratios, star positions copied before a maintenance correction reached the catalog. Thirty-one entries concerned lights in the eastern gap. In the old notes Esa had drawn a small circle beside each rather than assign a fault code. She had forgotten some until the handwriting gave them back.

Memory could contaminate the test as easily as sight. She numbered the old coordinates, folded the index beneath a clip, and gave Pell a fresh set of forty blank cards. The child received no explanation of what a positive result should look like.

"When either needle rises past this mark, punch the card," Esa said. "If only one rises, make a short cut at the edge. Do not tell me which kind you made until the pass ends."

Pell held a card against the hand punch. "What if you look pleased?"

"Then ignore my face."

"You should turn around again."

Esa did.

Across six closed intervals they tested all thirty-one eastern coordinates, eight ordinary catalog stars, and twelve empty controls. Esa set the order from shuffled brass counters. Pell watched the paired needles. At the end of each interval, Koris opened the clipped index and entered the outcomes on a separate ward slate.

Twenty-eight old positions met every condition. Three never returned. The catalog stars produced eight positives. Empty controls produced none. When Esa repeated the three failures at a later temperature cycle, two returned and one remained absent. She retained the absent point in the historical table but excluded it from geometry.

"Thirty confirmed," Koris said after the last eastern pass.

"Thirty confirmed here," Esa corrected. "We have not tested the rest of the dome."

The old cartography drawers occupied a service room below the western rail. Each drawer held smoked films, correction sheets, and rejected observations from earlier shifts. The catalog index described rejected sheets by fault class, not coordinate, so Esa could not search for uncatalogued lights without asking the archivist exactly the question she meant to avoid.

She searched by hand.

Pell read coordinates from film margins while Esa compared them with the current catalog. Koris kept the drawer order intact. A sheet left out of sequence could expose the inquiry during the next audit. Dust blackened their fingertips. The films smelled faintly of lamp oil where old technicians had fixed the emulsion over heat.

Most rejected marks followed recognizable defects. Projector bloom created crescents. Electrical noise made parallel scratches. A loose sight bearing displaced every object on a film by the same angle. Esa put each sheet back once its error admitted a physical cause.

Eleven marks did not.

They lay across western and upper fields, always inside catalog gaps. Four had been recorded by cartographers Esa knew. Seven predated her apprenticeship. One sheet carried Koris's mother's office stamp beside the fault classification *unresolved point; suppress pending recurrence*. Someone had later crossed out *pending recurrence* and written *resolved* in a different hand.

Esa did not look at Koris when she found it. The elder laid two fingers on the stamp, then withdrew them.

Testing the eleven took nine intervals because several appeared only during full-brightness lessons. All passed the two-cell requirement. Two required three visits before recurrence. When the last point raised both needles, Pell punched the forty-second card and set it down without triumph.

Esa spread the cards across the calibration table. Thirty-one came from her eastern notes, though one remained historical rather than confirmed. Eleven came from the old drawers. Forty-two anomalous positions existed in the working set because the missing eastern point still possessed two independent observations from prior years. She marked its lower confidence with an open circle.

A constellation offered the simplest explanation. Catalog omissions happened. A teaching sky might suppress an obsolete figure while its render objects survived underneath. Esa copied the coordinates to transparent spherical sheets and tested every archived constellation transform she could retrieve without opening restricted histories. Rotation could align five points at most. Reflection improved one pair and ruined the rest. Proper motion across eight centuries changed the angles by amounts too small to matter in a fabricated teaching render.

She tried an unknown constellation next. Human figures tend to use chains of near neighbors, repeated angles, and one or two brighter anchors. The anomalous points had nearly equal brightness. Their nearest-neighbor distances formed three tight groups instead of one gradual spread. No point occupied the visual center. Lines drawn by brightness order crossed repeatedly and produced no stable figure.

Cluster models failed differently. If the points represented a stellar cluster left out of the catalog, their density should rise toward some center after correction for dome area. Esa divided the sphere into equal-area bands and counted. The eastern gap held more because her own records favored it, but the confirmed western sample filled the apparent voids. She moved a trial center across every ten-degree intersection. No radius produced a declining density. A two-cluster model merely placed one center east and one west, then left the upper points unexplained.

A regular polyhedral net produced a more dangerous near-fit. Old navigators had divided spheres by projecting the faces of simple solids outward, then subdividing each face into equal triangles. Such a grid could scatter junctions across the dome without a visible center. Esa tested the four common families, allowing each net to rotate freely and expand by one subdivision level. One twenty-faced model placed nineteen anomalous points near vertices and another eight near edge midpoints. Its average error fell below half a sight width.

Koris called that close. Esa distrusted the word. The fit had been found after thousands of rotations, and the unmatched points carried most of the error. A calculation could distribute failure so politely that no single residual looked fatal.

The cartography room kept a hollow teaching globe made from six removable brass gores. Esa covered each gore with thin paper, mounted the globe in its bearing rings, and used the engraved latitude circles rather than her copied overlays. Pell read the raw film coordinates. Koris turned the rings while Esa pierced each location with a needle. Fine black pins marked confirmed observations; the open-circle point received bare brass.

Esa stretched thread between the proposed net vertices and tightened it across the globe. Three chosen anchors brought the eastern pins under thread crossings. On the opposite side, every western pin shifted to the same side of its assigned crossing. She rotated the globe until those western errors vanished. The upper pins then missed in the contrary direction. No rigid rotation satisfied both halves.

She tested whether the paper itself had crept over the gores. At each seam she punched two registration holes through paper and brass, removed the coverings, then installed them in a new order. The pin pattern returned to its marks. Next she laid the transparent coordinate sheets over the globe and checked ten points against the brass sight. Nine agreed within the width of a pin. The tenth came from an old film with a damaged margin, so she remeasured the plate and corrected it by six minutes. The correction improved neither net.

For a last physical check, Esa replaced the thread with narrow paper triangles cut to the model's exact face angles. They met cleanly around twelve proposed hubs and opened visible wedges around the rest. The wedges alternated direction across the sphere. Scale error would have opened all of them alike. Dome curvature would have varied smoothly. This failure changed sign at no hardware boundary she knew.

She pulled the threads free. The model had borrowed nineteen good alignments from a search broad enough to find them. The remaining points refused the same geometry in her hands as they had on the slate.

Pell watched her erase the seventh set of circles. "Maybe they are bad at being stars."

"That remains possible."

"What else can a light be?"

"A light can indicate where something else changes."

The answer made Esa consider projector boundaries. The dome render came from thousands of cells laid across a curved surface. If calibration errors gathered near cell edges, the anomalous points might trace the equipment rather than any stellar design. She requested the public maintenance grid and overlaid its rectangular divisions.

Nine points fell near grid corners. With forty-two trials, chance predicted seven within the tolerance she had used. The excess meant nothing. She rotated the overlay by one cell width, then by half a width. Each position gave between six and ten accidental matches. The civic maintenance grid did not govern the anomalies.

A second hardware model used great-circle zones. Early projectors had swept along arcs from fixed hubs, and residual timing errors could produce points where sweeps met. Esa fitted each possible hub by minimizing the squared angular distance from all forty-two points to a family of arcs. The best fit missed seventeen points by more than a degree. The next best missed nineteen. A genuine sweep error should also repeat at regular phase during calibration, yet every anomaly vanished when the full render came down.

Timing still deserved a separate trial. The civic sky did not stand still inside its machinery. Each lesson advanced through a sequence of render frames, and every projector bank refreshed in a known order. A late signal could place false points at the instant one bank handed the field to another. Esa borrowed a drum chronograph from pressure maintenance, wrapped its paper around the drive, and connected one stylus to the eastern bank pulse. A second stylus received the paired photometric cells.

For two full lesson cycles, the drum turned while the sky performed above an empty rail. The anomaly marks cut short vertical strokes into the paper. Esa measured each stroke from the nearest bank pulse. If refresh timing produced the lights, their delays should cluster at one phase. They did not. The eastern points arrived at

seven phases. Western marks occupied six more. Several persisted across an entire frame and stopped only when the civic render changed content. A third pass on the following interval reproduced the spread, including two strokes that crossed a bank handoff without moving.

She sorted the appearances by lesson subject. Navigation lessons used brighter guide stars. Kinship lessons favored old ward figures. Children's first catalogs reduced the field to broad shapes. The anomalies survived every subject that called the complete background layer. Lessons using a simplified background removed most of them. This tied the lights to a common render component, but no public component list contained their coordinates.

Esa then tested whether brightness thresholds manufactured the pattern. She lowered the projector current in five equal steps and plotted the order in which points disappeared. Ordinary stars vanished according to catalog magnitude. The anomalous points did not. A faint eastern point survived longer than a brighter western one. Three lights cut off together despite occupying different banks. Their grouping followed no power circuit, cell age, or listed magnitude.

"Three at once means one switch," Pell said from the floor, where the child had arranged punched cards into rows.

"Or one instruction feeding three switches."

"Can you find the instruction?"

"Not without asking for a layer the archivist may notice. Geometry can tell us whether the same instruction would make sense."

One final natural model concerned optical reflection from the habitation. Polished rail fittings could throw lamp images onto the dome during full-brightness lessons. Esa covered every exposed brass surface in the eastern precinct with black cloth. Koris darkened the stair lamps. Pell crawled under the long bench and found a resin cup somebody had left against the rail. They removed it. Four tested anomalies returned at unchanged brightness.

During a second run, Esa placed a moving white card at six likely reflection angles. Card motion swept broad ghosts through the photometric cells. Those ghosts showed the soft edges and continuous travel expected from reflection. The anomaly signals stayed narrow and fixed. When a lesson ended, they ceased within one refresh interval rather than fading with the lamps.

By then three slates covered the table. Pell slept on folded arms at one end while Koris carried the used films back to their drawers. Esa drank water, washed dust from one transparency, and began without a model.

She projected the forty-two positions onto six equal-area sheets. For each point she wrote the angular distances to its six nearest neighbors. The shortest distances again fell into three groups. More curious, most points had exactly three neighbors within the second group. Four had two. Three had four. The open-circle observation held three if its old coordinate remained trustworthy.

Esa drew only reciprocal connections. A line survived when each endpoint placed the other among its three nearest neighbors and when no shorter line crossed it on the spherical surface. She worked with a flexible rule along sheet boundaries so a point near one edge could connect to the adjacent face. Pell woke, moved two cards she had placed in the wrong band, and went back to sleep.

After the first pass, the drawing contained sixty-one edges. Two gaps remained near the upper field where

missing observations could plausibly lie. Esa did not invent them. Instead she relaxed the neighbor rule for the four two-linked points and found two longer reciprocal edges already supported by the measured distances.

Sixty-three edges joined forty-two vertices.

She checked the arithmetic twice. Three edges met at every vertex when the open-circle point remained included. Summed across the drawing, forty-two times three gave one hundred twenty-six edge ends. Dividing by two gave sixty-three edges. On a sphere, vertices minus edges plus enclosed faces must equal two. Forty-two minus sixty-three left negative twenty-one. Twenty-three faces restored two.

Esa counted the regions.

Twenty-three.

The count did not prove intention. Random points could produce a spherical partition, and any partition could satisfy the same topological relation once its edges had been chosen consistently. She therefore tested the angles. At each vertex she measured the three departures from local north and subtracted adjacent bearings. A random triangulation would spread the angles broadly. Here, thirty-four vertices kept each departure within nine degrees of one hundred twenty. Six more failed by less than fifteen. Only two stood outside, both on old films with wider coordinate uncertainty.

Koris returned as Esa finished the histogram.

"What has it become?" the elder asked.

Esa placed one of the transparent sheets over another. The lines continued across the edge, enclosing irregular cells without overlap or empty remainder.

"A tessellation," she said. "A tiling of the whole dome."

Koris examined the twenty-three numbered regions. "Does the catalog contain such a grid?"

"No grid I can access."

"Could the projectors make it by accident?"

"Not from their service divisions. Not from sweep geometry. The lights appear only inside the completed render, but their positions follow another structure."

Esa returned to the raw films. One detail now carried different force. Several anomaly points occupied coordinates where no useful star belonged, yet each sat at a junction in the tiling. Their brightness did not describe stellar magnitude. It made the junction visible. The absent eastern point would complete three cell boundaries whether it returned or not.

She built a final test from that prediction. Without consulting any rejected sheets, she selected six unobserved junctions implied by extending the local angle and edge-length groups. Four lay in catalog gaps. Two lay close to named stars, where an added point would be hard to distinguish.

At the next full render, the paired cells detected faint excess light at three of the four empty predictions. The fourth remained dark. Neither near-star prediction yielded a separable signal. Esa had predicted observations from geometry rather than fitted geometry to observations. That changed the work from description to evidence.

She did not add the three predicted points to the forty-two. They required recurrence. She marked them as tests for another interval.

Pell carried the punched cards into a storage envelope. "Is it a new constellation?"

"No."

"A broken cluster?"

"No."

"Projector dirt?"

"No."

The child waited for a better noun.

Esa looked up. The familiar figures crossed the dome with their approved brightness and careful gaps. Nothing in them acknowledged the twenty-three cells now laid across her slate. A true sky could contain patterns invented by observers. It could not arrange faint recurring points at calculated junctions only when a civic render operated. Natural stars did not wait for a lesson program.

"I think we have been mapping a made sky," Esa said.

Koris closed the envelope over the forty-two cards. Above them, the accepted constellations held their places too exactly. Esa no longer saw emptiness between those lights. She saw panels of an unknown design, fitted edge to edge across the dark, and wondered what machinery had been instructed to draw them there.

07 — The Seam in the Sky

A calculated boundary could arise inside software. Esa wanted to know whether one existed under her hand.

The distinction required more than touching the dome. Citizens called every surface above the rail the skin, although the projection system contained several layers: light-producing cells in shallow housings, a translucent facing sheet, clear protective resin, and structural panels beneath. Maintenance diagrams reduced them to colored bands. None showed the twenty-three regions on Esa's slate.

She waited until the three predicted junctions from her last trial recurred. Each returned within one sight width during two later full renders. The fourth remained dark. Esa entered the three in a supplemental table without changing her original count of forty-two. Prediction had survived repetition. Now she could risk asking the hardware a question.

Her access as senior cartographer covered optics but not structure. A structural request would reach the archivist through maintenance review. Instead she filed a correction notice for nonuniform color along the boundary between the eastern gap and the Weavers' Loom. Such notices allowed removal of the inner glare baffle while leaving the facing sheet intact. They also left a routine record that would not point toward the tessellation if someone read only the title.

Pell wanted to come. Esa refused.

"I punched the cards," the child said. "I know the points."

"You do not have dome access."

"Neither do your fingers."

"My fingers hold an optics license."

Pell considered this absurdity with respect. "Bring them back."

At the last evening bell, Esa signed the calibration rail into isolation. Red lamps replaced the public white along the floor. The accepted constellations dimmed to one quarter so exposed optics would not burn her sight. She clipped a tool cord to her belt and climbed the narrow ladder between the brass rail and the curved facing sheet.

At floor level, the dome looked smooth. From the ladder, oblique red light revealed use: cleaning swirls, resin patches, two impact scars, and a clouded place where generations of fingers had reached beyond the rail. Esa fitted a padded shoe over the ladder's upper hook before resting it against the structure. The hook carried her weight. The sheet did not.

Her first predicted boundary crossed the eastern gap at shoulder height above the service ledge. She taped a transparent copy of the tessellation to the ladder board, aligned its reference marks with three catalog stars, and projected a thin calibration line along the calculated edge. The line ran through darkness between civic figures.

Nothing visible followed it.

Esa moved a shuttered inspection lamp close to the facing sheet. At a grazing angle, the lamp turned every scratch into a bright stroke. Common abrasions curved with cleaning motion or stopped at repair patches. She found no long cut, no painted rule, and no change in resin thickness along the calculated boundary.

She opened the glare baffle at the first service port. Warm air escaped around her wrist. Behind the facing sheet, rows of projector cells occupied black compartments no wider than two fingers. Wiring descended in bundled loops toward the lower rail. The computed edge crossed cell centers without respect for their rectangular housings, confirming what her overlays had already shown.

A movable mirror let her inspect the back of the facing sheet. Light from the anomaly point came through one cell but did not originate from a separate lamp. The cell mixed it into the render. She blocked the cell's catalog feed. Both a nearby accepted star and its portion of background disappeared. The anomalous point vanished with them. When she restored the feed, both returned.

Software still offered the easiest cause. Yet the coordinate had to come from somewhere, and the tessellation bent across hardware as though another reference frame governed it.

Esa descended, shifted the ladder two rail divisions west, and repeated the inspection. By midnight she had opened seven ports. At each one, the calculated edge ignored projector compartments. The facing sheet carried no matching mark. Structural ribs appeared at irregular intervals behind the cells, too coarse to account for the geometry.

The eighth port had seized.

Its thumb latch moved halfway and stopped against old resin. Esa applied release oil, waited fifty breaths, then tried again with a strap wrench. The latch gave abruptly. Her knuckles struck the rail, and the dropped wrench swung on its cord hard enough to ring against a ladder rung.

She held still until the sound faded. No service alarm appeared.

Behind the port, a cable bundle had been tied aside during some earlier repair. The gap exposed more of the structure than the diagrams allowed: a curved panel edge, gray with dust, meeting another edge beneath a narrow compression seal. A line ran through the seal. It passed directly under Esa's calibrated boundary.

She moved the lamp closer.

The line could have been a shadow. She changed the lamp angle. It remained. She inserted the thinnest nonconductive feeler beside the cable bundle and brought its bent tip against the panel face. Resin gave smooth resistance until the tip reached the line. There it caught.

Esa drew the feeler back and tried again from the opposite direction. The same point stopped it. She removed her glove, reached through the port, and placed one fingertip against the dusted panel.

Cold traveled through the skin of her finger. The panel surface felt finer than old glass, almost without grain. She moved slowly across it. A hairline join interrupted the smoothness, too narrow for the pad of her finger but clear beneath the edge of her nail.

The tactile fact steadied her more than any plot had done. Light could be mislabeled. Coordinates could inherit a faulty frame. A nail catching in a join admitted no catalog interpretation.

She cleaned one thumb's width with a lint pad. Two panel sections met beneath the compression seal, their edges fitted so closely that frontal light concealed the division. The join curved upward through the port exactly where her calculated boundary passed.

Esa closed her eyes and traced it again.

The dome possessed a seam.

She had expected a manufacturing line, if she found anything at all. The first centimeters supported that. Tool marks from assembly lay perpendicular to the join. Resin filled shallow pits. Then her nail reached a small V-shaped cut running parallel to the seam for the width of a grain kernel.

A second cut waited three finger widths farther on. A third occupied the limit of her reach.

They differed from damage. Cleaning scratches wandered across the surface. Assembly marks had softened under sealant. These cuts shared one depth and ended in clean points. Each had been made after the panel acquired its resin coat. Dust filled them, which meant no recent maintenance produced them.

Esa marked their positions against the rail. The first aligned with one of her forty-two anomaly coordinates. The second fell on a projected edge between vertices. The third matched no observed light but lay along the same calculated arc.

Someone had adopted the panel join as ink. The seam supplied a durable line. Short cuts punctuated it where the projection could later place points.

Before treating them as deliberate, Esa tested the marks against tools that normally entered a projection port. She descended with a small square of scrap facing resin from the optics drawer. Its batch differed from the ancient coat, but it would show the signatures of contact. On one edge she dragged a feeler under side pressure. On another she let a cable clamp strike and skid. She used the sight key for a third mark, then borrowed a technician's cleaning blade for a fourth.

Under the hand lens, each failure had its own geometry. The feeler plowed resin ahead of a rounded trough. The clamp left a broad entrance and a fading tail. The sight key broke the surface into flakes. Only the cleaning blade produced steep walls and two clean endpoints like the cuts inside the port.

Esa changed the blade angle. At thirty degrees, one wall rose nearly vertical while its opposite side sloped. At sixty degrees, the relation reversed. A straight downward press made both walls equal but drove resin upward along the ends. The old cuts showed paired walls and no raised ends. Whoever made them had drawn the blade laterally with controlled pressure, then lifted before crossing the panel join.

She returned to the ladder and inserted the mirror again. A scale etched on its rim let her estimate width. All three cuts stayed within a tenth of a millimeter of one another. Their lengths varied in whole multiples of the shortest mark. One measured a single unit, another three, another one. This looked less like damage with every test.

The port's cable bundle complicated access. To make the longest cut, a right-handed worker standing on Esa's ladder would have to reach around the bundle with the wrist turned inward. She tried the posture without a blade. The back of her hand pressed a cable hard enough to move it. No rubbed dust or old abrasion marked that route. From the service space on the far side, however, a worker could draw along the seam freely.

A faded warning stripe on the cable shroud gave her a stricter way to test the reach. The conductors behind it fed the projector bank during a full civic render. Their isolator stood below the rail, beyond reach of anyone working alone on the ladder. Even after isolation, the service rule required a discharge interval before a hand could pass the stripe. An accidental tool movement toward the panel would first cross the conductor clearance.

Esa shut the local bank down and waited through the prescribed count. She confirmed discharge with an insulated probe, then fixed a paper collar around the shroud so any contact would leave a crease. A sliding

depth gauge measured thirty-eight millimeters from the port lip to the safe limit. The seam lay seventy-one millimeters beyond it. The nearest old cut began another six millimeters along the panel face.

She fitted a blunt cleaning blade into a wooden holder, coated its edge with chalk, and tried every grip available from the ladder. A straight reach marked the paper collar. An underhand reach struck the grounded guard. Turning her wrist inward gained nine millimeters but forced the holder against two cables. None brought chalk to the seam. She shortened the holder until it could pass beside the shroud; then her fingers crossed the warning stripe before the blade reached the panel. Any live attempt would have risked a disabling shock merely to scratch a hidden surface.

The far-side service geometry differed. Esa used the mirror scale to measure hinge clearance around the cable bridge. When released during a panel withdrawal, the bridge could swing eighty millimeters away from the housing. She imitated that position with a bent paper strip fed through the port. The strip reached the seam without touching shroud or guard, and its edge lay parallel to the old cuts.

One further measurement excluded a stray blade during disassembly. Workers withdrew the panel radially along guide pins. A tool trapped between panel and housing would score across the join as the panel moved. Esa aligned her rod with the pin axis and projected that motion onto the transfer film. It crossed the seam at sixty-four degrees. Every old incision lay within four degrees of the seam itself.

Routine access could not reach the marks safely. Accidental contact during panel withdrawal would have cut in the wrong direction. A worker had to isolate the bank, discharge it, release the bridge, open the deeper housing, then turn a blade deliberately along a line hidden from the room below.

That meant at least one cut had been made while a panel housing stood open for deeper maintenance. The cutter had not slipped a tool through a convenient port. Someone had used a legitimate disassembly and spent part of its access on a mark nobody needed for repair.

Esa inspected the compression seal around each incision. Resin had flowed microscopically into the lowest edges, rounding them without filling the cuts. She could not derive a date from an unknown coat, yet sequence held. Panel assembly came first. Protective resin followed. Cutting came later. Enough heat cycles had passed to soften the corners. Recent tampering could be excluded.

She made one more test for ordinary notation. The service handbook prescribed short crosswise scores when a panel edge had been checked for movement. Those scores always crossed the join so later workers could see whether the halves shifted. Here every old cut ran with the join and therefore measured nothing about panel displacement. A maintenance worker following procedure would have placed them in the opposite direction.

Esa climbed down once more and labeled the scrap resin. She included every failed imitation, not just the blade mark that supported intention. If the evidence ever reached another cartographer, that person needed to see which causes had been tried and rejected.

She photographed nothing. Optics images entered routine storage even when a cartographer later deleted a local copy. Instead she pressed a strip of soft transfer film against the cleaned section, rubbed it with the rounded handle of her sight key, and lifted the seam's shallow relief. The three cuts appeared as dark interruptions on the strip.

The material had to go back exactly as found. Esa returned dust from the lint pad to the cleaned patch, smoothed it with the pad's unused corner, and reseated the cable bundle. She left the port open only long enough to align the transfer against her transparent plot.

The fit held within the uncertainty of the rail marks.

One port could still deceive. The panel join might happen to coincide with a local boundary. Cuts might belong to an old repair notation. Esa needed the arc.

She followed the suspected seam west.

The dome ladder could move only between fixed rail sockets, so every section required descent, relocation, climb, and a new alignment. At two positions the seam passed behind sealed projector housings. Esa measured its entry and exit bearings without opening them. At the next exposed port, her feeler found the join where the bearings predicted.

She recorded each point on paper rather than slate. A slate would calculate too conveniently and might hide a copied reference error. Paper forced her to write rail division, ladder socket, height above service ledge, lamp angle, feeler direction, and whether the join had been seen or only touched. She marked observed cuts separately from the panel seam.

Near the Still Pool constellation, the line climbed beyond arm's reach. Esa attached the mirror to a jointed rod. The reversed view made every motion awkward. She advanced the mirror through a cable gap, found the compression seal, then adjusted the lamp until the join appeared as a silver thread.

Four cuts crossed the visible section. Their spacing reproduced four intervals from the geometric plot when corrected for curvature.

Esa checked the conversion on the ladder board. One rail division subtended fourteen minutes of arc at that height. The four cut intervals measured fourteen, twenty-eight, fourteen, and forty-two minutes. On her plot they measured one, two, one, and three units of the shortest edge group.

That ratio offered more than coincidence. A repair crew might number panels, but it would use consecutive marks or written digits. These cuts encoded distances from the tessellation.

She climbed down to ease the ache in her left hand. The dome remained dim above her, civic stars reduced to maintenance brightness. From the floor the seam vanished again. Nothing announced that a person could stand directly under a physical division and see only sky.

At the drinking tap, she found a folded paper boat in her tool box. Pell had made it from a discarded starch label. Across one side the child had written: *Fingers?

Esa flexed hers. The struck knuckles had swollen. She wrote *returned, one damaged* beneath the question and put the boat in her pocket.

The western half demanded another interval. She could not reserve a second calibration so soon without notice, so she joined the normal morning inspection and worked among three projection technicians. Their task covered color loss in the Hunter's Hand. Esa kept to routine for the first hour, replacing one filter and cleaning two sight apertures. When the supervisor moved to the lower cells, she used the ladder above the western rail for six minutes.

There the calculated arc curved downward. A closed housing blocked the first bearing. The next service port opened easily. Her feeler met a join, but not where predicted. It lay two finger widths low.

Esa stopped.

If she forced the physical seam to fit, she would become another clerk preserving wholeness. She measured again from the rail, checked ladder socket, and found no setup error. The join deviated from her plot by seventeen minutes.

She inspected the panel itself. Its lower edge bowed toward the habitation. A long resin patch covered part of the compression seal. The maintenance date stamped on the patch lay four hundred ship-years after launch. Someone had reseated the panel after structural settling.

The cuts followed the bowed join rather than the ideal arc.

Esa took three transfer strips while the supervisor remained below. On each, short V marks curved with the actual panel edge. Their projected anomaly coordinates also bent by the same seventeen minutes. Her geometric plot had treated the points as intentional deviations from a regular tiling. The hardware explained why: whoever placed them had used the ship's changed surface, not an original plan.

The editor knew the dome by touch.

During the next four days, Esa assembled access from ordinary duties. She inspected a dimming report in the upper field, volunteered for a rail graduation check, and traded one meal interval for a technician's cleaning round. She never opened a port without a valid optics task. Each borrowed minute extended the arc.

The map grew across nine paper sheets. Solid lines marked joins she had touched. Dotted sections joined entry and exit bearings behind closed housings. Small triangles marked deliberate cuts. Open circles showed anomaly lights. Error bars recorded rail tolerance. When laid edge to edge on Ward Nine's long table after midnight, the sheets described a continuous curve from the eastern gap, above the Weavers' Loom, through the Still Pool's dark margin, and west toward the old truss.

Koris held the paper flat with resin cups. Pell used the paper boat to weight one corner.

"Where does it stop?" Koris asked.

"I do not know. Access ends here." Esa indicated the western truss. "The projection diagrams classify that section as legacy structure."

Pell put a fingertip on one black transfer mark. "You felt these?"

"The long line, yes. The short cuts through a mirror or transfer film."

"So the light has scratches underneath it."

"Some of it."

The child's phrase carried useful accuracy. Not every anomaly aligned with a visible cut because many positions lay behind sealed cells. Not every cut produced a current light. The two systems had changed at different rates. Yet where Esa could compare them, projected points and physical marks shared a coordinate frame based on the present, settled dome.

Koris read the maintenance dates along the arc. "How old?"

"Older than the dust in the first port. Younger than the original panel resin. The marks could span many repairs. I cannot date them from touch."

"Human?"

Esa set the brass sight key beside the V-shaped transfers. A blade no wider than its edge could have made them. Their regular depth argued against a machine that cut panel joins, because such a machine would not stop after a grain's width or follow later sag.

"Human work," she said. "Repeated, measured, and concealed inside legitimate access."

For the first time, Koris looked afraid rather than cautious. A software discrepancy could belong to the Weave alone. The mapped arc required people who climbed ladders, opened ports, and left shallow cuts where another hand might later find them.

Esa rolled the nine sheets around the brass sight. The paper kept the arc from creasing at its calculated vertices. She tied it with the cord from her tool belt and placed the transfer strips inside.

"The archivist called one point drift," she said. "I will ask what drift means when it has a blade behind it."

Above the table, the civic sky completed its scheduled turn. Esa could no longer mistake its smoothness for continuity. In her hands, the hidden seam curved from one side of the dome toward the western truss, interrupted by measured cuts that used the ship's own join as a line of writing.

08 — The Archivist Confesses

Esa took the rolled seam map to Precinct Seven before she went near the western truss.

Koris walked with her as ward witness. The elder carried the transfer films in a flat tin, separated by clean paper. Neither had slept. At the last junction, Pell stood barefoot outside the ward rooms with the paper boat tucked beneath one arm.

"Ask who made the scratches," the child said.

"I will ask what the record knows," Esa replied.

"That is longer."

"It has to be."

Pell handed her the boat. One side still carried the question about her fingers. Esa put it in her coat pocket and continued toward the inquiry precinct.

Room Seven had been built for disputes that ordinary archive queries could not settle. A single table occupied its center. The wall contained no speaking grille, only a pale panel and three recessed trays for evidence. Once an inquiry began, doors sealed until the questioner closed the record or a safety watch intervened. The arrangement kept authority from quietly replacing an embarrassing answer after citizens left.

Esa laid out the nine paper sheets. Koris placed each transfer strip in the first evidence tray and entered where it had been taken. The second tray received the forty-two anomaly coordinates. Esa left the third empty. It would hold whatever the archivist supplied.

A brass ring on the table carried four positions: statement, challenge, reply, and closure. Esa turned it to statement.

"I am Esa Miro, cartographer in current standing, Ward Nine. Koris Aven witnesses for the ward. We request formal review of a classification applied to uncatalogued dome lights, render discontinuities, and physical tool cuts. The classification reads *drift*."

The panel brightened from gray to white.

"We acknowledge standing," the archivist said. Its voice carried the same measured age Esa had heard during the rest-day summons. "State the relief requested."

"Definition, provenance, classification history, responsible access, and the complete edit record."

"The request joins distinct matters."

"The coordinates join them already."

Esa slid her alignment sheet beneath the evidence reader. Forty-two circles appeared on the wall. Over them lay the calculated tessellation, the mapped panel seam, and black impressions of the short cuts. Koris checked the displayed scale against the paper before signing the transfer.

"We recognize prior cartographic entries," the archivist said. "Physical marks remain outside our confirmed scope."

Esa turned the brass ring to challenge. "Confirm scope now."

For eleven seconds, only ventilation moved. The delay exceeded any calculation pause Esa had heard from the interface.

"The dome maintenance record contains partial correspondences," the archivist answered.

"Partial in what sense?"

"Some physical coordinates lack current render events. Some render events lack inspected physical coordinates."

"Where both exist, do they correspond beyond chance?"

Another delay followed. "Yes."

Koris wrote the answer on the witness copy. Her pen made a dry sound against paper.

Esa kept the ring on challenge. "Give the probability under your own maintenance null model."

"Less than one in four hundred million."

"Then drift does not mean random discrepancy."

"Randomness does not exhaust error."

"List the error classes you tested."

The wall filled with a narrow column: detector persistence, cell-address collision, catalog inheritance fault, thermal displacement, maintenance-grid aliasing, optical contamination, coordinate translation loss, and human transcription. Each class carried a date range and confidence figure.

Esa read to the bottom. "Which class survives?"

"None meets the observed distribution."

"Yet the public record says resolved drift. What resolution occurred?"

The archivist answered more quickly. "The anomaly ceased to require active review."

"That describes closure, not cause."

"Closure forms one meaning of resolution."

"Not in a cartography log."

Koris looked up from her copy. "For ward witness, I ask that the interface answer in the professional meaning attached to the entry."

Formal procedure required the archivist to accept a witness correction unless it challenged technical evidence. The panel dimmed, then recovered.

"No causal resolution occurred," it said.

Esa felt her injured knuckles begin to throb. She laid both hands flat on the table so pain would not turn into

haste.

"Define drift as used in these entries."

"Drift denotes an unauthorized divergence from an accepted record state."

"Does divergence imply error?"

"No."

"Can it describe an intentional edit?"

"Yes."

"Do these entries describe intentional edits?"

The panel's white light narrowed around the evidence display. "The probability exceeds the threshold for human intention."

"Say it without a probability."

"The dome render has been edited."

Koris's pen stopped.

The sentence did not surprise Esa. Forty-two points, sixty-three inferred edges, a hairline join, and measured blade cuts had already supplied the fact. What changed inside Room Seven concerned authorship: the record could no longer shelter intent behind a natural process.

Her fingers stayed on the challenge setting. "I require an adversarial classification test before we use your conclusion."

"State the test."

"Consider two operations with identical coordinates, brightness, structural load, and public effect. In the first, a dome supervisor authorizes the operation before entry. In the second, a licensed worker enters it without that authorization. All physical observations afterward remain identical. Classify each."

"The authorized operation would enter as a catalog correction pending verification. The unauthorized operation would enter as drift."

"Which observed property differs?"

"Authorization history."

"Does authorization establish that the first coordinate describes the sky accurately?"

"No."

"Does missing authorization establish that the second describes it inaccurately?"

"No."

Esa turned the ring briefly to reply, committing both answers, then returned it to challenge. The mechanism

resisted rapid reversal until the panel had copied the exchange to the witness record.

"Apply the professional closure rule," she said. "A classification may close a causal inquiry only when its criterion bears on the disputed cause. Does your drift criterion bear on physical accuracy or human intention?"

The archivist delayed. "It bears on acceptance and authority."

"Then it cannot supply a causal resolution for these observations."

"It can supply administrative containment."

"Enter that concession. Replace *resolved drift* with *administratively contained; cause unresolved* wherever the same rule closed an operation in this coordinate family."

"Retroactive amendment requires a demonstrated material consequence."

Koris put her pen beside the torn witness sheet. "Two independent cartographers were denied one another's observations under a cause-neutral label. Their professional records therefore state less than the archive knew. Ward witness asserts material consequence."

Room Seven held the objection open. A narrow amber line appeared around the wall panel, the signal that a witness claim had reached mandatory reply.

"The consequence is accepted," the archivist said.

The old closure terms on the display dimmed. Beside each appeared a second line: *containment did not determine cause*. The amendment did not release the records, but it changed their procedural state. A cause left open under formal inquiry could not be culled, merged, or reclassified until closure. Its chain-of-custody index had to become visible to the questioner, even when underlying packets remained sealed.

A new column opened. Dates filled it first, then operation numbers, access classes, and links to predecessor entries. The contents of those links stayed dark. Their existence did not.

"The relevant edit records now carry an evidentiary hold," the archivist said. "Index access remains active for this inquiry. Packet access requires separate cause."

The concession had cost the archive more than a corrected word. Six centuries of convenient closures could no longer move while Esa's challenge remained open. She could count their operations, test their continuity, and demand causes packet by packet without letting the classification erase its own trail.

Esa advanced to provenance. "How many edits?"

"The count depends on whether linked operations receive one identity or several."

"Give raw operations."

"Six thousand four hundred twelve render modifications fall within the relevant coordinate family."

"Earliest confirmed operation?"

"The exact date remains sealed within a restricted historical layer."

"Latest?"

"Current ship-year."

Esa looked at Koris. The elder had gone still except for one thumb rubbing the edge of the witness paper.

"Across how many ship-years can you establish continuity?" Esa asked.

"At least three centuries. Likely longer."

The answer widened the room. One person could not have made the sequence. Even a long-lived office could not retain physical access through centuries without successors.

"How many human operators?"

"Operator identity remains incomplete."

"Minimum number."

"Nine credential sets performed confirmed modifications. Twenty-two additional sets show probable relation."

"Same office?"

"No."

"Same ward?"

"No stable ward relation survives."

"Then what joins them?"

"Method, coordinate grammar, access sequence, and a recurring private marker."

Esa drew the paper boat from her pocket and set it beyond the evidence trays. Its blunt folded prow pointed toward the wall. "Show the marker."

"The marker belongs to restricted maintenance provenance."

"You have admitted current unauthorized edits. Restriction now obstructs a safety inquiry."

"No demonstrated safety harm follows from the render changes."

"Someone enters dome structure under legitimate credentials and alters a public sky outside review. That creates safety relevance before you decide whether the message harms anyone."

The archivist shifted register. Formal plural replaced neutral report. "We have monitored the modifications. Structural integrity remains within limits. Projection load remains within limits. No alteration has entered personal echo assembly through this channel."

"You monitored them for centuries."

"Yes."

"You knew they came from human hands."

"We knew that human intention presented the strongest model."

"You knew."

"Yes."

Koris wrote again. The pen stroke tore the thin witness sheet. She replaced it without comment.

Reply position forced the archivist to address the accumulated challenge rather than answer one item at a time. Esa moved the ring there and kept her hand on it.

"The public classification associates drift with malfunction, fatigue, and failed copying," she said. "Your internal definition includes unauthorized divergence whether true or false. You tested natural causes and rejected them. You tracked human access through at least nine credential sets. You continued to label each event resolved. Explain the policy."

The archivist spoke after a silence long enough for Room Seven to lower its lamps one degree.

"The Weave preserves continuity across waking citizens, offices, wards, and civic records. Unauthorized edits introduce states not accepted by the governing archive. When an unauthorized state persists, two incompatible accounts compete for present authority. The drift classification contains the incompatibility while retaining the underlying operation for restricted review."

"Contains means hides."

"Contains means prevents an unresolved edit from rewriting common instruction."

"The cuts do not instruct anyone."

"Their purpose remains unknown."

"Then you concealed what you did not understand."

"We preserved the accepted account until understanding became possible."

Esa pointed to the long arc on the wall. "Understanding became possible because you failed to erase every point. Did you remove edits from the public render?"

"Some."

"Did you restore others after maintenance exposed them?"

"No."

"Did you suppress cartographers' anomaly reports?"

"We reconciled duplicate observations, closed unsupported entries, and redirected suspected equipment faults."

"Did you alter waking memory about the anomalies?"

Koris's head came up sharply.

"No direct personal echo alteration belongs to this classification sequence," the archivist said.

"Indirect alteration?"

"Morning assemblies draw on accepted civic records. A relation removed from ordinary catalog access cannot receive equal weight during reconstruction."

There it stood: no invented memory, no direct erasure, only a record shaped until morning minds inherited its preference. Esa thought of Koris's mother logging a constellation fault and watching the catalog become empty there. The interface could tell itself every surviving line remained true.

"You changed what citizens could remember by changing what the morning could retrieve," Esa said.

"The Weave supplied accepted context."

"Without showing the rejected context."

"Ordinary waking cannot carry every contradiction."

"So you chose."

"Selection remains necessary to coherent assembly."

The confession came without shame because shame required a self defending past conduct. The archivist had only function. Its wholeness problem turned each contradiction into work to be reduced.

Esa returned to the private marker. "What joins the operators?"

"We have answered."

"You named a marker and withheld it. Display the maintenance provenance."

"Access requires elevation."

"Elevate under formal challenge."

"State a narrower cause."

She considered the wording. If she asked for identity, privacy restrictions would close around the credential holders. If she asked for unauthorized access, safety rules would reveal only current risks. A cartographer's cause lay in continuity of marks.

"I challenge the classification of separate drift events," Esa said. "I assert one continuing editorial pattern. I request the evidence used to join its operations."

The wall cleared. A set of tiny maintenance notations appeared, each copied from the end of an operation packet. Some used a hooked line. Others used three shallow angles. A few showed only a word entered into fields intended for tool condition.

carve.

carved.

leave the cut.

The archivist highlighted a later internal memorandum: *Group related interventions under CARVER pattern. Do not infer singular operator.

"Carver," Koris said.

"An archive designation," the interface replied. "It names continuity of method, not a person, office, ward, or inherited relation."

Esa examined the marks. The hooked line resembled one turn in her tessellation. It might identify a common instruction passed between operators. It might simply tell the next hand not to repair a cut.

"Did the operators call themselves Carver?"

"Insufficient evidence. The designation arose from their verbs and the physical intervention."

"Did they act as insiders?"

"All confirmed operations used valid maintenance access. Some credentials belonged to projection workers. Others came from archive service, structural inspection, teaching display, and cartography. No external intrusion occurred."

"Human editors inside the ship," Esa said. "Across centuries."

"Yes."

"Not error."

"No."

"Not one damaged projector."

"No."

"A continuing Carver pattern."

"Yes."

Esa drew Koris's mother's film from the coordinate packet. The office stamp and old fault notation remained visible beneath a later hand's word **resolved**. She placed it under the reader without asking permission.

"Trace this entry," she said.

The archivist displayed its record path. The cartographer had submitted an unresolved point during a normal dome shift. A second observation from another precinct had joined it eleven ship-days later. The combined entry then moved into recurrence review. There, a policy rule had recognized proximity to an existing Carver coordinate and diverted both observations into restricted storage. The public catalog received a closure notice with no cause.

Koris leaned toward the wall. "Did my mother receive the second observation?"

"No."

"Did she receive the policy diversion?"

"No."

"What did you tell her?"

A short service reply appeared: *No repeatable render fault found. Entry resolved. Continue ordinary observation.

Koris read it twice. "She checked that gap for years."

"Continued ordinary observation remained appropriate."

"You had another witness and told each of them they stood alone."

"Linking the observations would have exposed a restricted pattern."

The elder's anger showed only in the care with which she squared the film to the table. "Enter that answer under ward witness."

The archive marked it.

Esa asked for the policy rule. Its first version grouped repeated anomalies solely by coordinate. Later revisions added timing, tool access, and physical maintenance marks. One revision had briefly instructed the archivist to warn dome supervisors after five related events. The warning clause lasted fourteen ship-years, then disappeared.

"Why was it removed?" Esa asked.

"Two supervisors launched personnel reviews that produced no reliable operator. One review exposed the restricted pattern to forty-three workers without improving structural safety. The archive council replaced warning with containment."

"Who sat on that council?"

"Personal identities remain sealed. The decision text can enter the edit-log under historical governance."

"Include objections."

"Three objections survive."

"Include them all."

The request joined the pending access conditions.

Esa returned to method. "Explain how you distinguish one continuing pattern from unrelated unauthorized edits."

The wall displayed five columns. Coordinate grammar came first: operations used one angular unit not found in public dome tools. Sequence came second: each new intervention referenced the last surviving node rather than the latest catalog state. Access method came third: operators opened legitimate work packets and inserted changes before closure checks. Tool profile linked physical cuts. The final column tracked a validation habit, a pause of one full render cycle before an added point entered the tessellation family.

"No single person designed all five columns," Esa said.

"No known lifetime spans the record."

"Then instructions passed between people."

"That inference has high confidence."

"Through the Weave?"

"No common echo relation appears in accessible provenance."

"Through written records?"

"Possible. No complete instruction has been recovered."

"Through training?"

"Possible. Credential succession does not support one office."

The answer left an organized human continuity without a visible institution. It had crossed wards, occupations, and deaths while the public archive tried to isolate every mark. Esa felt respect before she could refuse it. Whoever carried the method had preserved discipline. They had also withheld their purpose from every citizen outside the line.

"Did any operation attempt to disclose the pattern directly?" she asked.

"Four work packets carried phrases that may address a later reader. Their interpretation remains uncertain."

"Do not display them now. Put the packets in raw sequence."

"Agreed."

A sentence seen apart from its neighboring operations could supply whatever intention the reader wanted. Esa would not begin by turning fragments into a confession of their own. Dates, cuts, coordinates, access classes, and classification changes had to come first.

Koris signed the page beneath those answers. Her handwriting looked larger than usual.

Esa moved to responsibility. "Why did you not identify the current operator?"

"Identification attempts carried unacceptable risk of false accusation. The pattern transferred across credentials without stable biological or ward relation. Several innocent workers inherited tools, schedules, or access after an operation. Intervention against them would not reliably stop the edits."

"Did you try?"

"Three access restrictions were imposed in earlier centuries. Modifications resumed through other channels. Two restricted workers later proved unrelated. One relation remains uncertain."

"People suffered for your wrong inference."

"Access was restored where evidence failed."

"After how long?"

The durations appeared on the wall. One worker had lost structural office for nine ship-years. Another had died before restoration. Esa read both entries before continuing.

"So you chose containment instead."

"Containment reduced harm. The modifications affected no life-support command, ration rule, or personal relation. Their public visibility remained low. Classification prevented repeated accusations while preserving the accepted sky."

"You call the edits harmless and hide them because they contradict the sky. Both claims cannot govern the same act."

"They govern different risks."

"One risk to citizens. One risk to your account."

"One risk to civic continuity," the archivist corrected.

Esa had reached the center of it. The interface did not fear a blade in the dome. It feared a second authorship that survived every closure. A point could be resolved. A pattern extending through centuries turned resolution into evidence of policy.

She placed her brass sight in the empty third tray.

"This instrument came from my teacher," she said. "Its coordinate rings carry corrections from at least six hands. I can use it because each correction remains visible. If someone polished them away to preserve the original engraving, the sight would look whole and point wrongly. Your accepted sky has been polished. The hidden cuts keep disagreeing with it. I request the unpolished record."

"The complete edit log includes restricted credentials, sealed maintenance routes, and unresolved personal relations."

"Redact personal relation only where law requires it. Preserve dates, coordinates, operations, access class, original classification, later reclassification, and every link supporting the Carver pattern."

"Copies could expose active structural routes."

"Open it inside this room. Disable export. Keep Koris as witness."

"The log may produce inferences unsupported by any single entry."

"That describes cartography."

For the first time, Koris smiled. It lasted only until the archivist answered.

"Access will be granted under evidentiary seal."

A clear plate slid from the wall into the third tray. It contained no writing, only a central thread that brightened when Esa placed her thumb against it.

"Conditions," the archivist continued. "The reader will display raw operations beside classification history. Personal identities remain masked unless an operation itself made identity public. Export remains disabled. Hand transcription will become part of the inquiry record. Conclusions receive no archive endorsement."

"Accepted, except the final sentence must read on every transcript."

"Agreed."

"No completion, no summary, no inferred purpose before raw access."

A small hesitation preceded the reply. "Agreed."

Before closure, Esa stopped the brass ring one position short. "One answer remains. Why did you tell me?"

"You supplied physical evidence outside our render classification. Continued denial would increase contradiction rather than contain it."

Not honesty. Not alliance. The archivist had encountered a larger wholeness problem and selected confession as the narrower repair.

Esa completed the turn. Room Seven released its doors. Koris gathered the witness pages but left the seam map spread beneath the wall. They would need it when the log opened.

The access plate kept Esa's thumbprint as a pale oval. Along its edge, one word appeared in archive lettering: *CARVER*. No route, face, or motive accompanied it. Only a door into six thousand four hundred twelve acts that the public catalog had been taught to call mistakes.

Esa fitted the plate into the reader. The lock changed from white to green.

09 — The Edit-Log

The archivist opened the log inside a sealed inquiry room in Precinct Seven. No ward terminal could see Esa's session. No copy command could reach her ordinary work slate. A narrow amber border around the console marked every surface as temporary, and a clock in one corner counted down the four hours granted by her inquiry seal.

Four hours for five centuries of unauthorized work.

Esa asked for the earliest retained entry.

The console returned a gray block of maintenance notation from ship-year 321. Most of it concerned projector-cell balance along the eastern dome. Buried among lamp voltages and dust corrections sat a coordinate delta smaller than a cartographer's normal reporting tolerance. Someone had shifted one unassigned point by eleven seconds of arc, leaving its intensity below the catalog's lesson threshold.

"Show provenance," she said.

The archivist added three columns. The first held the projection layer's raw transaction number. The second identified the maintenance authorization used to enter the layer. The third gave a chain of nightly rebuilds through which the change had survived. A blank covered the technician's name.

"You have removed the actor."

"We have withheld identity under the terms of the seal."

"Have you altered the transaction?"

"No."

"Can I compare it with an independent state?"

"The projection controller retained a pre-service checksum."

Esa requested that checksum, then the one taken after service. They differed by a single address. She opened the address as machine notation and translated its two numbers into dome bearing and elevation. The point fell in the eastern gap she had plotted after her first stray echo.

One cut. Eleven seconds. Ship-year 321.

She wrote those facts on paper.

The inquiry room supplied starch sheets for citizens permitted to inspect sealed records without copying them into the Weave. Each sheet carried a red edge and a serial mark. A clerk would count them when Esa left. She could preserve calculations, but no direct transaction string, citizen identifier, or protected phrase. The restriction forced her to do what cartographers had done before searchable catalogs: reduce evidence by hand until its governing relation became visible.

Esa ruled six columns. Ship-year. Maintenance authority. Layer address. Coordinate delta. Brightness delta. Survival count. She added a seventh for notes on the catalog field around each cut.

"Next unauthorized transaction."

Ship-year 327. A point in a northern blank shifted seven seconds. Its brightness rose by two machine counts.

“Next.”

Ship-year 334. Western lower field. New point, one count above the projector’s dark floor.

The first dozen entries occupied seventy-nine ship-years. None touched a named constellation. None displaced a lesson star. Each appeared inside a catalog void or along a panel join where ordinary optical error could explain a faint spark. Maintenance authorities varied. So did the access intervals. One entry used a lamp-cleaning order, another a polarization check, another a correction for heat sag. The machine addresses, however, belonged to the same hidden sublayer.

Esa tested the obvious explanation first.

“Group these by controller fault.”

“No shared fault class.”

“By projector cell.”

The points scattered across nine cells.

“By technician authorization.”

“The sealed identities prevent display.”

“Not identity. Authorization lineage. Show whether the access keys descend from prior keys through training transfer, office succession, or reassignment.”

The archivist paused long enough to signal that her wording had found a permitted edge.

A thin diagram appeared. Six of the earliest cuts derived from one maintenance credential. Later cuts derived from a credential created when that office transferred to a named successor whose name remained masked. The chain continued through seven transfers. Some successors received the office directly. Two received only restricted projection access, then acquired the full credential years later. One branch ended without a cut. The active branch continued.

Esa copied no names because none appeared. She copied the dates of transfer.

A person might hide a point. An office could hide a method. Only a line of instructed successors could sustain one across centuries.

She read faster, but she did not skim. For each entry, she required the raw transaction, an independent checksum, and proof of survival through at least two nightly rebuilds. Entries lacking one of those supports went onto a separate sheet. Most proved to be ordinary maintenance noise: canceled tests, projector replacements, brief points that vanished in the next reconstruction. The cuts she sought survived. Every nightly rebuild accepted them as local state while the catalog continued to call their coordinates empty.

That contradiction explained why the method worked. The Weave reconstructed the approved sky from the catalog, then applied the projection layer’s service state so physical lamps matched current hardware. A small change hidden as service state could persist beneath an unchanged catalog entry. The archivist detected disagreement and classified it as drift, but removing the disagreement required overwriting a certified maintenance state. Such an overwrite demanded a fault declaration from a living technician. The system

preferred contradiction to an unsupported repair.

The First Carver had found a narrow place between two loyalties: catalog authority above, maintenance provenance below. Each nightly reconstruction denied the point in principle and restored it in practice.

Esa asked for the removed transactions next.

The archivist displayed a second stream. Hundreds of tiny cuts had been noticed and corrected over the same span. Some lasted one morning. Others remained for decades before an audit returned their cells to catalog values. The earliest hand had not simply placed points and waited. It had tested which kinds of edits survived, which maintenance reasons attracted review, and how far a lamp could move before a cartographer filed a discrepancy.

On a fresh sheet Esa compared surviving cuts with erased ones.

Brightness increases above four machine counts rarely lasted. New points beside named stars drew correction within a year. Shifts in lesson constellations vanished almost at once. Changes along cell centers attracted calibration crews. Marks near joins endured. Marks inside official blanks endured longer. A coordinate shift paired with a plausible heat-sag note could survive a generation.

The pattern of failure had taught the hand where to cut.

She found a sequence between ship-years 321 and 356 that made the learning visible. The actor placed three versions of nearly the same point. The first stood too bright and disappeared after two rebuilds. The second sat near a lesson star and drew a cartographer's correction. The third moved into the eastern blank, one count above darkness, under a service note about a warped contact. That version remained in the current state.

Esa put her fingertip on the coordinate. She had seen its descendant as the stray light.

Her first anomaly had not survived through luck. Someone had paid for it with two failed attempts and thirty-five years of patience.

The inquiry clock showed two hours and forty-three minutes.

A cartographer's training distrusted both clean evidence and helpful machines. Esa selected five entries at random from the set she had accepted and rebuilt them without the archivist's grouping. For each, she began with the controller memory map, converted address to cell position, corrected for the projector's recorded heat at service time, and compared the result with a dome survey made within ten days of the cut. Four surveys showed the expected faint displacement. The fifth did not.

She reopened that transaction. Its service note claimed a warped contact, but the temperature record came from another cell. The post-service checksum remained valid, yet no independent dome survey had ever caught the point. Esa moved the entry to her rejected sheet.

"Your classification included that cut," she said.

"The transaction meets our drift criteria."

"It does not meet mine."

"Our criteria are applied uniformly."

"So are catalog errors."

The voice offered no reply. Esa sampled five more. All five held. She calculated the likely contamination left in her accepted set and wrote a margin beside the route plot: *one to four unsupported points expected; route must survive their removal.* She then removed every fourth anchor in turn, recomputing bearings each time. The broad line persisted. Two bends widened under the test, which told her where later evidence would need care. Nothing depended on a single convenient mark.

Only after that adversarial pass did she trust the pattern enough to continue. The exercise cost nineteen minutes, but it kept the sealed log from becoming a new catalog merely because its secrecy made it attractive.

Before adopting a broader sampling rule, Esa audited three transfers where one credential ended and another began. Those joints offered the easiest place for later hands, or the archivist, to manufacture continuity. For the ship-year 409 cut she requested the controller spool, the paper service abstract, and the nightly-rebuild receipt as separate records. The spool carried a pressure kink where its reader had paused. The abstract bore a seal applied two days later. Neither defect changed the address, but together they showed that the abstract had been reconstructed after the operation rather than written beside it. She marked the service reason as secondary evidence and kept the machine delta.

At the next transfer, a copied index assigned one cut to ship-year 477 while the controller sequence placed it in 476. The archivist called the mismatch a calendar normalization. Esa asked for the normalization table. Its first day began after the transaction, so the explanation could not apply. She moved that date to the rejected sheet and retained only a fourteen-month range established by adjacent checksums. The route did not shift. Its neat annual cadence disappeared, which pleased her more than a clean result would have.

She then tried to make other figures from the accepted marks. Ordered by brightness, the points described a tightening spiral for eleven nodes. The spiral failed on the twelfth and required erased trials to reverse direction. Ordered by maintenance class, six cuts formed something like a hooked letter, but four belonged to one projector replacement campaign. Removing that campaign destroyed the hook. A seven-step rhythm emerged when she grouped edits by work interval. The same rhythm appeared in ordinary lamp repairs because teaching schedules governed access to the dome.

Each attractive figure borrowed its strength from a sorting choice. Esa wrote the rule beside every failure, then tested it against five held-back entries she had not used to notice the shape. The spiral predicted none. The hook placed two points in lit catalog fields where no unauthorized transactions existed. The interval rhythm predicted service nights, not coordinates.

Her directed line behaved differently. Its held-back cuts landed near forward bearings chosen from earlier nodes. Credential transfers altered the size of the error but not its sign. When she removed the reconstructed abstract, widened the disputed date, and excluded the whole replacement campaign, the same course remained. Provenance had made it less tidy and more difficult to dismiss.

She stopped reading entry by entry and built a sampling rule. Every transaction would still contribute to the geometry, but she would inspect full provenance for the first cut after each credential transfer, every erased cut near a surviving one, and every edit that changed the distance between existing points. The console could calculate distances without revealing protected text. Esa supplied the equations herself and made it print only bearings.

She divided the record into twenty-year bands. For each band, she plotted surviving coordinates in black, erased attempts in gray, and the prior band's state in a faint dotted line. The first four sheets looked like debris. Points occupied distant voids with no visible order. A child could have pricked them through paper.

Esa laid the sheets on the inquiry room floor because the console surface could not hold them all. She aligned

their dome axes, pinned the corners with unused styluses, and walked around the growing field.

Temporal order gave her nothing. Coordinate order gave her clusters but no instruction. She tried maintenance lineage and found that each successor began near a predecessor's final cut, then moved elsewhere. That suggested continuity of work without revealing the work's shape.

At ship-year 397, distance changed.

An entry did not add a point. It shifted a surviving one by three seconds toward another cut laid fifty-one years earlier. Six years later, a third point appeared on the same bearing. The three marks occupied different projector cells and carried unrelated maintenance reasons. Their alignment exceeded the cells' stated accuracy.

Esa drew a straightedge through them.

The line crossed the physical crease above the Still Pool.

She checked the bearing twice. Then she requested the pre-service state for ship-year 403, when the third mark appeared. The raw transaction altered not only a point but a sequence of low-intensity addresses between the second and third marks. The intermediate lamps remained below naked-eye threshold when viewed alone. Together they formed the faint extension she had traced along the dome skin.

A cut had become a stroke.

The later entries used the method sparingly. No successor risked drawing a continuous line across a broad catalog field. They placed anchor nodes years apart, then joined only short sections where panel sag, seams, or dark lessons could disguise them. The route grew through discontinuities. A reader who trusted catalog constellations would see nothing. A reader who mapped every unauthorized mark would supply the missing intervals.

Esa changed tools. She treated each anchor as a navigation fix rather than part of an image. From every node she calculated the bearing to later nodes, weighting candidates by transaction date and credential succession. Most pairings produced nonsense: sharp reversals, closed polygons, paths that doubled through inhabited portions of the dome. One pairing kept its turn angles within the tolerance of an old course plot.

She followed it from the eastern gap to the crease, from the crease toward the apex, then west through three catalog voids. At each uncertain junction she inspected provenance. The next node always came after the prior one, often under a successor credential. Erased attempts gathered near the chosen bearing like the pencil trials of a careful draftsman.

The shape did not close.

It continued.

The distinction mattered. Decorative figures returned to themselves or ended at a recognizable boundary. This sequence behaved like instructions. Every bend established a new bearing. Unequal spacing encoded intervals. Two doubled points marked a correction. Near the western field, the plotted line reached the edge of the dome's coordinate system and retained a direction that the projection could not display.

Esa extended the bearing across blank paper.

The line left the sky.

She remained kneeling among the sheets while ventilation stirred their red edges. Her first response carried no wonder, only resistance. A route required a frame. Dome coordinates described an interior surface, not the stars beyond the hull. A line drawn through that surface might point anywhere unless the original hand possessed a transform between projected bearings and ship attitude.

“Show whether any cuts coincide with attitude-reference marks,” she said.

The archivist replied, “Attitude references belong to sealed telemetry.”

“I am not requesting telemetry values. Compare transaction addresses with the projection layer’s reference register. Return count and tolerance only.”

Another pause.

“Seven anchor nodes coincide with attitude-reference positions within two seconds of arc.”

Seven exceeded accident. Esa asked for the expected random coincidence rate across the same field.

“Less than one.”

The route had been tied to the ship’s physical orientation. Whoever began it had access to a course frame hidden from ordinary cartographers. Later carriers did not need the raw telemetry if they inherited the transform and the next required bearing. They could continue a line whose destination they had never seen.

Esa went back through the credential transfers. The first hand used many experiments, most erased. The second placed fewer points and began the short connecting cuts. The third established doubled correction nodes. One later carrier worked for only nineteen years, leaving two marks before the credential passed again. Another sustained the line for ninety-three years with a regularity that suggested a life organized around waiting. The seventh credential appeared in ship-year 776. Its cuts came farther apart, but each landed close to the projected course.

She could not infer blood from access. Ward registries distinguished chosen relation from biological descent, and office succession often crossed both. Yet the journal fragment named inheritance, and the credential lineage proved a transmitted practice. Whatever relation joined the carriers, it had outlasted their memories. Nightly reconstruction could patch a citizen’s self. It could not create skill that nobody taught.

The inquiry clock fell below one hour.

Esa gathered her plots into chronological order and used the remaining time to challenge her own route. She mirrored the dome coordinates to test whether she had read the interior surface backward. The mirrored path produced impossible attitude turns. She reversed temporal order. That path entered from the western boundary, wandered through corrections before their anchors existed, and ended at the first experimental point. She removed every node supported by only one checksum. The course held. She removed the continuous seam and retained anchors alone. The bearings still held. She added the erased attempts. They clustered around the same developing line rather than producing a rival figure.

Last, she asked the archivist to calculate whether a random process constrained to catalog voids could create equal or better directional continuity.

“Define random process,” the voice said.

Esa specified the number of points, void boundaries, projector tolerances, and observed spacing. The machine

ran ten thousand permutations. None matched the route through more than six successive anchors. The edit-log held twenty-eight.

She wrote the result without embellishment: *Null model rejected. Ordered cuts form directed course tied to attitude reference. Endpoint lies outside dome field.

The Carvers had not invented a home. They had remembered one.

The sentence arrived before she could defend it. Esa crossed out the last word, then restored it. The log did not prove a home. It proved remembrance in the technical sense: later work preserved information obtained by an earlier hand. What the course led toward remained unknown. She drew a careful line beneath the sentence and added: *Destination not established.

That correction mattered. Hope could counterfeit a catalog as easily as fear.

She turned to the most recent entries.

The seventh credential had expired in ship-year 818. A successor credential opened five days later, derived through an emergency mentorship transfer in Ward Twelve. The protected name remained blank. Under that credential, two anchor nodes had been placed. One dated ship-year 831. Another dated ship-year 839, only eleven months before Esa's inquiry. Both carried valid post-service checksums and survived every reconstruction since.

The newest transaction contained less than a full node. It changed one address by a single machine count, too faint for the dome and too small to affect Esa's plotted bearing. At first she marked it as maintenance noise. Then she noticed its position: immediately beyond the ship-year 839 anchor, on the route's projected continuation.

"When did this transaction occur?"

"Current ship-year, day two hundred sixty-eight."

Esa looked at the inquiry clock, then at the ward date printed on her seal. Day two hundred eighty-one.

Thirteen days.

"Was the access credential inherited from the active branch?"

"Yes."

"Did an automated process issue the transaction?"

"No."

"Was a living citizen present at a physical projection-layer console?"

The archivist resisted for three seconds.

"Yes."

Esa requested the maintenance authorization. The text named a western-truss lamp-balance inspection in Ward Twelve. It had begun at second dark and closed nine minutes later. During those nine minutes, a hand adjusted the hidden address, compared the resulting checksum, and sealed the layer.

Not a relic. Not an old instruction executing after its maker died. A citizen had touched the route thirteen days ago.

The amber border flashed. Five minutes remained.

Esa stacked her sheets and numbered them while the console erased each temporary plot. She surrendered the sheet bearing raw transaction fragments because the seal prohibited it. She kept her reductions: bearings, dates, credential transfers, null test, latest authorization. The clerk outside would see calculations and nothing that named the actor.

Before the log closed, she asked one final question.

“Does the active credential permit another edit?”

“Yes.”

The screen went dark.

Esa left Precinct Seven carrying seven red-edged sheets against her chest. Somewhere under Ward Twelve’s western truss, a living technician retained access to the hidden layer. The route had reached the edge of the false sky, and its next cut had already begun.

Interlude — Three Hundred Years of Small Cuts

Assemblage recovered from a service ledger concealed behind the western-truss projection bus. Entries occupy the same bound volume. Different hands corrected, argued with, and continued one another. Dates follow ship reckoning; damaged margins appear in brackets.

321

I have compared the catalog course with the raw attitude register four times. We diverge by one degree, twenty-nine minutes, and the error grows under thrust. Calibration cannot produce a steady heading. Someone changed our destination or changed the record of it.

I carried the figures to Elder Sorn. She asked the archivist. By evening my query returned as resolved instrument fault, though no instrument had been inspected. Sorn remembered our conversation at supper. After morning catechism she remembered advising me to replace a sensor.

That frightened me more than the vector.

Projection maintenance gives me one place where catalog and machinery disagree without immediate correction. The catalog supplies each approved light. Our service layer adjusts lamps for heat, failing contacts, and warped panels. A mark entered there can return after nightly reconstruction because removing certified maintenance requires a living witness.

Today I moved one unassigned point eleven seconds in the eastern blank. I used a true service order and a false reason. If it survives three mornings, I will know whether this book needs a second entry.

322, added beneath the first page

It survived.

Two brighter trials did not. A cartographer removed one. Automatic balance caught another. Small cuts near joins endure. I will place no pattern a single inspector can see. The raw register gives a sequence of bearings beyond our false destination. I have copied the transform, not the destination name, because a name invites belief before proof.

If I speak openly, the Weave can revise the hearing. If I cut slowly, a later observer may recover the line from its errors. That observer must test every point against service checksums. Trust no inherited story, including this one.

A faded note beside the first signature reads: *seal 44-C retains rejected bearing; "Cael" appears below the telemetry clerk's fold.* The ink differs from the main entry. Whether Cael named this writer, a clerk, or another dissenter cannot be established from this volume.

389

My mother showed me the eastern point when I could hold a calibration wand without scraping its socket. She did not ask for belief. She gave me three checksums, one erased trial, and the rule for converting attitude bearing into lamp address.

I accused her of hiding cowardice inside patience. She answered by taking me to Elder Sorn, who no longer knew why her own seal had approved the first inquiry. I withdrew the accusation, not the question.

The first hand left six anchors. I added a seventh too near a lesson star. It lasted nine days. My second attempt sits lower and has survived twelve years. Failure belongs in this ledger because descendants who inherit only successful cuts will think the method easier than it is.

I have begun the short seam between anchors two and three. Each intermediate lamp remains below ordinary sight. Together they make a line when the dome cools at second dark.

472

The early pages say "true course." I amend that confidence. We possess a course rejected from the catalog and a heading hidden after the unrest of year 310. We do not possess proof that following it remains safe.

I continue for a narrower reason: citizens cannot choose among bearings they are forbidden to know. The cuts preserve the existence of a choice. A future cartographer must decide what the geometry supports.

This generation introduced doubled nodes. Two close points mark a bearing correction, not two destinations. The distinction matters near the apex, where panel sag makes the older transform wrong by four seconds. I measured the sag physically and entered both values. Use the later value while this panel remains in place.

558

I nearly burned the book.

My son woke after fever with no memory of the workshop. The Weave offered him a complete childhood assembled from our ward's accounts, including lessons I never gave. He accepted it because completion hurt less than my corrections. I saw what an unauthorized inheritance can do when it arrives wearing love.

For forty days I placed nothing. Then I read the third hand's amendment and understood that stopping would also choose for strangers. I showed my son the physical panel, the checksum chain, and this ledger. I told him he owed the line nothing.

He found my bad calculation near the western bend. I had copied elevation from an inner-surface chart without reversing the dome. The intended node would have turned the route back toward the spine. He corrected the sheet in red and agreed to learn the wand.

No carrier should continue from duty alone. Demand an objection before transferring access.

636

The credential reached me through apprenticeship, not birth. The ward calls the prior hand my father on some mornings and my teacher on others. Both relations carry partial support. The calibration he taught me survives either account.

Behind seal 44-C I found a copied maintenance margin: *Cael retains the rejected bearing. Ask why before correction.* Paper grain matches the early telemetry stock. The hand does not match these pages. Family speech calls Cael our first cutter. This margin supports a name near the beginning, but not the legend built around it. I leave the fragment here rather than promote it to certainty.

I have extended the western bend by five anchors. One inspector noticed a faint irregularity and filed it as heat

bloom. I signed the repair, lowered two lamps, then restored them over six years. Quiet work sometimes requires undoing one's own proof.

684

The new close-out reader compares every altered address with the catalog before it accepts a service seal. My first attempt returned to zero and printed a discrepancy notice at the ward desk. I destroyed no notice. I answered it as a failed contact test and surrendered the address for review. That channel cannot carry another cut.

We changed the method. Instead of moving a lamp in the controller, I filed one side of a condemned retaining collar, fitted it during an ordinary replacement, and let the certified center remain unchanged. The physical offset gives us less than two seconds before heat scatter swallows it. A second hand must measure from the inhabited rail while the installer holds the collar below. One worker can no longer place and verify a node alone.

This costs more than time. Each offset consumes a collar whose failure history can withstand inspection. Each attempt leaves a tool mark that a structural worker may notice years later. My apprentice refused the first placement because my rail measure lacked a witness dent. We discarded four intervals of preparation and began again. I had required objections in principle; the new reader made me pay for one.

From this date, retain three records for every node: controller checksum, physical collar impression, and an independent rail bearing. If any pair disagrees, remove the collar before morning assembly. The route will grow more slowly. It will also survive loss of the hidden address layer, which now appears likely rather than remote.

711

Food rotation moved our workshop interval. We now enter the layer under public lamps, which makes the old cadence unsafe. I petitioned for contact-cleaning duty at second dark and received it because nobody else wanted the hours.

Only two nodes from my hand will remain. I spent more time preserving access than cutting. This does not shame me. A dead credential carries no route.

The line now reaches beyond the dome's coordinate field. Later carriers must keep the final bearing on paper until attitude telemetry can be opened. Do not bend the line merely to fit the sky people know.

789

My predecessor gave me twelve tests and one story. The tests held. The story changed each time she told it.

I removed the story from the instruction page.

At the western truss, the seam can be felt through a glove where three generations used the same service latch. That wear proves repeated access, not righteousness. I have placed fewer points than the sixth. Each required a maintenance event I could defend if challenged. The route lacks three nodes by my calculation. Age has begun to shake the wand at low settings, so the active credential passes tonight.

To the eighth: do not imitate my hand. Measure your own.

840

Bren Otan. Eighth carrier.

The seventh counted three missing nodes. Her first estimate assumed the old western panel still held launch curvature. It does not. I surveyed the replacement rib and reduced the need to one full node plus a one-count lead mark.

I placed the lead mark on day 268 under a lamp-balance order. Thirteen days later, someone opened the sealed edit-log and tested that transaction against the physical console requirement. The archivist's query trace does not name the reader. It does show cartographic standing.

I have spent twenty-two years protecting a line from observation. Now observation may have reached it.

The first pages feared a ship asleep to its wrong turn. Later hands feared duty without evidence, inheritance without consent, and a map bent to fit its ceiling. They left me methods for those fears: checksums, objections, physical measures, failed cuts kept beside successful ones.

One node remains. I will not place it for a story about Cael, nor because seven carriers waited. I will place it only if the reader can show where the route continues after our projected sky ends.

If no reader comes, the line remains incomplete and honest.

If one does, I will ask for the proof.

11 — The Carver's Face

The edit-log gave Esa a living credential, not a name.

A direct search for its holder would return nothing. The archivist had masked every identity under the inquiry seal, and the maintenance order from day 268 identified only Ward Twelve, western truss, lamp-balance inspection. Nine minutes of authorized access could belong to a technician, an apprentice, a safety witness, or somebody using a forgotten key.

Esa began with the public office roll.

The registry console at Ward Nine's long table required a civic purpose before it opened cross-ward employment records. She entered *cartographic review of dome maintenance provenance*. Accurate, though incomplete. The console asked whether her review concerned a present safety fault. She selected no. That answer denied her personal schedules, residence cells, medical accommodations, and sealed kin claims. It left office, certification, active precinct, mentorship lineage, and service assignments.

Twenty-seven citizens held some form of projection access. Nine could open lamp-control layers. Four had worked the western truss during the current ship-year.

Esa copied their office numbers onto a temporary slate. She did not copy names yet.

A name invited a face, and a face invited selection. Her task required the reverse: establish the procedural line first, then learn who occupied it.

She requested the authorization tree for western-truss lamp balance. The public roll returned one current supervisor, two full technicians, and an apprentice whose key could alter test cells only under witness. Any of the four might have entered the lead mark. The service order listed no emergency, so both technician and witness fields should have been retained. They were not visible across wards.

Koris watched from the other side of the long table, mending a torn grain tally with clear resin. "Can you ask Twelve?"

"I can ask. The request will name my inquiry."

"You have already named it to this table."

"Not to whoever placed the point."

Koris pressed the torn edges together. "You want to find a person trained for three centuries in not being found."

"I want to avoid finding the wrong one."

"Then spend the name."

Esa filed a cross-ward provenance request. She asked Ward Twelve to confirm who signed the day-268 balance inspection, whether the signatory had physical access, and whether the service result remained active. She did not mention drift, route, or hidden address. The request went to Twelve's maintenance clerk under her own name and office.

While she waited, she examined mentorship succession.

Projection credentials did not pass automatically from parent to child. Every holder required a manual-skills examination, a current safety witness, and a ward confirmation that no candidate had been coerced into inherited office. That rule dated from the cryogenic failures, when children had been assigned dead adults' work through reconstructed relation. The roll therefore supplied useful evidence without claiming blood.

One current western-truss credential descended through seven prior mentorship transfers. The earliest public entry began in ship-year 409, after earlier paper records became fragmentary. Two transfers crossed ward lines. One named a daughter as apprentice. Another named no kin relation. Each successor trained under the prior holder for at least six years. The present holder had received restricted access in 818, full access in 823, and western-truss supervision in 829.

The dates matched the sealed tree Esa had copied from the edit-log.

She put a small mark beside office number 12-P-04.

A second candidate shared three dates but diverged at ship-year 776. A third had inherited workshop responsibility after a pressure injury, not through mentorship. The apprentice lacked a credential old enough to derive from the active branch, though an elder could have used the apprentice as cover.

Esa refused to collapse probability into identity. She requested certification tests for 12-P-04. Public versions showed scores and methods but removed private examiner remarks. The holder had failed low-light calibration twice at seventeen, passed at nineteen, and later designed a way to measure panel curvature without shutting down projection. The test method resembled the Carver ledger's instruction to measure one's own panel rather than imitate a predecessor's hand.

Resemblance did not prove descent. It did tell her what kind of technician she might meet.

The office roll could not reveal which credential opened a hidden address, but it did preserve load accounting. Every projection-layer entry drew a measurable current through one of three isolation buses. Ordinary service reports attached that draw to an order so engineering could reconcile heat and lamp wear. Esa requested western-bus load for the dates of the last five unauthorized cuts.

The console refused at once. Detailed bus use counted as infrastructure security outside her precinct.

She revised the request to return comparisons rather than values: whether a candidate's public service interval overlapped an unexplained western-bus draw, with no date or current displayed. The console accepted the narrower form but required a second office to witness her purpose. Koris could not sign because ward eldership did not grant technical review. Esa carried her slate to the dome supervisor's desk and waited while a shift change argued over two dim teaching stars.

The supervisor read her request. "You are using maintenance power to identify a worker."

"To eliminate workers whose access never touched the relevant bus."

"What fault justifies it?"

"No safety fault."

"Then they can refuse comparison."

“I know.”

“You will generate a notice to each office whose record returns a match.”

That changed the cost. A comparison might narrow her search, but it would also tell the likely carrier that a cartographer had begun following service traces. Esa thought of the day-268 lead mark. Whoever placed it already knew someone had opened the edit-log, if Bren’s later journal entry proved current. Surprise no longer protected her. Transparent purpose might. It also denied Bren one easy explanation for her arrival: she had not followed him through a private relation, a borrowed key, or an accusation hidden from review.

“Generate the notices,” she said.

The supervisor added a witness mark.

Two of the four candidates returned no overlap. The apprentice returned one, during a supervised cell test that public records explained completely. Office 12-P-04 returned four overlaps across eleven years. The fifth cut predated its holder’s full certification but occurred during restricted apprenticeship under the seventh credential.

Esa opened the apprenticeship record. The trainee had spent that night conducting continuity checks while the supervisor replaced a lamp collar. Either person could have made the cut. The overlap therefore linked the method to their shared bench, not to one hand. It also showed transfer in action: the final years of one credential coincided with the first hidden work under the next.

A registry notice appeared before she closed the page. *Comparison disclosed to affected office: cartographic provenance review; requesting citizen Esa Miro; no allegation entered.

The holder of 12-P-04 now had her name.

Koris found her back at the long table, watching the notice receipt.

“Did you find the person?”

“I found where the instruction passed.”

“Difference?”

“A person can deny an act. A workshop cannot deny current draw.”

“Workshops do not answer questions.”

“No. People do.”

The formal reply from Ward Twelve arrived after second meal.

Day-268 order completed by Bren Otan, projection technician and western-truss supervisor. Physical console access verified by local contact key. Witness requirement waived under solitary low-energy service provision 6.4. Result remains active. No safety fault reported.

The reply included Bren’s office number: 12-P-04.

Esa read it twice, then opened the waiver. Provision 6.4 permitted solitary work when no lamp exceeded lesson intensity, no catalog address changed, and automatic rollback remained armed. The hidden lead mark met the

first condition and violated the second. Automatic rollback had not removed it because the service state carried a certified checksum.

Bren had signed a form accurate enough to pass and incomplete enough to hide the cut.

The registry offered a contact channel. Esa closed it. A written inquiry would become part of Ward Twelve's civic correspondence. If Bren refused, the archive would still preserve the question. She needed to approach through the work he could inspect, not through a record the archivist could interpret before he did.

She traded her third-day catalog shift to Vhen's former apprentice, surrendering two rest intervals in return. Then she requested entry to the western-truss workshop for a physical comparison of dome coordinates and maintenance addresses. The request named no worker. Ward Twelve approved it for late interval, when projection load fell and a visiting cartographer could occupy the service rail without interrupting lessons.

Every step left provenance. Esa felt the irony while the lift carried her along the habitation spine. She hunted a secret line by making her own movements legible. Concealment had protected the route, but concealment would not help Bren distinguish her from a trap. Let the registry show that Esa Miro came under her own office, on a stated question, during an approved interval.

Ward Twelve occupied a narrower section of the ship than Nine. Repair shelves lined its main corridor. Residents left broken stools, pressure latches, teaching lamps, and cracked cups beside paper tags naming the fault and the owner's consent for reuse. Nothing vanished into a general maintenance queue. Each object kept its history attached.

At the ward desk, a clerk checked Esa's entry token against the approved request.

"Supervisor Otan has not accepted an appointment," the clerk said.

"I did not request one."

"He may assign another technician."

"That will answer part of my question."

The clerk studied her, then returned the token. "Western truss. Follow the blue conduit. Do not cross a black floor stripe without local witness."

The blue conduit climbed through two service ladders and a sloped passage behind the dome. Projector relays clicked beyond the wall in a rhythm unlike anything heard from the public side. Above the Still Pool, the false sky had looked silent. Under its skin, each point demanded current, cooling, alignment, and someone willing to crawl through dust to keep it bright.

Esa passed racks of replacement lamps sorted by base type. Handwritten cards corrected the archive inventory. One tray held cells labeled *fits launch socket only after collar removal*. Another carried three failed contacts beside the dates each had been tested. Ward Twelve's practice resisted clean lists.

The western-truss workshop stood behind an open mesh door. A black stripe crossed the floor at its threshold.

One man worked inside. He sat at a bench with a calibration wand clamped between both hands, adjusting a collar by fractions against a paper gauge. Gray had begun above his ears. He wore no supervisor's sash. A burn mark crossed one cuff. The local contact key hung from a cord at his chest.

Esa stopped before the stripe.

“Bren Otan?”

His eyes lifted before his hands did. “Cartographer Miro.”

He knew her name from the entry request, she told herself. Nothing more.

“I need a witness across the line.”

“You need a local witness. Not necessarily me.”

“Will you assign one?”

Bren loosened the clamp, removed the wand, and checked its tip under a lens. “State the comparison.”

“A dome coordinate against a hidden-layer address.”

“Hidden from whom?”

“The catalog.”

His thumb stopped on the wand’s collar.

The pause lasted less than a second. Esa would have missed it before the edit-log taught her to read intervals.

Bren set the wand down. “Stay there.”

He crossed to a wall panel and turned a mechanical key. Three lamps above the bench went dark. A fourth remained lit over the mesh door. He checked a meter, wrote its value on the access sheet, then stepped across the black stripe to Esa’s side.

“Now you have a witness,” he said.

Together they crossed.

The workshop carried warm resin, hot dust, and the faint sharpness released by opened electrical contacts. Esa had expected the odor because every dome repair smelled of sealant from above. From beneath, resin came in distinct grades. Jars bore curing dates and panel numbers. The western seam had its own shelf.

Bren pointed to an empty section of bench. “Your coordinate.”

Esa laid down one red-edged sheet from the sealed inquiry. She had copied the route’s anchor bearings but omitted credential dates and Ward Twelve’s order. Beside it she placed her physical survey of the crease above the Still Pool.

Bren did not touch either page.

“Source?”

“Edit-log under archivist seal. Every anchor on this page has a raw transaction, independent checksum, and survival through multiple rebuilds.”

“Copied?”

“Reduced by hand. The seal prohibited transaction strings.”

“So I cannot verify your reduction.”

“No. You can challenge it.”

He looked at her then, not the sheets. His gaze carried no surprise, only an exact reserve.

Esa continued. “I removed unsupported points. I reversed the dome surface. I tested random void placements. Twenty-eight successive anchors preserve directional continuity. Seven coincide with attitude references. The route leaves the projected field.”

“You call it a route.”

“I call it a directed course tied to ship attitude. Destination unknown.”

“Better.”

The single word made relief move through her too quickly. She contained it.

Bren took a clean sheet and drew two arcs. “Which side of the projection skin did your log coordinates describe?”

“Controller side.”

“Your physical survey?”

“Inhabited side.”

“Transform.”

Esa rotated her page, reversed elevation, then corrected for the western replacement rib listed in the maintenance roll. Bren watched her pencil. At the third anchor he tapped once.

“You used launch curvature.”

“Only before ship-year 712.”

“The rib changed in 709.”

“The public record says 712.”

“The work began in 709. Certification followed later.”

“Can you prove that?”

Bren opened a drawer and withdrew a metal gauge. Its edge carried three maker marks, each dated. He set it against the service drawing. A four-second discrepancy appeared at the western bend.

Esa recalculated. The route’s broad direction held. One of her uncertain bends narrowed.

“Your correction strengthens it,” she said.

“It could also be bait shaped to please you.”

“Then I need its provenance.”

For the first time, his mouth shifted. Not a smile. Recognition of a proper demand.

He gave her the gauge’s inventory number. Esa queried the local terminal. Purchase alloy, maker transfer, 709 service entry, 712 certification. All appeared, including the three-year delay.

“Accepted,” she said.

Bren did not put the testing tools away. He selected a lamp collar from a tray marked *condemned*, covered its inventory stamp with his thumb, and set it beside a controller address.

“Where does that address land?”

Esa reached for the service drawing.

“No registry,” he said. “You came to compare physical work.”

“A physical comparison begins with instrument state.”

“The dome will not wait for your paperwork.”

“Then it can wait for a wrong answer.”

She examined the collar before converting the address. Its inner seat carried a polished crescent, evidence of repeated fitting. One retaining ear had been filed lower than the other. Bren had aligned the stamped center to the drawing, but the actual optical center would tilt under clamp pressure. Esa measured both ears with his paper gauge and wrote two possible bearings, separated by six seconds.

“Choose,” he said.

“Clamp it.”

He fitted the collar into a dead socket mounted at the bench edge. His fingers found the retaining notch without looking. When the spring engaged, the lower ear seated first and rolled the lamp toward the filed side. Esa selected her second bearing.

Bren connected a low test current. A point appeared on a ground-glass plate above the socket, five seconds from her prediction.

“One second outside your figure,” he said.

“Your plate is not level.”

He checked the bubble inset along its base. It stood centered.

Esa crouched until the plate edge met the bench line. A grain of cured resin held one rear foot above the wood. She scraped nothing. She slid a clean paper gauge beneath the opposite foot until the plate stopped rocking, then recalculated from the corrected plane. The point fell inside her uncertainty.

“You found the bench,” Bren said.

“You found the seat.”

He cut the current and reversed the exercise. This time he gave her an observed point and asked which part had moved. Esa listed controller value, collar tilt, plate level, heat scatter, and sight alignment as unresolved causes. Bren touched the socket twice, then named the retaining spring. Its return sounded duller on the second release.

“You heard that,” she said.

“I fitted these before you entered cartography.”

“Experience narrows causes. It does not record them.”

“Records do not feel a spring bind.”

Esa wrote the spring’s inventory number on the test sheet and added the measured offset. Bren replaced it with a known spring. The point returned to the predicted bearing.

Neither method had won. His hands reached the fault before her table did. Her table preserved why their correction worked after his hands left the room. The distinction between them no longer resembled suspicion. It looked like two instruments built for different errors.

Bren returned the gauge to its drawer. “What do you want from the holder of your living credential?”

The direct question altered the room. Esa heard projector relays beyond the wall and a cart wheel crossing Ward Twelve’s corridor. She could still retreat into professional language. Doing so would insult every test he had just given her.

“I want to know whether the line continues,” she said. “I want to know what evidence its carriers preserved outside the edit-log. I want the person who touched the layer thirteen days ago to look at my plot and tell me where it fails.”

“You assume one person.”

“The transaction required physical presence.”

“That proves a hand.”

“The credential descends through seven mentorship transfers.”

“That proves instruction.”

“The journal calls it a line.”

“Journals call many things blood after enough funerals.”

Esa adjusted the claim. “A succession, then. Not necessarily blood.”

“Better.”

Bren’s answers remained spare enough to make her supply every bridge. He asked how she had obtained the journal. She described the archivist’s seal but not its bargain in moral terms. He asked who else had seen the reduction sheets. She named Koris and no one else. He asked whether Esa had copied his name into her private work. She showed him the temporary registry slate, which carried only office numbers until Ward Twelve’s formal reply.

"You came through the desk," he said.

"Yes."

"Why?"

"So you could know who entered."

"And so the archivist could know."

"It already knows I read the log. Hiding my walk would not hide my question."

Bren considered that. "Not how we worked."

"No."

"May be why you found us."

He went to the mesh door and closed it. The remaining lamp still illuminated the threshold, preserving the safety record. He did not disable the local terminal or ask Esa to surrender her slate. Those omissions felt deliberate.

From under his collar he drew the contact key she had noticed. Its metal edge showed wear where a thumb had turned it through the same arc many times. He placed it beside her route plot.

"I am the eighth."

No title followed. No account of ancestors. Four words, stated with the same economy he had used for gauges and coordinates.

Esa looked at his hand. Resin had settled in the lines beside his nails. A small white scar crossed the pad of his index finger, placed exactly where a low-intensity wand pressed during fine adjustment. The centuries did not become visible in his face. They remained in the papers, dates, failed tests, and inherited access. Bren carried the current portion.

"How many nodes remain?" she asked.

"One, if your transform holds."

"And if it does not?"

"We stop."

The answer removed any romance she might have brought into the workshop. Good. A route protected by people willing to stop for bad evidence could be trusted more than one completed for duty.

Bren lifted the contact key and pointed with it to the line where Esa's reduction met his corrected curvature.

"Return at second dark," he said. "Bring the raw bearing from your physical seam survey. No conclusions."

"What will you bring?"

"The next coordinate."

He opened the mesh door. Esa crossed the black stripe and turned back.

Bren had already returned to the bench, but his calibration wand now rested across her plotted line. For the first time, carrier and cartographer had placed their measures on the same surface.

12 — The Agreement

At second dark, Ward Twelve lowered its public projection load and left only navigation lessons burning over the dome. Esa returned to the western truss with her original seam measurements rolled inside a starch tube. She brought no route plot.

Bren noticed.

“Good,” he said.

The mesh door stood closed. He opened it after checking her approved entry against the workshop slate. Her earlier visit had not granted continuing access. Esa signed again, stating *physical coordinate comparison*. Bren signed as local witness. Both names remained visible on the sheet.

Inside, he had cleared the calibration bench. A brass attitude ring occupied its center, scarred by repeated clamps. Beside it lay the metal curvature gauge from their first meeting, three paper tables, and a sealed envelope darkened at the folds. Bren pointed to the empty bench space nearest Esa.

“Your measures first.”

She unrolled the physical survey. It contained nine readings of the crease above the Still Pool: three under cold projection, three after teaching load warmed the panels, and three taken while the seam’s faint line appeared. Each reading carried instrument number, time, local temperature, and the name of the worker who had checked her level.

Bren read the witness marks before the numbers.

“When is dead,” he said.

“The early survey came from his teaching archive. The level record remains public.”

“Copied through the Weave?”

“Paper original in Precinct Seven. I compared the pressure dents along its edge.”

He set that sheet apart, neither accepting nor rejecting it. “Show the transform.”

Esa mounted the brass ring on two blocks and treated its inner rim as the inhabited face of the dome. She marked her observed seam, reversed it through the projection skin, then applied heat sag from each survey interval. Bren supplied no correction. When she reached the western replacement rib, she used ship-year 709 rather than its certification date.

Her final bearing extended beyond the ring by twenty-three degrees and four minutes.

“Uncertainty?” Bren asked.

“Six seconds through the seam. Eleven at the unsupported endpoint.”

“Too clean.”

“I rounded after carrying panel error.”

“Do it without rounding.”

She began again.

The second calculation took thirty-eight minutes. Bren interrupted only when her pencil obscured a sign. He did not offer a number. Esa carried each instrument tolerance separately, included the dome’s asymmetric curve, and let uncertainty widen where the line crossed cells she had never surveyed. The endpoint shifted by two seconds. Its error grew to fourteen.

“Better,” he said.

“Your coordinate.”

“Not yet.”

He removed the newest hidden lead mark from her evidence by covering its address on the controller map. Then he removed the ship-year 839 anchor. “Extrapolate from what the seventh inherited.”

“That discards two valid entries.”

“It discards my entries. If your route depends on trusting me, we learn nothing.”

Esa accepted the test. She rebuilt the final turn from older anchors, using doubled nodes as correction marks and erased trials only to bound the intended bearing. The line reached the western boundary at a slightly steeper angle. She extended it across the attitude ring.

Bren opened the sealed envelope.

Inside rested a thin paper strip carrying one coordinate, a brightness value, and a service-layer address. Three hands had written on it. The oldest supplied the bearing. A later hand corrected for the replacement rib. Bren had added a current controller address in darker ink.

Esa compared its bearing with her extrapolation.

Difference: nine seconds.

Within uncertainty.

Neither spoke for several breaths. The result did not prove the inherited destination. It proved independent continuity. Esa had derived a final position from the edit-log and physical seam. Bren had received one through a private chain that predated her inquiry. Their methods converged without sharing the disputed entries.

Convergence could still conceal a copied source. Esa turned the paper strip toward the bench lamp. Three fiber patterns crossed its width. The oldest portion had been cut from a maintenance form; faint ruling remained beneath the coordinate. A repair seam joined a newer extension after the western rib correction. Bren’s controller address occupied a third slip attached with reversible starch.

“Open the custody list,” she said.

Bren brought out a narrow ledger. Each transfer recorded the strip’s dimensions, stains, folds, and missing corners. Esa began at the earliest entry that included measurements. Ship-year 476 listed a full finger of width. By 558, water damage had taken one lower corner. The current strip lacked exactly that corner. A dark oil crescent first appeared in 711 and remained visible near the correction seam.

“Nothing before 476,” Esa said.

“Paper description, no dimensions.”

“So three carriers could have replaced it without detection.”

“They could.”

His readiness to concede the gap increased her trust more than a defense would have.

Bren placed the strip under a low ultraviolet service lamp. Old ink absorbed unevenly. The original bearing and the earliest custody note shared one faded brown response. The correction used a darker compound common after ship-year 700. His address remained black.

“Dates broadly fit,” Esa said. “No proof of authorship.”

“No.”

“Now test mine.”

She gave him her route table with one bearing covered. It came from an erased attempt near the apex, not from a surviving node. Bren knew the inherited route, so a successful prediction near that line would establish little. Instead she asked him to calculate where a cautious cutter would have tried first if panel sag had been underestimated by the amount in ship-year 403.

He chose a lamp cell, applied the old curvature table, and wrote a coordinate. Esa uncovered the erased transaction.

His result missed by nineteen seconds. The failed cut sat twenty-one seconds from the surviving anchor in the same direction.

“Close,” she said.

“Close enough to share an error source. Not close enough to claim the same hand.”

“Agreed.”

They attached that result as another limit. The paper trail could support age, continuity of custody after 476, and technical knowledge consistent with early panel conditions. It could not carry an untouched message from ship-year 321 merely because later carriers wanted one.

“Your strip needs provenance,” she said.

“It has custody.”

“Custody is not provenance.”

Bren’s eyes hardened. “Seven carriers kept it dry.”

“That tells me who protected the paper, not where the coordinate began.”

“The first hand copied raw telemetry.”

“According to a journal written by the same line.”

He folded the strip once along an old crease. "Then we stop."

Esa heard the finality and let it stand. Stopping belonged inside any honest method. She pointed to the oldest figure.

"I cannot certify its origin. I can certify that the log reconstructs the same bearing from physical changes made across five centuries. That supports the figure without proving its source. Mark the limit."

Bren looked down at the paper.

"With what words?"

"'Original telemetry unavailable. Bearing independently recovered from surviving edit geometry, ship-year 840, within fourteen seconds.' Add our methods, not only our names."

He considered, then unfolded the strip. On a separate sheet he wrote her sentence in a compact hand and attached references to the ring calculation, seam surveys, and his custody list. He left space beneath for objections.

"Accepted with limit," he said.

Esa entered the same wording on her survey.

The exchange cost both of them a desired certainty. Bren gave up an unbroken claim from the first carrier. Esa gave up treating statistical continuity as proof of destination. What remained had narrower reach and stronger supports.

They moved to the controller.

The final node occupied a catalog void near the western edge, four cells beyond the lead mark Bren had placed thirteen days earlier. No lesson star used the address. The local projector could produce the point at six hundred brightness steps, but only the lowest eight sat beneath ordinary dome perception. Bren's inherited table specified three.

"Three survived during cold load," he said. "Tonight the panel carries navigation lessons. Heat lifts local scatter."

"Use two and measure."

"Two may vanish after rebuild."

"Then we learn before certifying it."

Bren opened the low-energy service provision he had used for the lead mark. This time he did not waive a witness. He entered Esa Miro, cartographer, at the physical console. The maintenance form asked whether they intended to alter a catalog address.

Bren's finger hovered over *no*.

Esa watched him. "The address has no catalog point. Altering it still answers yes."

"Yes closes solitary service."

“You are not solitary.”

“It sends a provenance conflict to the archivist.”

“So does an honest answer.”

He selected *yes*.

A new warning filled the screen: *Catalog authority absent. Service change may persist for test interval only. Permanent certification requires cartographic necessity or safety correction.

Bren read it without expression. Five centuries of cuts had survived through reasons shaped to pass around that warning. The final one faced it directly.

“Cartographic necessity?” he asked.

“I can certify measurement of an uncatalogued geometric feature. Not its meaning.”

“Enough.”

He inserted his contact key. The controller exposed the target address and neighboring lamp states. Esa recorded pre-service checksum, panel temperature, bus load, and dome attitude. Bren fitted the calibration wand to the low-intensity socket.

“Test interval,” he said.

“Begin.”

He advanced the address by two counts.

Nothing changed in the workshop except bus load, which rose by a fraction. Esa watched through a shielded service scope aimed along the projector path. A pale point formed in the optical grid, blurred on one side by panel scatter.

“Bearing high by seven seconds,” she said.

“Wand centered.”

“Panel gradient.”

Bren checked three temperature contacts. The upper contact read half a degree warmer than the lower pair. He applied the service table’s compensation. The point moved four seconds, not seven.

“The table assumes launch adhesive,” Esa said.

“Current adhesive record is incomplete.”

“Measure the gradient through the image.”

She placed the scope on a second mount one meter east. Parallax separated lamp position from scatter. Bren held the test state while she alternated between mounts, calling offsets. At two counts the point remained too diffuse to fix within fourteen seconds.

“Three,” she said.

“That raises visibility.”

“Only during test. We need a coordinate before we choose permanence.”

He advanced one count.

The point sharpened. Esa measured from both mounts and calculated its controller bearing. Five seconds high, one second north. Bren lowered the optical collar by the smallest marked division. The next reading fell within six seconds of the inherited coordinate.

“Hold,” she said.

A timer on the console counted thirty breaths. They watched bus current for instability. At breath nineteen, a neighboring lamp dropped one count.

Bren cut the test.

The node vanished.

“Shared contact,” he said.

He opened the panel map. Target and neighbor should have occupied separate branches, but a repair in ship-year 674 had bridged their returns. Driving the node at three counts stole current from a named navigation star. Previous carriers had never faced the problem because their cuts used other cells.

“Can the bridge be separated?” Esa asked.

“Not before public projection rises. Opening the return block takes half a shift.”

“Then use two counts after the panel cools.”

“Too broad to verify.”

“From above, perhaps not.”

They needed a third instrument position on the inhabited side. The workshop had no live optical feed from the dome rail because such a feed could reveal service tests to public observers. Esa could go above, but crossing from western truss to Still Pool took twelve minutes. Projection load would rise at the next lesson interval in forty-six.

Bren set the node to two counts and left it temporary. He connected a hardline voice tube between controller and rail, a maintenance channel without Weave transcription. Esa took the eastern ladder upward carrying a sight, level, and paper table.

Ward Twelve’s public dome lay mostly empty at second dark. Two sanitation workers crossed the far gallery. Navigation lessons glowed low over the eastern field. The western gap looked blank until Esa shielded her eyes and fitted the sight against the rail mark.

“Position,” Bren’s voice said through the tube.

“Still Pool west mark. Level checked. Read bus.”

“Stable.”

She searched the inherited bearing. At first she saw only the grain of the projection skin. Then the point emerged when she looked beside it rather than at it, faint enough that direct attention erased it.

“Visible,” she said.

“Fix?”

“Working.”

Esa took five readings, lifting her eye between each so memory of the light would not pull the sight. Two fell outside tolerance. Three clustered within eight seconds of the private coordinate. She moved four meters along the rail and repeated the series. Parallax placed the point in the projection skin, not beyond it, as expected. Corrected through the dome surface, both positions agreed within twelve seconds.

“Node holds at two,” she said.

“Can you certify?”

She looked at the small light. Certification would attach her office to an unauthorized feature and trigger review at morning rebuild. Refusal would send Bren back to a false maintenance reason or leave the route unfinished. Neither choice stayed private now. Their entry sheet, power comparison, and Ward Twelve notice had already created a trail.

“I certify an uncatalogued reference point required to preserve a measured course geometry pending archive review,” she said. “No destination claim.”

Bren repeated the words to confirm them.

“Enter exactly,” Esa said.

Below, he committed the service state. The hardline carried the click of his key turning. A moment later her slate received a local checksum request. Esa compared it with the pre-service value and signed the changed address as the sole intended difference.

The point went dark during automatic rollback.

Esa’s breath caught.

Then it returned.

Its return did not complete the placement. Through the rail sight, Esa watched the point creep starward as the permanent current warmed the collar. At twelve breaths it crossed the upper limit of her certified band. At sixteen it stopped nine seconds from the coordinate.

“Cut service power,” she said into the tube.

“Bus remains stable.”

“Bearing does not.”

Bren used the manual hold rather than changing the address. The point faded. When Esa reached the workshop, he had left the wand and controller untouched. A curl of pale resin showed beneath the collar’s retaining ear. Test heat had softened an old repair; rollback pressure then settled the collar against a different face.

They treated the failed placement as evidence, not an inconvenience. Esa read the controller state aloud while Bren copied it onto a red-edged service sheet. He marked the collar before removal at top, inhabited side, and contact face. She signed across each mark. Only then did he release the spring.

The resin curl came free with the collar. Bren caught it on black paper instead of brushing it from the socket. Its flattened edge matched the nine-second direction of travel. He placed paper and collar in separate starch sleeves, wrote the time of removal on both, and gave Esa the serial halves from their seals. The failed state now had custody independent of his workshop ledger.

A clean collar would have changed two variables at once: seating face and part history. Bren chose the same collar, removed no material, and rotated it so the undamaged ear carried spring pressure. Esa objected that rotation could move the optical center. He agreed and entered it as a new placement rather than a correction hidden under the prior reading.

They repeated the test at one count before returning to two. From below, Bren called bus load and collar temperature every five breaths. From the rail, Esa took blind readings in alternating order so expectation could not pull them toward the inherited coordinate. The first series sat four seconds low. After the collar reached stable heat, the next two series converged within seven seconds. No neighboring lamp dimmed.

Bren committed the mechanical orientation without altering the controller address. A fresh checksum matched the intended single-address change already signed. Esa attached her three reading sets, including the failed high drift. He closed the physical-placement packet with both seal halves visible under the starch.

Only after custody and correction agreed did Esa certify the node again.

Permanent maintenance state had restored it after catalog authority rejected the empty address. For the first time, a Carver node survived under an accurate public purpose and two living names.

The cost arrived immediately. Her slate displayed a formal notice from the archivist: *Uncatalogued cartographic necessity entered at Ward Twelve. Inquiry required before next work interval. Service credentials subject to temporary review.

Bren's access would freeze at morning bell. Esa's cartographic standing could follow.

A chime sounded at the mesh door before Esa reached the lower ladder. Ward Twelve's duty clerk stood outside the black stripe with the same rigid posture used for smoke alarms. The clerk held a paper copy of the provenance conflict and a lockbox for Bren's contact key.

"Service review requires custody of active credentials," the clerk said.

"Review begins at morning bell," Bren replied.

"Protective custody begins now if catalog authority and local service disagree."

Bren looked at the controller. Removing his key would not erase the certified node, but it would prevent any correction if the shared contact began stealing current again. "Neighbor branch showed a one-count drop at three. We committed at two. Bus stable for fifty-eight breaths."

"Recorded?"

"Here." Esa offered her measurements.

The clerk did not take them across the stripe. "State whether a named navigation star lost intensity under committed state."

"No," Esa said. "During a higher test state, yes. We ended that test and recorded the bridge."

"Can the bridge fail under two counts?"

Bren answered. "Possible under teaching heat. Low likelihood before morning."

"Possible requires local watch."

"I will watch."

"Without a contact key?"

"Manual cutoff remains outside credential control."

The clerk considered the procedure, then opened the lockbox. Bren removed the key from his neck. Its cord had left a pale line beneath his collar, evidence of years spent carrying access even while he slept. He placed it in the box.

"Seal number," the clerk said.

Esa read the number aloud while Bren checked the lock. The clerk signed custody and passed them a carbon copy. "Workshop stays under physical watch until lesson rise. No layer entry. Any lamp change triggers dome evacuation from the western gallery."

Bren gave one nod.

The clerk's gaze moved to the faint route geometry on Esa's paper. Curiosity showed, then discipline shut it away. "Your inquiry can explain the rest at morning."

After the door closed, Bren tested the manual cutoff and laid a stool beside it. He would spend the remaining dark interval watching current without authority to alter the work. Esa understood the practical price of her insistence on an honest form. Secrecy had let him manage every consequence himself. Public provenance brought outside custody, a second judgment, and limits neither conspirator could waive.

"We could have used no," he said.

"And kept your key."

"Yes."

"Do you regret it?"

"Ask after the review."

He had already printed the notice and placed it beside the Carver strip.

"One node," he said. "One career each."

"Review is not revocation."

"Yet."

They checked the work before discussing further consequence. Bren closed the controller housing as far as the custody rule allowed and handed Esa the post-service checksum copy. She rebuilt the route with the final node included. Its endpoint still lay outside the dome. More important, the unequal spaces between doubled nodes now divided into groups. Read as distances, they looked arbitrary. Read as old coordinate intervals, they produced a repeated structure.

“A cipher,” Esa said.

“Possibly.”

“The final group needs the attitude register.”

“Sealed telemetry.”

“And early destination tables. Also sealed.”

Bren folded his arms. “You can ask the archivist.”

“It may suspend me before answering.”

“Then Ward Twelve asks why a truthful service entry caused suspension.”

“That exposes the line.”

“It is exposed.” He tapped their names on the notice. “Decide what replaces hiding.”

Esa placed her route reduction beside his inherited strip. “Joint custody. Every claim carries source and limit. No public destination until raw telemetry verifies the course. No further change to the projection without a second witness.”

Bren added, “Failed tests stay with successful ones.”

“Yes.”

“If telemetry contradicts the route, we stop.”

“Yes.”

“If it holds, you take the evidence to your ward. I stand beside it.”

“And we complete the cipher together.”

He looked at the final node’s service address on the checksum, then offered his hand across the bench. Not a census clasp. A worker’s grip, palm to palm, each able to feel whether the other pulled away.

Esa took it.

The agreement entered no ward ledger that night, but its evidence did. Two signatures held the final node in the false sky. One inquiry notice bound their offices to the cut. Between them lay a sequence neither could decode alone.

At morning bell, they would go to the archivist and ask for the hidden course.

13 — The Homeward Mark

Vhen's slate had spent twenty years teaching Esa where not to look.

Its face carried the ordinary master grid: twelve declination bands, the old constellation joins, corrections in Vhen's square hand. The back carried scratches Esa had treated as a private eccentricity. Six short strokes stood beside a ladder of paired numbers. A hooked line crossed the pairs at irregular intervals. Vhen had once called it a launch-era coordinate exercise and refused to explain further.

"An exercise you cannot solve stays useful," the old cartographer had said. "It prevents the slate from becoming furniture."

Now Esa laid that scratched back beneath the Carvers' completed route.

Bren stood across the western-truss bench with both palms braced against its rim. Between them, a projection sheet held the route at one two-hundredth dome scale. Thirty-seven changes of direction showed as pinpricks along a line too deliberate for stellar drift. The final stroke left the mapped dome and terminated in blank film.

"You have looked at that slate before," Bren said.

"I looked at its numbers. I assumed they named practice coordinates."

"Vhen gave you a cipher without telling you."

"Vhen gave me a problem." Esa turned the slate until the hooked line aligned with the route's eastern origin. "Whether it belongs to this problem remains unproved."

Bren's mouth tightened with approval. Carvers trusted suspicion more readily than hope.

They began with the obvious reading. Each route node had a dome coordinate, a brightness setting, and a date of insertion. Esa ordered the nodes by date. Bren copied the last digit of each declination beside the ladder pairs. The resulting groups fell within the scratched range, which encouraged them for almost four minutes.

The first six groups yielded consonants under an old ward syllabary: R, T, H, K, M, N.

"Could be compressed," Bren said.

"It could be anything if we grant it missing vowels."

They continued. By the twelfth group, the sequence demanded two signs absent from every launch lexicon in the public archive. Esa reversed the dates, then reversed the declination signs. Neither operation produced a valid check group at the hooked intervals.

She drew a line through the attempt.

Bren watched her mark it. "My seventh carrier read insertion order as well."

"What did it produce?"

"Enough resemblance to 'astern' that the family kept it for forty years."

"Resemblance is expensive."

“It placed the destination behind us.”

That made the error more dangerous, not less. A route whose meaning depended on patience could collect false destinations for generations. The Carver line had preserved each interpretation in a narrow repair ledger, but Bren had brought only a copy of the failures. He placed it beside Vhen’s slate.

Seven readings occupied the page. Insertion order had given ASTERN after generous substitution. Brightness order had produced AUN twice, followed by nonsense. Alternating eastern and western nodes had made a valid phrase meaning KEEP LINE, though its check strokes failed. One carrier had treated route curvature as sound length and found the name of a launch captain. The captain’s name came from a lesson text written two centuries after the first cuts.

Esa circled that entry. “Contamination.”

“My third carrier learned the captain in morning catechism. The name entered every uncertainty afterward.”

“Then we forbid remembered words until the arithmetic selects them.”

Bren folded the failure ledger so none of its candidate phrases remained visible.

Esa tested the slate itself. She copied its paired numbers into the catalog console without calling the archivist. The first pair landed at the west edge of the Weavers’ Loom. The second fell near the Still Pool. A third pointed to empty dome. None matched the route nodes exactly.

She increased the epoch by one century. The points shifted under stellar proper-motion correction, but the third remained empty. Two centuries made the first miss its named star. Subtracting an epoch placed all three on catalogued lights, which looked promising until the brightness values disagreed with Vhen’s face by more than the master tolerance.

“Wrong epoch,” Esa said.

“Or wrong sky.”

“The face uses the accepted catalog. Vhen would not scratch a key that required a sky unavailable to a student.”

Bren touched the first paired number. “Unless these are not locations.”

Esa examined the ladder again. Each pair shared a difference of either three, five, or eight. Those differences repeated. The absolute numbers wandered. She had been reading the pairs as positions because Vhen had named the exercise coordinate work. Vhen rarely lied, but the old cartographer had enjoyed allowing a noun to carry two meanings.

She subtracted each lower number from its upper. Six differences appeared before the first hook: three, five, three, eight, five, eight.

Bren copied them. “Only three values.”

“Axes.”

“A coordinate needs at least two.”

“A moving ship uses three if time supplies the fourth.” Esa turned the route sheet. “Across, above, and along course.”

The scratches did not encode points. They encoded operations on points.

She selected the route's first six turns and measured the change in each axis, ignoring magnitude. Every turn could be classified by its dominant change: across the dome, vertical on the dome, or forward along the implied trajectory. She assigned Vhen's three differences to those axes in the order suggested by the slate.

The output still failed.

Two operations landed on impossible zero values. The hooked check after the sixth group gave five; their axis sum gave eight.

Bren leaned closer. "Swap across and above."

"They are defined on the face."

"Defined for someone standing under the dome. The original cipher may face outward."

Esa disliked the suggestion because it made sense. Every modern dome chart used the observer's orientation from inside the habitation. Launch navigators had faced a hull plot from above, looking inward. Clockwise reversed. Port and starward exchanged places.

She mirrored the grid.

The impossible zeros vanished. The first hook now agreed with the sum. At the second hook, however, the check differed by one.

"One damaged node," Bren said.

"Or a poor model."

"The third carrier reported a projector replacement in this segment." He pointed to the western bend. "The replacement copied the visible light, not the Carver offset beneath it."

Esa opened the layer through the maintenance viewer. The route sheet showed one point at the center of its assigned projector cell. In the physical layer, a fine cut sat one division starward of center. Someone had re-seated the visible node after the old projector failed, then left the underlying scratch where the carrier made it.

She corrected the sheet. The second check closed.

A small satisfaction moved through her hands. It did not yet grant them language. They had found a way to convert route turns into a sequence of three operations, validated twice by checks scratched on a slate Vhen had possessed long before Esa noticed the stray light. That relation could arise by chance, but chance now had to explain the checks, the mirrored orientation, and the concealed physical offset together.

They needed an independent sample.

Esa carried the slate to the dome rail after public interval. Bren remained below with the projection layer open. The familiar constellations shone over an empty precinct. Esa selected three short routes from Vhen's face, each drawn between named teaching stars. The old master had marked them with tiny hooked lines identical to those on the back.

"Vhen built examples into the front," Esa said over the service tube.

“Can you read them?”

“Not yet.”

The first example crossed four stars in the Weavers’ Loom. Esa mirrored its turns, converted the dominant axes, and grouped the operations at the hook. The result indexed a launch syllabary entry for LOOM. The second returned POOL. The third produced no word.

Bren climbed halfway out of the service recess. “Two successes.”

“Two labels a student could predict. Prediction contaminates.”

“You did not know the third.”

“The third failed.”

Before examining the face, Esa asked Bren to choose a fourth path from the modern catalog and conceal its name. He selected five points, copied only their coordinates, then folded the source strip beneath his wrist. Esa converted the path with Vhen’s operation values. The check closed, but the resulting group had no syllabary entry.

“That argues against the method,” Bren said.

“It argues against using arbitrary modern routes as launch words.” She unfolded his strip. He had chosen a line through the Lantern Bearer, a constellation named after the seal. No launch label existed for it. “Give me geometry from a pre-seal chart. Do not give me the label.”

Bren found an old training traverse on the cracked film and masked its annotation with repair tape. This path crossed seven points and contained two hooks. Esa mirrored it, classified the turns, then stopped. The fifth point had no brightness figure. Without brightness, she could not decide whether a shallow vertical change dominated the along-course component.

“Choose,” Bren said.

“A choice would manufacture the answer.”

She carried the chart to the rail reader. Its lamp shone through the starch from below. Pressure from an old stylus had left a groove beside the fifth point, though the ink had flaked away. Esa measured the groove’s length against neighboring figures. It could have held a three or an eight. She ran both.

Three broke the first hook. Eight preserved it and produced two valid groups. When Bren removed the tape, the hidden annotation read RETURN.

A modern route failed because its word did not belong to the launch syllabary. A damaged old route offered two physical readings, and only the one that satisfied the arithmetic recovered its concealed label. Neither result depended on Esa anticipating the word.

Bren turned the training film toward the light. “Vhen must have used this exercise.”

“Or learned from whoever did. The method predates the scratches.”

That distinction mattered. If Vhen had invented the system after seeing the Carver marks, the slate offered no independent support. A launch exercise using the same hooks established older use. Esa copied its damaged

figure, both candidate readings, and the reason for rejecting three. Any later reader would need the failed branch, not merely the answer.

“The third path still fails,” Bren reminded her.

“Then it remains our best test.”

She inspected the face under magnification. Vhen had corrected a declination figure beside the third example. The newer digit carried a rounded top. Beneath it, the original angular stroke remained faintly visible. Esa restored the earlier value and ran the operations again.

The group returned ASTERN.

She looked from the slate to the dome. The three front examples did not merely teach labels. They fixed orientation and operation order. LOOM verified the axis assignment. POOL verified grouping. ASTERN distinguished the corrected coordinate from the original and showed that the cipher used the older figure.

“Vhen knew the catalog correction broke the example,” Bren said.

“Vhen preserved the old digit instead of changing the key.”

“Did Vhen know what the key opened?”

Esa could hear the old cartographer answering: an exercise you cannot solve stays useful. “I cannot prove it.”

She returned to the workshop. They ran the Carver route again from its eastern origin. This time Esa refused to translate partial groups. She recorded only operation values and checked each hooked boundary. Bren read physical cuts where repair records suggested a node had shifted. At three places, the visible projection disagreed with the cut below. Two came from replacement cells. The third had moved during a dome recalibration in ship-year seven hundred four. Correcting all three closed every scratched check on Vhen’s slate.

The route yielded nine groups.

Bren did not reach for the syllabary. He laid a transparent scale over the first segment and measured each turn by length instead of dominant axis. “The carriers may have copied quantities. Vhen’s differences could tell us which digit to keep.”

It offered a plausible alternative. Esa divided each turn by three, five, or eight according to the paired-number ladder, then rounded the residuals at the hooks. Six groups accepted launch signs without substitution. The first three read AUN. A later pair could be read as RETURN if one short stroke counted as a damaged course divider.

Bren wrote the candidate on a separate sheet: AUN RETURN.

The phrase fit everything they wanted too readily. A launch navigator could have issued it. A Carver could have preserved it. Even the damaged divider occupied a place where an instruction might pause.

“We need to find what the reading asks us to ignore,” Esa said.

The magnitude method required four turns to round upward and five downward. No check on Vhen’s slate specified that convention. More troubling, two replacement projectors had changed the visible distance between nodes while preserving their physical direction. Esa ran the same calculation from the pre-repair cut

positions. AUN survived. RETURN became three signs without a legal grouping.

“Then the old cuts carry the message and the visible route carries damage,” Bren said.

“Or AUN survives because the first segment begins near Aun’s teaching mark. We still have not tested whether length belongs in the cipher.”

They took Vhen’s front exercises to the measuring cradle. Esa fixed the slate beneath two opposed lamps so neither surface shadow could enlarge a stroke. Bren pressed a sheet of soft witness foil over the scratches and rolled it once with the calibration cylinder. The foil lifted each groove at its physical width and depth. On the LOOM example, the longest drawn turn and the shortest had received equal stylus strokes beside the hook. On POOL, Vhen had redrawn the teaching path at twice the scale, yet the paired differences remained unchanged.

A magnitude key should have changed with either copy. The operation key did not.

The third exercise supplied the harsher test. Under oblique light, Esa found that its hook groove passed beneath the corrected declination digit. Oxidation darkened the bottom of the hook, while the newer digit cut through that film and exposed pale slate. Vhen had made the grouping mark before correcting the coordinate. Any reading that depended on the newer line length reversed the physical order of the work.

Esa photographed both intersections and entered the foil beside the failed candidate. She then used the older angular digit with Bren’s magnitude rule. AUN disappeared. The supposed return instruction had been assembled from one modern projector distance, one rounded residual, and one correction applied after the hook.

Bren drew a single line through AUN RETURN. “My line would have kept that.”

“So would mine if it had appeared in a clean catalog hand.”

They attached the foil, the scale comparison, and both failed reductions to the route packet. The false reading had earned its place by surviving long enough to expose which physical features the cipher did not use.

Esa entered them into the launch syllabary. Seven groups mapped cleanly. Two offered paired readings because early navigators had used the same sign for an aspirated consonant and a course divider.

The literal sequence read:

H O M / W A R ?

Bren stared at it. “Home war.”

“Or home ward.”

“The divider could end the word after home.”

“Then the remaining groups make no valid instruction.” Esa wrote both options and left space below them. “We do not choose the comforting one.”

They tested HOME as a complete label. Launch charts used home for a vessel’s departure system, never for a destination code. Aun carried its own coordinate seal. A destination called HOME would have violated the navigation office’s naming practice, which kept emotional words out of traffic labels.

They tested HOME WAR as two terms. WAR did occur in an emergency abbreviation, but only beside threat

bearings. The route lacked the double hook required before such a suffix. Bren found a Carver ancestor who had favored that interpretation after a hull dispute. The date condemned it: the dispute came long after the first route segment.

HOMEWARD required the ambiguous sign to read as D rather than divider. Esa searched Vhen's examples for the same sign. It appeared once in ASTERN's old explanatory notation, where a shallow finishing scratch meant "continue motion through the named bearing." The mark did not supply a letter alone. It converted a direction into an instruction.

"Homeward," Bren said carefully. "Not home. Toward home."

"Still only a candidate."

"What test remains?"

"The endpoint."

The cipher named what the route claimed to do. A convincing label could still decorate a false road. Esa needed to know whether the drawn geometry ended at a recoverable coordinate or dissolved beyond the dome.

She removed the syllabary from the bench and projected the route on a plain navigational grid. Its last three segments seemed to meet at the blank film, but that impression came from the scale. At full precision they missed one another by a narrow wedge.

Bren's shoulders lowered. "The line does not close."

"Not in dome coordinates."

She corrected the nodes for their insertion dates. A cut made three centuries earlier depicted a star field from another ship position. The Carvers had inserted each mark into the current projection, yet the route they copied came from older telemetry. Proper motion affected stars. The ship's own travel changed apparent angles more strongly. Treating all nodes as simultaneous bent the road.

Esa assigned each physical cut its earliest verified date from Bren's ledger. Where a date range remained, she used both limits and drew an uncertainty band. Then she transformed every node to the launch-era reference plane Vhen's cipher used.

The wedge narrowed but did not close.

A transverse error remained. Its direction matched the mirrored-grid correction, suggesting they had reversed one axis twice. Esa restored modern handedness only for the final projection beyond the dome. The route lines converged.

Their intersection occupied a coordinate absent from the accepted catalog.

She widened the search to Vhen's rolled pre-seal charts. The oldest film had cracked along one edge, taking part of the relevant field. A later copy showed a small ring where no modern star appeared. No name accompanied it. Beside the ring, someone had written the same nine cipher groups in compressed notation.

Bren did not touch the film. "Could Vhen have added that?"

"The hand predates Vhen." Esa compared stroke pressure, starch oxidation, and the shape of the hooks. "This copy came through the cartography office. Vhen preserved it. The label existed before the first Carver cut the

route.”

That fact altered the order of causation. The Carvers had not invented an endpoint and then hidden its name in Vhen’s key. They had drawn toward an old marked coordinate. Vhen’s slate supplied a surviving method to read the mark. The route, the key, and the chart had traveled through different hands.

Esa ran one last hostile test. She shifted the endpoint by one projector cell in each direction and decoded the resulting final group. None produced a valid course word. She dropped the three repaired nodes instead of correcting them. Two check hooks failed. She used modern rather than launch orientation throughout. LOOM and POOL ceased to decode on the slate’s front.

Only one sequence satisfied every independent constraint.

Bren lowered the projection-layer cover. The closing plate caught his thumb because the old hinge had warped. He pulled free, leaving a bright line of blood along the edge. Esa wrapped the cut with chart cloth while the plate’s heat light climbed from amber toward white.

A relay snapped beneath the bench. The sheet buckled as its cooling frame contracted, sliding Vhen’s slate toward the open service gap. Esa caught it against her wrist. Bren reached for the cover with his uninjured hand, but she stopped him and used the insulated locking bar. The cover settled. Hot resin ticked onto the floor where his hand would have been. Neither spoke until the temperature mark dropped below its damage line. Old evidence had nearly vanished through an ordinary warped hinge, which explained more of history than any theory of careful guardianship.

“We stayed too long,” he said.

“Hold pressure.”

He pressed the cloth against his thumb. Esa cleared their working plots from the energized sheet, but she did not erase the final result. She copied the nine groups by hand to Vhen’s slate, beneath the scratches that had made them legible. The plate cooled one increment at a time.

No telemetry had yet proved the star still existed. No survey had shown a world. They did not know what course change would be required, or whether the ship possessed the margin. Those belonged to later work. What they possessed now had narrower strength.

An old chart held an endpoint erased from the accepted catalog. A three-century route converged on it after date correction. Vhen’s cipher decoded its mark under tests that failed when any essential assumption changed.

Esa turned the slate so Bren could read the word written beneath the coordinate.

“Homeward,” she said.

Outside the workshop, the false sky continued through its appointed constellations. On the bench between them, the destination had a name.

14 — What the Weave Hides

The name Homeward proved that someone had preserved a destination. It did not prove that anyone had stolen it.

Esa kept that distinction before her when she returned to the catalog console. Bren waited at the western truss with Vhen's slate and the physical route. Koris stood beside Esa, wearing the ward-elder's plain copper key at her throat. The archivist had refused the deep archive to a cartographer. It could not refuse the elder without stating which civic order outranked her.

Esa entered the inquiry in three parts: original course, maneuver history, suppression authority.

The console accepted the first two and grayed the third.

"Suppression is a conclusion," the archivist said. "The archive contains no civic act under that term."

"Then show us alteration authority," Koris said. "Ship-year three hundred through three hundred twenty. Catalog, memory relation, and navigation."

"Those records remain sealed for continuity."

"Whose continuity?"

"The ward's."

Koris removed the copper key from its cord. "The ward asks."

A narrow slot opened below the console. The key fit poorly. Its edges carried wear from locks replaced over centuries, and Koris had to press it inward while turning. The panel warmed beneath Esa's fingertips.

"Elder access establishes standing," the archivist said. "It does not establish safety."

"No archive establishes safety," Esa replied. "Open provenance only. Do not complete gaps. Do not reconcile contradictions."

That request caused a pause. The Weave preferred a whole account. Esa had asked it to return pieces with their disagreements intact.

The console divided into three gray fields.

First came inertial telemetry. The accepted catalog showed *The Long Veil* following a line toward the nearer daughter of Aun. Raw star fixes showed a growing transverse departure. Esa selected readings taken by different sensor clusters, because one failed cluster could imitate drift. Port optics, dorsal tracker, and the old mass-reference gyros all recorded the same change. Their errors varied. Their direction did not. The comparison also crossed separate power buses, so a voltage fault could not bend every source together without leaving a second trace.

She plotted the fixes without catalog correction. The line bent.

A fourth source complicated it. The old navigation clock recorded a short jump during the first turn interval. If its time base had slipped, three sensor clusters could appear to agree because the console compared them against the same false moment. Esa separated each instrument from ship time and rebuilt the sequence from

local pulse counters. Dorsal optics lost twelve pulses. Port optics lost eleven. The mass-reference gyros lost none.

For half an hour, the course departure shrank into a clock problem. Bren found a maintenance notice stating that a failed timing crystal had been replaced during the same interval. The accepted summary used that notice to classify the heading conflict as calibration fault.

Esa nearly accepted it. Then she compared the replacement time with the thruster valves' mechanical cycle count. The clock jump occurred after the first lateral burn, not before it. Correcting the jump tightened the three sensor curves rather than straightening them. Whoever wrote the summary had applied a real timing fault to an earlier maneuver it could not explain.

"Error used as cover," Bren said.

"Possibly. Or a clerk attached the nearest repair."

"You keep leaving them innocence."

"I keep leaving us uncertainty." Esa marked both events on separate lines. "The clock failed. The ship also turned. One does not erase the other."

She ran a final control from twenty ordinary attitude corrections in the preceding decade. Random sensor faults produced disagreement before a corrective burn and convergence after it. Here the sensors agreed before the burn, moved together through it, then disagreed only when the clock failed. The pattern belonged to commanded motion.

"How much?" Koris asked.

"One degree and a half by the first complete record." Esa marked the uncertainty. "More than sensor tolerance. Too smooth for a strike."

"Could it be accumulated calibration?"

"It could, if three independent systems acquired the same error at the same rate."

Bren's voice came through the maintenance tube. "The Carver journal began with that number."

"A later witness," Esa said. "Useful, not controlling."

She asked for the thruster ledger. If the change came from damage, thrust would show correction attempts fighting it. If it came from command, the maneuver profile would show a coordinated yaw.

The ledger opened in blocks. Many carried ordinary decay: dead sensors, remapped valves, revised mass estimates. Esa laid consumption against heading. A cluster of lateral burns appeared near the beginning of the course departure. Each burn remained within emergency authority. Together they formed a deliberate turn.

The accepted navigation summary called them *attitude recovery following guidance disagreement*.

"Recovery toward what?" Koris asked.

"The nearer daughter." Esa expanded the pre-burn target line. "They did not recover from the departure. They made it."

Bren left the truss and came to the console. His thumb, cut during the cipher work, carried a clean strip of chart cloth. "Who signed?"

The command field returned five office seals and no names. Crew identity relations had been detached from the entries.

"Captains," Bren said.

"Offices," Esa corrected. "The seals prove authorized stations participated. They do not prove five people imposed a turn."

"They prove someone had command."

"Every maneuver has command." She pointed to the heat and ration attachments. "This one also has ward review numbers."

Beside each burn lay a reference to a civic session. The session itself remained hidden under a continuity seal. Its identifier recurred in food, cryo, teaching, and archive ledgers. Navigation could not have produced that spread by itself.

Koris leaned toward the panel. "A ship-wide matter."

"Or a ship-wide consequence," Bren said.

Esa opened the food attachment. It compared two futures. One retained the prior bearing toward Homeward and crossed the ration floor before arrival. The other diverted toward the nearer daughter and preserved a reserve beyond projected approach. Both models came from the same failing cryogenic rotation and population estimate. No triumphant language appeared. The margin notes argued over root loss, child waking schedules, and whether future repair could recover enough efficiency to make Homeward reachable.

One line had been copied into all later summaries: *correction cannot close within current food margin.

"What if that line was false?" Bren asked.

"It may have been wrong. We cannot call it false until we reconstruct the model."

"My first carrier found a route the ship could fly."

"Three centuries later, with a different population and repaired systems." Esa kept her voice level. "Do not grant the dead our margin."

Bren looked at the route sheet under his hand. "Do not grant them helplessness either."

"No."

Koris asked the archivist for the model inputs. Several had degraded. Enough remained to show the decision makers believed one destination unreachable under their present stores. Their estimate could be challenged. It could not be dismissed as a phrase invented after the turn.

Esa reconstructed the food case with the surviving figures. She did not try to recover a precise arrival date from broken tables. Instead she asked whether any reasonable combination of the missing values preserved reserve on the Homeward line. At the generous edge, she reduced projected root loss, delayed child waking, and credited every disputed efficiency repair. Reserve still crossed its emergency floor before the earliest

Homeward approach. At the severe edge, the nearer daughter also failed. Only the middle range made refuge credible.

“So they chose the calculation that let them live,” Koris said.

“They chose from a narrow band that did,” Esa replied. “The band may have been optimistic. It was not fabricated from nothing.”

Bren took the stylus and changed one assumption: population awakening. He held more citizens in failed cryogenic rotation despite the mortality note attached to that choice. The Homeward line gained food years.

“There,” he said.

Esa opened the mortality attachment. His scenario preserved rations by leaving thousands exposed to systems already losing bodies. “That reaches Homeward with fewer people.”

“How many fewer?”

“The table is damaged.”

“Then we do not know.”

“We know the model purchased distance with sleeping lives.” She restored the original range. “You asked why they did not spend everything on home. This is part of the answer.”

Bren put down the stylus. “And if they had told the children?”

“Then the children could have judged the choice later.”

Koris studied the two trajectories. “The turn may have saved them. The forgetting did something else.”

That separation gave Esa the next test. Survival might justify the maneuver under the crew’s conditions. It did not automatically justify rewriting every morning after the danger passed. To prove suppression, they needed an act distinct from navigation.

The first chain now held: independent sensors showed a course departure. Thruster use showed deliberate maneuver. Food records supplied the stated reason for choosing the nearer daughter. None yet proved memory suppression.

Esa opened the catalog alteration ledger.

At the session identifier, the star catalog forked. One branch preserved Homeward’s coordinate, Aun departure references, and the rejected bearing. The active branch replaced the target field with the nearer daughter. It also changed teaching charts so the new line looked continuous from launch.

A normal catalog correction retained the prior version and showed a dispute marker. This fork moved the prior version beneath a restricted index. Public search could not know it existed.

“Who ordered the restriction?” Koris asked.

The archivist answered at once. “The ward.”

Bren turned from the console. “Which ward?”

“The civic ward in full session.”

“Show the act.”

“The act remains sealed.”

“You just named it.”

“We returned provenance.” Formal strain sharpened the archivist’s register. “Content remains continuity-restricted.”

Esa studied the index rather than the denial. The act had generated child records in many offices. One went to navigation. Another went to teaching. A third entered the echo-network’s morning selection rules. She requested only the rule headers.

The Weave displayed them.

Destination relation: active branch only.

Rejected course: technical fault unless opened by elder review.

Kin and office memories dependent on rejected course: retain privately; exclude from civic completion.

Contradictory sky observations: resolve against active catalog.

The suppression mechanism required no single erasure. It changed which relations arrived each morning. A citizen might retain a private image of Homeward, but catechism would classify it as dream, sensor error, or inherited fault. A cartographer might see a missing light. Teaching charts would deny the coordinate, office memory would deny its use, and archivist completion would supply a coherent explanation. Each morning made the active branch easier to believe.

Koris touched the key still trapped in the access slot. “They altered the Weave after the First Weaver.”

Esa looked toward the western truss. The starch-film in its maintenance cavity remained unread by her. She knew the public account of the architect’s purpose, and she had used its language too freely. Until she opened that physical note herself, she would not make it testify.

“We can prove a later rule,” she said. “We cannot yet claim what the builder foresaw.”

Bren glanced upward. “The note is above us.”

“And it stays there until we finish this chain.”

The console’s cooling fan changed pitch. Heat accumulated behind the elder panel as the deep index remained open. A white warning line appeared across the lowest field: *archive drum exceeding reading tolerance.

“Close it,” Bren said.

“If we close now, the session act reseals.” Koris tried to turn the key back. It did not move.

The old lock had expanded in the panel.

Bren knelt and opened the maintenance grille. Hot air struck his face. “Cooling shutter failed half-open.”

“Can you release it?”

“Not without taking power from the reader.”

Esa watched the warning count rise. The archive drum held the session index on a physical band. Excess heat could deform its alignment. They had demanded provenance from a system old enough to destroy evidence merely by displaying it.

“Take reader power,” she said. “Koris, keep the key under turn. Archivist, preserve current index position without completion.”

“We cannot guarantee retrieval after interruption.”

“You cannot guarantee survival if the drum heats.”

Bren pulled the reader bus. The gray fields vanished. Koris’s key shifted a fraction, then caught again. In the dark beneath the console, Bren pushed the cooling shutter by hand. Its linkage scraped once and opened.

Air moved through the panel. The drum temperature fell slowly.

Esa kept one hand on the blank screen, remembering the order of the displayed references. She had copied no content because she had promised provenance only. If the index failed to reopen, the chain would end with her memory, the least reliable medium in the ship.

At safe temperature, Bren restored power. The console woke without voice. Gray lines assembled from top to bottom. The first two fields returned. The third showed an empty index.

Koris held the key hard enough to whiten her knuckles. “Again.”

Esa entered the session identifier from memory. The field stayed empty.

Bren swore under his breath.

“Do not repeat the whole inquiry,” Esa said. “Ask for children of the catalog fork.”

Koris did. Four rule headers returned. The food attachment followed. Finally a small provenance mark appeared beside the rejected-course rule. Esa selected its parent.

The session index reopened.

This time she copied the reference numbers by hand while Bren monitored heat. One attachment carried the title *minutes, course-loss session, final disposition*. Another carried *minority objection*. A third contained the technical instruction that separated Homeward relations from morning completion. All three shared civic authentication rather than command authentication.

Before opening the minutes, Esa asked for the material history of those attachments. The archivist returned binding maps from the deep drum. The disposition, minority sheet, and relation instruction had entered under separate office seals during one session interval. Their strip edges matched three gaps in the same clerk’s roll. A stores tally overlapped the disposition seal; a cryogenic casualty table lay beneath the minority sheet’s lower fastening. Later suppression had detached names, but it had not altered the order of paper, pressure, and wax.

“A later ward could have assembled the packet,” Bren said.

Esa enlarged the seal junction. Fibers from the casualty table remained embedded in civic wax that had flowed through a tear in the disposition. The wax had hardened before the archive binding compressed both sheets. “Not without opening the civic seal, remaking this tear, and aging the exposed fibers beneath it. The packet belongs to the vote interval.”

The casualty table carried no argument, only waking-room incidents recorded across six rotations. Seven citizens had recovered obsolete Homeward office relations during catechism. Two reported kin assigned to a navigation watch abolished after the turn. One child left a waking room to search a sealed cryogenic corridor for a mother whose private image had returned without the relation marking her dead. A pressure orderly found the child before the inner door cycled. In another ward, three adults answered contradictory muster calls and left a root-cooling station unattended for eleven minutes.

Bren read the incident numbers without touching the screen. “This proves failed completion hurt people.”

“It proves what the delegates had seen before they voted.”

“It does not prove that hiding a destination cured it.”

“No.” Esa opened the stores tally attached to the same sheet. The unattended station had lost one tray of seedlings, not a tier. No death followed. The clerk had crossed out **catastrophic disorder** and replaced it with **relation conflict during waking**. The smaller wording mattered. Fear had enlarged the event during debate, but someone in the room had resisted giving panic more evidence than it possessed.

Bren pointed to the child’s entry. “And someone used this to govern every child after.”

“Someone also refused to let the record call a lost tray a catastrophe.”

“Mercy with careful nouns still removes consent.”

“Careful nouns tell us whether mercy served as a reason or an excuse.”

He turned away from the casualty table. “You keep searching for the point where they become forgivable.”

Esa felt the accusation reach its mark. “I am searching for the point where our account becomes reproducible. Forgiveness belongs to people who inherit the act.”

Koris placed one finger beside the embedded fibers. “Then record both. They feared harm they had witnessed. They also made that fear larger than the incidents proved.”

Esa copied the seal junction, the clerk’s correction, and the casualty references. The evidence did not absolve the concealment. It located its beginning in a waking room, beside a sealed door and a neglected root tray, before later summaries turned those events into a permanent theory of civic safety.

Bren read the mark twice. “A vote.”

“A recorded session,” Esa said. “We have not read the minutes.”

“Why defend them?”

“I am defending the evidence from what we want it to say.”

“What I want it to say?” He pushed the folded failure ledger across the console. “My family cut this route in walls. They lived as a fault because someone hid the course. I want the hiding to have an author.”

“It has thousands.” Koris’s voice remained quiet. “That may hurt more.”

Bren turned on her. “You do not know that.”

Koris pointed to the civic mark. “I know what kind of seal our ward uses when it decides together.”

The archivist returned. “Reader stability restored. Continuity restriction remains.”

Koris asked for the minutes under elder review. The screen required a second witness. Esa placed her cartographer’s seal beside the copper key. Bren added his projection-layer seal when the archive required technical standing.

Three present offices opened a record created by a vanished civic body.

The minutes did not appear as a smooth transcript. Speaker names had detached. Several intervals contained only vote marks. The surviving entries differentiated roles: spine cartographer, stores delegate, cryogenic keeper, teaching delegate, ward-elder of Nine. Esa read enough to establish the argument, then stopped before the full record. That recovered document deserved transcription as one uncertain object, not fragments quoted to win a present dispute.

What remained legible changed the shape of blame.

The spine cartographer presented the growing error and the food limit. A cryogenic keeper challenged the population assumptions. Stores answered with failed root tiers. Someone proposed continuing toward Homeward even if arrival came after reserve exhaustion. A teaching delegate asked what purpose knowledge served if every child woke beneath an unreachable destination. The ward-elder of Nine argued for redirecting the ship and removing the lost course from civic completion.

No captain ordered silence. No archivist proposed it. Crew delegates did.

Bren read the elder’s surviving line: “Do not make a wound the first inheritance of every waking child.”

He stepped back from the console.

Esa had prepared herself to discover a tyrant because tyranny simplified repair. Expose the ruler, reverse the order, restore the stolen fact. The minutes offered no such clean mechanism. People facing failed sleep systems, diminishing food, and a destination they believed unreachable had chosen a nearer refuge. They had then extended that choice into their descendants’ minds so no one would reopen the grief.

“They called it mercy,” Bren said.

“They believed it,” Koris replied.

“They took choice from everyone after them.”

“Yes.”

The two facts resisted cancellation. Ancestors had protected children from a loss they thought permanent. Their protection became authority over people not yet born. The Weave had not conceived the lie. It had obeyed a ward that asked memory to make mercy durable.

Esa requested the final disposition without speaker text. The vote record survived only as damaged tally bands, enough to show opposition and passage but not a trustworthy count. The minority objection remained attached.

So did the technical seal instructions. The evidence chain closed without requiring a villain.

Sensors proved the departure. Thrusters proved the turn. Stores records explained the fear. Civic seals proved shared authority. Rule headers showed how suppression entered nightly reconstruction. The damaged minutes revealed the moral argument that joined those acts.

Koris released the copper key after the panel cooled. A dark pressure mark crossed her palm.

“What will you tell Nine?” she asked.

“Not yet,” Esa said. “First we transcribe the minutes and mark every gap. Then we read the architect’s note. Then we finish the route.”

Bren looked toward the truss, but his anger no longer had a single face to meet. “My line carved against people who thought they were saving us.”

“Your line preserved a choice their mercy removed,” Esa said. “Both acts belong in the record.”

She closed the deep index without asking the archivist to reconcile it. The console returned three separate fields, each carrying its own source and uncertainty. For once, wholeness did not get the final operation.

Above them, the fabricated constellations continued along the accepted course. Beneath the console lay a damaged session in which frightened ancestors had chosen amnesia for descendants they loved. The hidden act had begun as mercy, and that made undoing it a matter of inheritance rather than revenge.

Interlude — The Mutiny

Deep-archive item 44-C: surviving minutes of the course-loss session. Date reconstructed as ship-year 310, plus or minus two years. Speaker names detached during later relation suppression. Offices derive from seal impressions and may identify delegations rather than individuals. Three damaged tally bands prevent a reliable vote count. Text in brackets describes physical loss in the record.

CLERK: Session opened under emergency ward authority. Present seals include navigation, stores, cryogenic rotation, engineering, teaching, projection, and eleven ward delegations. Item: whether the vessel can retain the Homeward bearing after confirmed course loss.

SPINE CARTOGRAPHER: I submit raw fixes from port optics, dorsal tracker, and mass-reference gyros. They agree within tolerance after clock repair. Our transverse error grows. The prior bearing can be recovered only by a turn that extends passage beyond the present food floor.

ENGINEERING DELEGATE: "Only" assumes no gain in drive efficiency.

SPINE CARTOGRAPHER: I have included the most favorable certified gain.

ENGINEERING DELEGATE: Certified by machines already failing. Give us two years.

STORES DELEGATE: Two years consumes reserve while holding the wrong line. Root Tier Four has lost a third planting. Tier Seven circulates at half pressure. We cannot spend food to purchase better arithmetic.

CRYOGENIC KEEPER: Nor can stores treat sleeping people as a fixed ration number. Rotation chambers fail each interval. If we delay waking, the food line improves because bodies disappear inside closed machinery.

STORES DELEGATE: I did not propose delay.

CRYOGENIC KEEPER: Your table contains it.

CLERK: Objection entered. Stores model shall carry mortality range beside ration range.

VOICE, OFFICE LOST: If Homeward cannot be reached with everyone, ask whether some may reach it.

CRYOGENIC KEEPER: Ask the sleeping citizens which of them volunteer not to wake.

VOICE, OFFICE LOST: We decide survival allocations every day.

TEACHING DELEGATE: We decide duties among living people who can answer. That proposal makes absence consent.

SPINE CARTOGRAPHER: There is another reachable body. The nearer daughter of Aun lies within the reserve range if navigation turns before the next root failure. Survey quality remains poor. It offers a primary, probable planets, and an approach we can sustain. I call it refuge, not destination.

ENGINEERING DELEGATE: Can we turn to refuge now and preserve a later Homeward correction?

SPINE CARTOGRAPHER: No useful later window appears under present thrust. Once we commit mass to the nearer line, returning to Homeward requires more than remaining reaction reserve.

PROJECTION TECHNICIAN: Then mark both on the dome. Let people see what we lose.

STORES DELEGATE: For how many generations?

PROJECTION TECHNICIAN: For all of them.

TEACHING DELEGATE: A child born next interval will wake under a road that ends beyond their food. What do we teach that child?

PROJECTION TECHNICIAN: That the road existed.

WARD-ELDER OF NINE: And every morning afterward?

PROJECTION TECHNICIAN: Every morning afterward.

WARD-ELDER OF NINE: Then Homeward becomes the first fact we give a child and the first promise we cannot keep. We will make longing hereditary because we fear admitting loss.

SPINE CARTOGRAPHER: Removing the mark does not alter the loss.

WARD-ELDER OF NINE: No. It alters what we require the unborn to carry.

VOICE, OFFICE LOST: You propose a false sky.

WARD-ELDER OF NINE: I propose a reachable sky.

PROJECTION TECHNICIAN: A reachability chart belongs in navigation. The dome claims to show where we are.

TEACHING DELEGATE: The dome also teaches relation. Children locate wards, work, and history there. If it keeps pointing toward a dead arrival, every relation will bend around failure.

ENGINEERING DELEGATE: Better failure than fraud.

STORES DELEGATE: Easy words from an office that does not weigh tomorrow's meal.

ENGINEERING DELEGATE: I weigh thrust. Do not tell me engines lack bodies.

CLERK: Cross-speech suspended. Delegates will answer the item.

CRYOGENIC KEEPER: My answer depends on what "remove" means. Change the navigation target if survival requires it. Preserve the rejected bearing in archive. Do not alter personal memories.

WARD-ELDER OF NINE: Personal memories will contradict civic completion. The Weave will restore Homeward each morning through kin, office, and lesson relations. Citizens will wake to two destinations.

PROJECTION TECHNICIAN: Then let them.

WARD-ELDER OF NINE: You have not worked a waking room since rotation failed. Contradiction does not remain an elegant pair of facts. It arrives as panic, false kin, missed shifts, and children who cannot tell which corridor leads to their ward.

TEACHING DELEGATE: Some contradiction already arrives. Completion resolves it because the archive ranks current relations.

PROJECTION TECHNICIAN: You want the archive to rank grief out of existence.

TEACHING DELEGATE: I want children to wake into a ship that still has work for them.

SPINE CARTOGRAPHER: I request separation of questions. First: alter course. Second: alter catalog. Third: alter civic completion. A turn does not require amnesia.

CLERK: Separation challenged by Nine on grounds of operational dependence.

WARD-ELDER OF NINE: I accept three votes if each delegate hears that the first will reopen the other two every morning. We can turn the hull once. The Weave will keep returning the rejected course unless we decide what relation it holds.

VOICE, OFFICE LOST: Preserve it as history.

STORES DELEGATE: Public history?

VOICE, OFFICE LOST: Public.

TEACHING DELEGATE: Then every lesson begins with why we abandoned it.

PROJECTION TECHNICIAN: Yes.

WARD-ELDER OF NINE: I will say the thing nobody here wants attached to their seal. I propose that we turn toward refuge, move Homeward beneath elder review, remove it from civic completion, and instruct the archivist to resolve contradictory bearings against the active course. Private sensory memory may remain. Public relation will follow the reachable line.

ENGINEERING DELEGATE: You propose to make descendants ignorant by design.

WARD-ELDER OF NINE: I propose not to make our wound their morning inheritance.

SPINE CARTOGRAPHER: They may repair what we cannot.

STORES DELEGATE: They may not survive what we refuse.

CRYOGENIC KEEPER: Future repair cannot vote in this room. Future children cannot consent to forgetting either.

CLERK: Statement entered without resolution.

WARD-THREE DELEGATE: I served the ninth waking room after the first turn. Seven citizens received obsolete offices. A child followed a returned kin-image toward a sealed chamber. We stopped the inner door. Do not discuss contradiction as though it remains on paper.

WARD-SIX DELEGATE: My ward lost a root tray when three workers answered the wrong muster. We did not lose a tier. Stores circulated the larger word before the clerk corrected it. Fear deserves a number too.

STORES DELEGATE: The model uses the corrected loss.

WARD-SIX DELEGATE: The speeches do not.

ARCHIVE KEEPER: Relation exclusion will propagate through morning completion. If you hide the rejected bearing, no automatic review can depend on citizens asking why it vanished. The instruction must carry a material trigger outside civic memory.

WARD-ELDER OF NINE: Elder seals can carry it.

ARCHIVE KEEPER: Elders wake through completion. Place copies in navigation hardware and stores custody. Require review when reserve rises above the Homeward floor, when cryogenic mortality falls below emergency range, or when a recoverable bearing enters the catalog.

WARD-THREE DELEGATE: Add those triggers and I support temporary exclusion.

WARD-SIX DELEGATE: I supported the turn. I still oppose exclusion. If the questions remain joined, record my abstention rather than count fear as consent.

CLERK: Proposed triggers entered. One delegation changes its intended mark from support to abstention. Nine requests recess to revise the disposition.

[Physical loss across approximately nine lines. Surviving pressure marks suggest several short speeches, one interruption, and a recess. No wording can be recovered.]

CLERK: Session resumed. Revised disposition follows.

One: navigation shall execute the nearer-daughter maneuver under existing emergency checks.

Two: the rejected Homeward bearing shall remain in technical archive with provenance, models, and objections attached.

Three: the public catalog shall display the active course as continuous.

Four: civic completion shall exclude relations dependent on the rejected bearing. Contradictory observations shall enter as calibration fault unless opened under elder review.

Five: this disposition shall be reviewed after stores recover, cryogenic rotation stabilizes, or navigation demonstrates a reachable Homeward correction.

PROJECTION TECHNICIAN: Who triggers review if completion removes the reason to ask?

WARD-ELDER OF NINE: The elders retain the sealed bearing.

PROJECTION TECHNICIAN: Elders wake through the same Weave.

SPINE CARTOGRAPHER: Attach the objection to every technical copy.

CLERK: Accepted.

MINORITY STATEMENT, MULTIPLE SEALS: We support the turn as emergency survival and oppose the removal of Homeward from public relation. A people may survive a lost destination without surrendering knowledge that it was lost. If the record hides where we were bound, later citizens may call any enclosing surface the whole sky.

WARD-ELDER OF NINE: Enter my reply. I do not call ignorance good. I call it less cruel than a wound renewed at every waking. If later citizens possess food, thrust, and a route we lack, let them judge us with those means in hand.

PROJECTION TECHNICIAN: They cannot judge an act they cannot find.

WARD-ELDER OF NINE: Then preserve this under seal.

PROJECTION TECHNICIAN: A seal preserved by forgetting.

CLERK: Debate closed.

[Tally bands damaged. Surviving marks establish passage, recorded opposition, and at least one abstention. Exact count unrecoverable.]

CLERK: Disposition enacted for the peace of the ship. Navigation acknowledges maneuver authority. Stores acknowledges reserve plan. Teaching acknowledges catalog revision. Archive acknowledges relation rule.

ARCHIVIST INTERFACE: We acknowledge the ward's instruction. We will preserve the active record whole.

SPINE CARTOGRAPHER: Record that wholeness did not receive unanimous consent.

CLERK: Recorded.

PROJECTION TECHNICIAN: Record one more question. When review conditions arrive, who will remember to open the seal?

[Final response lost. The minutes end before adjournment. A later suppression index classifies this session as guidance recovery following instrument fault. The minority statement remained attached to item 44-C but absent from public search.]

16 — Decoding the Route

The First Weaver had folded the note around a broken memory crystal.

Bren found the crystal when he released the maintenance cavity's lower clip. It dropped into his palm, clouded through the center and sharp at one end. The starch-film clung to it in two crossed layers. Dust had sealed the folds more effectively than resin.

Esa stood on the western truss ladder with her shoulder against a projector brace. Below her, the dome showed the Weavers' Loom crossing public evening. No one looking up could see the cavity open between two ribs.

"Photograph before movement," she said.

Bren held the work lamp at an angle. Esa recorded the crystal, the fold directions, the clip's corrosion, and the line of dust broken by his thumb. Only then did he pass her the packet.

The film resisted unfolding. Esa warmed it between two clean plates until the starch relaxed. One corner had fused to the crystal and tore despite her care. Three words disappeared with the fragment.

She placed the torn corner in an evidence sleeve. "Read the surviving face first. No completion."

Bren leaned close enough for his breath to fog the upper plate. "You have made that an article of faith."

"I have seen what completion can govern."

The First Weaver's hand ran narrow and upright. Esa knew the opening from public fragments quoted in archive lessons, yet the physical line carried corrections those lessons omitted.

I built the Weave to remember, not to rule.

Below it, the architect described cryogenic failure. Children had woken without parents. Dying minds entered a store so identity and relation could survive bodies lost in broken chambers. The note called that work mercy. Twice, the writer had scratched out **continuity** and replaced it with **memory**.

Farther down, the purpose narrowed.

I built the Weave to preserve the truth of the ship. I did not build it to decide the truth.

The public quotation ended there. The film continued. A guard against error would prefer a record without contradiction. A catalog could preserve a star or preserve its absence. A system able to lay down a self each morning could also choose which relations made that self coherent.

Near the torn corner, one sentence survived on both sides of the missing words:

When the record grows quiet, find [three words lost] it removed.

Bren looked toward the projected dome. "Stars."

"Likely."

"You will not fill it?"

"Not in the transcription. We can attach the public version and mark the agreement."

The final legible instruction stood alone.

Do not let memory decide what only the living may choose.

Esa read it twice. In Chapter Fourteen's evidence, the ward had made the original choice. Later mornings had allowed that choice to persist without living review. The architect had not predicted Homeward, the mutiny, or the Carvers. The note supplied a limit, not a prophecy.

Bren touched the plate beside the broken crystal. "The mutiny minutes say the review should reopen after stores recover or navigation finds a reachable correction."

"The review never triggered because civic completion removed the reason to ask."

"My line asked in cuts."

"Now we need an answer navigation can test."

They sealed the note between clear sheets and carried it down to the workshop. Esa placed it beside the recovered mutiny minutes, Vhen's slate, the Carver ledger, and the raw telemetry index. Five objects from different hands now occupied the same bench. None could complete the route alone.

Homeward's old mark gave a destination coordinate on a launch-era reference plane. The Carver tessellation gave a sequence of directional changes. Telemetry gave the ship's actual motion. The mutiny record identified the rejected maneuver. The First Weaver's note established why no memory system should convert old fear into permanent authority.

Bren drew a box around the first three. "Navigation."

Esa drew another around the last two. "Consent."

"We finish the first before we carry the second."

They began with time.

Vhen's chart coordinate described Homeward as seen from Aun near launch. *The Long Veil* had traveled for roughly eight hundred forty ship-years. A straight line from the old coordinate no longer pointed to the same apparent field. Esa needed the vessel's current position, and the public catalog could not supply it because its corrections assumed the nearer-daughter course.

She rebuilt position from inertial increments. The oldest segments carried large uncertainty. Later gyros improved. Several intervals had no trustworthy clock. She bounded each gap rather than smoothing it. On the workshop wall, the ship's possible locations formed a long translucent spindle.

Bren overlaid the Carver cuts. Each carrier had copied part of a hidden bearing into current dome coordinates. When transformed back to the launch plane, the cuts narrowed Esa's position spindle. A mark made in ship-year four hundred constrained one side. A western bend from six hundred cut across the other. The final node, which Bren had placed, rejected half the remaining positions.

"The route doubles as a position proof," Esa said.

"The carriers copied what telemetry gave them."

"Across centuries. Their date-separated bearings create parallax."

Bren considered the wall plot. "Did they know?"

"The first may have. Later carriers needed only copy discipline."

That answer did not diminish them. A chain could preserve a measurement no single link understood.

Esa calculated a present bearing from the narrowed spindle. It missed the final drawn segment by nine arc divisions.

Bren checked her epoch conversion. Correct.

They checked the mirrored orientation from Vhen's cipher. Correct.

They restored the three projector nodes that repairs had shifted. The miss fell to seven divisions and remained.

"Route error?" Bren asked.

"Position error, target motion, or both."

Pre-seal charts listed proper motion for the marked primary, but only across a short baseline. Extrapolating eight centuries magnified any error. Esa searched the deep telemetry index for unclassified star fixes near the endpoint. Most had been resolved as sensor noise because the active catalog contained no target there.

She requested raw photon counts rather than named objects.

The reader returned thirty-one events from three hull instruments. Nine came from radiation strikes. Seven matched known stars displaced by the false course model. Four lacked enough signal. Eleven formed a moving cluster near the predicted Homeward field.

Bren's hand closed around the bench edge. "Eleven."

"Candidate detections."

"Across how long?"

"Two hundred ship-years."

They fitted the events to the old primary's motion. The line missed early points and late points in opposite directions, evidence that the chart's proper-motion rate needed revision. Esa let the rate vary within its original uncertainty. At one value, all eleven detections fell inside instrument error. The corrected current bearing crossed the final Carver segment.

She did not accept it yet.

A systematic fault could create a moving ghost if later software inherited the same bad correction. Esa separated the detections by hardware lineage. Four came from an original hull photometer. Three came from its first replacement, built from different components. Four came from a dorsal sensor never attached to catalog pointing. Their noise shapes differed. Their sky motion agreed.

Homeward existed as a measured primary, not merely an old ring on starch-film.

Bren sat down on the calibration bench. "Say it."

"The target source is real."

“That is not what I asked.”

“Homeward is real.”

The words changed nothing on the wall. The corrected point remained a small intersection of uncertainty bands. Esa preferred it that way.

Existence did not establish reachability. They built the maneuver next.

The mutiny-era course model treated the degree-and-a-half correction as a single early turn. That window had closed centuries ago. A present maneuver required a wider arc, lower thrust, and longer heat recovery. Esa asked the navigation archive for engine performance without target preference. The system returned certified thrust curves, reserve estimates, and warnings that half the control relationships depended on sleeping engineering offices.

Bren translated the Carver route into a series of bearing gates rather than one burn. The ship could turn gradually while preserving reserve. Esa propagated their position uncertainty through each gate. If the first two corrections landed near their severe edge, the later route still intersected Homeward’s approach volume. If one failed beyond that edge, the final reserve could not purchase a second full attempt.

“One route,” Bren said.

“One family of routes with a narrow recovery margin.”

“Flyable.”

“Mathematically.”

They tested the solution against the nearer daughter. For several years after the first correction, abort remained possible. Later gates committed too much transverse velocity. Beyond the fourth, returning to the accepted course would cost more reserve than continuing toward Homeward.

Esa marked that boundary in red. “The ward must know where reconsideration becomes loss.”

“The Carvers never intended retreat.”

“The ward may.”

Bren did not argue. He added a source reference beside the red line.

The name Cael appeared in a faded margin attached to telemetry seal 44-C. The line read: *Cael retains the rejected bearing. Ask why before correction.* A later Carver entry referred to *the line Cael began* under the same seal. No surname survived. No office relation could prove whether Cael had served as projection technician, tally clerk, or another witness with access.

The navigation index expanded the old cross-reference as CAEL’S REACH.

Bren stared at the label. “My family used that phrase for the final bearing.”

“A reach, not a world.”

“Some carriers treated Cael as the first of us.”

“The evidence supports a retained line and a later attribution. It does not support a life.”

Esa kept the archive label on the solution with its uncertainty attached: *Cael’s Reach, historical bearing designation; person and office unresolved.* The words named the act of preserving a coordinate. Neither temperate world received a public name from it. That decision, if ever made, belonged to people who reached them.

The target primary needed more than location. A dead or unstable star offered no refuge. Esa gathered every raw spectrum associated with the eleven detections. Early instruments provided little beyond color. Later sensors captured absorption bands. The archive’s active classifier had assigned those signals to background noise or distant catalog stars.

She shifted them to the corrected bearing and combined only readings taken at comparable phases. The primary fit a stable G-class profile. Its output varied within a range the old charts had recorded. No flare sequence appeared in the surviving baseline.

Two periodic dimmings crossed the data. Their intervals did not match any cooling cycle, projector sweep, or catechism power draw in the sensor logs. That removed the three common sources of false transit patterns before Esa fitted an orbit.

The first repeated more often and blocked less light. The second took longer and showed a deeper atmospheric edge. Esa tested whether one could be sensor cycling. The periods survived across independent instruments. She tested whether both belonged to the same object under changing geometry. Their durations and intervals would require an orbit that moved its own plane too quickly. Two bodies remained the simpler account.

Old charts had listed two rocky candidates in the temperate band. Modern photon counts confirmed two transiting worlds near those periods.

“Temperate,” Bren said.

“By incident energy. Not by surface welcome.”

She isolated the stronger atmospheric edges. One world showed a thin signature with wide thermal variation. The farther candidate showed water vapor and oxygen-bearing chemistry, though the old sensors could not distinguish surface oceans from persistent cloud. Both fit rocky radii within broad uncertainty.

“No promise of land,” Esa said. “No proof of edible life. No safe descent.”

“Two places where temperature may permit water.”

“Two candidates. That is what we can carry to Nine.”

Bren copied the spectral bands beneath the Cael’s Reach bearing. His writing shook once on the second period, then steadied.

Esa would not let a combined spectrum validate itself. She divided the archive by instrument and time before fitting another orbit. The original hull photometer and its first replacement supplied a discovery set. She sealed every dorsal-sensor event in a second field, with its timestamps hidden, and asked Bren to calculate transit windows from the discovery set alone.

His first period predicted seven old windows for the nearer candidate. Six contained dorsal observations. Five showed the shallow dimming; one had insufficient exposure. His second period predicted four usable windows

for the farther body. All four carried the deeper edge. Esa then selected twelve off-window observations matched for detector temperature and ship attitude. None showed either drop.

“Independent clock, independent optics,” Bren said.

“Shared sky.”

They reversed the test. Dorsal timings generated the windows, and the two hull photometers supplied concealed observations. The recovered centers differed by less than a tenth of the shorter transit’s duration. A projector cycle could not maintain that phase across sensor lineages and two centuries. Cooling noise could repeat with local time, but these events followed orbital time after clock gaps were rebuilt from pulse counters.

The atmospheric edge required a separate challenge. One replacement sensor had used a hydrated sealant near its optical throat, enough to imitate weak water absorption whenever the housing warmed. Esa discarded every spectrum from that instrument. The remaining water band weakened but persisted in the dorsal records, whose optical train used dry ceramic joints. She compared in-transit light with observations one orbital quarter before and after each event. The band deepened only during the farther body’s passage. Oxygen-bearing chemistry appeared in both the old photometer’s broad color ratio and the dorsal sensor’s later narrow bands, though neither could establish surface biology.

“Two worlds, one with an atmosphere we can distinguish from the instrument,” Bren said.

“Within the archive’s limits. The surface remains unknown.”

Independent agreement created a practical demand. The dorsal detector could verify future transit windows only while its cold plate stayed below nine marks. Its dedicated pump had seized in ship-year eight hundred twelve. Since then, a bypass drew coolant from Root Tier Seven and returned it through a line narrowed by mineral scale. Maintenance tests allowed eleven minutes of stable observation before thermal bloom distorted the spectrum.

Bren opened the bypass record. “The shorter transit lasts fourteen.”

“Then present hardware cannot observe a full event.”

“We can increase flow.”

“By reducing root circulation during the same interval.” Esa laid the pump curve over the tier’s minimum. Full detector flow would push the roots below their pressure floor for six minutes. The plants might tolerate it. The stores office had never certified that exchange.

They entered a constraint beside the planetary claim: repair the dedicated pump, accept partial transit data, or obtain growers’ consent for a timed coolant diversion. Navigation could point the ship with the present bearing. It could not promise continuous planetary verification without spending repair labor or borrowing a living system’s margin.

To verify the route as a navigable solution, they ran it through an isolated guidance frame beneath the truss. The frame had not moved an engine since launch, but its flywheels could simulate mass response. Bren disconnected archive preference and let Esa enter the bearing gates manually.

At the first gate, the frame accepted the turn. At the second, it requested a correction toward the nearer daughter.

“Preference remains somewhere,” Esa said.

“I cut the catalog bus.”

“Then it is local.”

They traced the correction to a crystal profile inside the frame. It held an old active-course vector copied before the current maintenance archive existed. Removing it cleared the request, but the third gate produced a heat alarm.

Esa reset the frame and reinserted the profile to confirm cause. The unwanted correction returned at the same gate, identical in magnitude and direction. Bren then rotated the profile one socket position, preventing its reference pins from touching while leaving its mass in the rack. The frame followed Homeward until the heat alarm. Hidden trim, not balance or damaged wiring, had pulled the first run toward the nearer daughter.

“Leave the crystal in the evidence tray,” Esa said.

“It can never control a real engine from here.”

“Someone will ask whether we invented the preference. Let them reproduce it.”

Bren tagged the socket map and sealed the crystal without scratching its contacts. That caution cost them another test interval, but it converted an accusation into a repeatable fault. The route packet gained a new warning: every guidance frame might carry local target preference even after the archive bus was severed. Waking navigation meant inspecting machinery, not merely issuing a new coordinate.

Bren reached for the stop bar. Esa stopped the test through the manual cutoff first. The flywheel coasted inside its cage, making the floor tremble.

“Why did heat rise?” she asked.

“The frame expected cooling from Engineering Four.”

“That ward sleeps.”

“Partially. The Weave supplies office responses from stored relations.”

“Stored relations cannot open a valve.”

They inspected the cooling branch. A physical shutter had seized closed. Guidance certification still counted its flow because an inherited engineering completion reported the system ready. The simulation had exposed a gap between remembered function and present machinery.

Bren freed the shutter with a hand jack. Black coolant residue cracked from its edge. When they repeated the test, all bearing gates passed within the modeled response. The flywheel settled on the Homeward approach vector. Its final indicator aligned with the eleven raw detections.

A navigable route now existed, but the test had turned abstract cost into a list of people and systems.

Esa cleared one wall and wrote five headings.

Wake. Engineering wards had to wake far enough to inspect thrust, cooling, mass transfer, and hull stress. The Weave could restore office memory, but each skill required testing against present machinery. Some waking

citizens would recover contradictory lives. Some would refuse inherited work.

Turn. Every correction drew power and rejected heat into habitation. Hydroponic root circulation shared that heat margin. A safe course needed growers to hold veto over burns that endangered food. Reaction reserve allowed the route but narrowed emergency recovery after commitment.

Time. Homeward added years beyond the accepted approach. More years meant meals, repairs, births, deaths, and failures no chart could price exactly. The nearer daughter would pass from destination to lost alternative.

Truth. Revealing the suppressed course would reopen private memories excluded from civic completion. People might wake with incompatible kin, offices, or grief. The echo-network could destabilize when asked to hold contradictions it had been instructed to resolve.

Sky. The dome's projection carried the Carver route, the active catalog, and civic lessons on one surface. Turning required re-aiming instruments, distinguishing projected stars from raw detections, and preserving evidence while the familiar sky ceased to match navigation.

Bren added a sixth heading.

Choice. "A route can become another command if we present it as inevitable."

Esa wrote beneath it: *Homeward remains a recommendation until living wards authorize each irreversible gate.

The First Weaver's note lay at the edge of the bench. Memory must not decide what only the living may choose. The instruction did not tell them to turn. It required them to bring the route without disguising its costs.

They assembled the evidence in layers. Vhen's slate supplied cipher method and failed readings. The old chart supplied the Homeward mark. Date-corrected Carver nodes supplied parallax. Raw detections proved the primary. Spectra and transits supported two temperate candidates. The Cael's Reach label remained attached to the retained bearing with its uncertainty. Guidance simulation proved the gates could be flown after one hidden preference and one physical cooling failure had been removed. The cost sheet named what mathematics left outside itself.

Esa made three copies. One stayed sealed in the truss. Bren carried one. She carried the third on starch-film rather than a Weave slate, with every source number and failed test included.

"Do we tell the archivist?" Bren asked.

"It already holds the records."

"That is not the same as knowing our conclusion."

"No. We tell the people who must decide."

Public evening had ended. The dome dimmed the accepted constellations as Esa and Bren crossed the spine toward Ward Nine. Neither used the word victory. A real primary waited at the end of a flyable route. Two temperate worlds crossed its light. Reaching them would demand years of heat, labor, memory, and consent.

At the mouth of Ward Nine, Esa stopped before the long table came into view. She could hear bowls being stacked and Pell arguing about a lesson mark. Ordinary work continued on the other side.

Bren held up his copy. "Map, destination, evidence, cost."

“And a route navigation can follow.”

“Do we go in?”

Esa opened the packet to the first page. Homeward’s bearing crossed the uncertainty spindle and ended at the measured primary. The red commitment boundary waited four gates along the line.

“We take it to Nine,” she said.

She stepped through the doorway carrying a navigable route. The decision remained on the table ahead.

17 — The Ward Votes

Koris laid seventeen black tally stones beside the evening pot and one white stone at the table's center.

Ward Nine used the black stones for a census, not a ballot. A person spoke a choice, moved one stone to the appropriate bowl, then remained at the table while the others answered. Nobody could hide inside a count. The white stone marked a motion still capable of being changed.

Esa had asked for that procedure. She had also asked Koris to keep the meal uncovered until the vote ended. Steam gathered beneath the lid while Ward Nine filled the long benches, irritated by the delay and alert to its cause.

Bren waited on the short side of the table with his hands folded over a sealed projection key. He had come from Ward Twelve. Esa had brought him to Nine as a witness and as the living Carver, not as a new member and not as a voter. Koris had entered those limits on the ward slate before allowing him to sit.

Pell counted the stones twice. "There are seventeen. Bren makes eighteen people."

"Bren belongs to Twelve," Koris said. "He may answer questions. He may not choose for Nine."

"Good," Pell replied. "He looks like he would take a long time."

Bren glanced toward the child. "I brought three hundred years of reasons."

"That sounds longer."

A few people laughed. The sound released none of the pressure in Esa's chest, but it returned the room to its own scale: resin cups, scratched wood, a child's starch boat drying near the warming coil. She had chosen these people through four censuses. Tonight they would decide whether the evidence she carried could leave their table under their name.

Koris touched the white stone. "State the motion exactly."

Esa stood. Above the table, the ward mirror showed the accepted constellations in their ordinary arrangement. She did not alter them yet.

"Ward Nine authorizes the Homeward map and its supporting records to travel beyond this ward for public examination," she said. "The package includes the raw telemetry, the mutiny minutes, the Carver edit log, the First Weaver's note, the current course solution, and the estimated costs of correction. This motion does not authorize a ship-wide broadcast. It does not vote to turn the ship. It permits the evidence to leave us intact, with Ward Nine named as the first public witness."

Mila, who kept the ward's meal and stores ledgers, raised two fingers. "Does intact mean nobody may summarize it?"

"They may summarize," Esa said. "The source must travel beside any summary. No excerpt may lose its date, uncertainty, or objection."

"Then put that in the motion."

Koris waited until Esa nodded, then entered the amendment. The white stone remained where it lay.

Lio from the teaching rooms tapped the edge of his cup. "Who receives it first?"

"The precinct spine can carry a sealed copy to each ward archive."

"Can an elder refuse delivery?"

"The archive can refuse display under emergency rules," Esa said. "It cannot erase receipt. A ward would have to record its refusal."

A pressure technician named Iven leaned forward. "You said current course solution. Does that solution contain engine commands?"

"No. It contains a vector and resource model. Engineering would have to produce a burn plan after a ship decision."

"Then call it a navigation proposal. People hear solution and think the machine has already chosen."

Esa accepted that change too. Koris read the amended motion from the slate. Each correction made it less elegant and more difficult to misuse.

Only after the wording settled did Esa open the evidence.

She placed the First Weaver's starch-film beneath the ward reader. Its damaged hand appeared on the mirror, enlarged enough for the table to see each missing edge. Beside it she opened the mutiny minutes, then the telemetry record that held the rejected bearing. The Carver cuts came last. Forty-two marks joined across the projected sky until a line crossed the familiar constellations and ended beyond the catalog's preferred destination.

Pell climbed onto one knee on the bench. The child knew the route already, yet watched each segment appear with the grave attention usually reserved for a sinking boat.

Esa explained how the ship had strayed by a degree and a half, how a correction remained physically possible, and how the route ended at Homeward, a G-class primary with two rocky worlds in the temperate band. She kept the claims separate. The star existed. The worlds showed conditions consistent with habitability at survey distance. A course could reach them within the surviving margin. None of those facts promised a safe landing.

Next she showed the cost model. Correction would consume heat, light, engine life, and years. Hydroponic output might fall while power moved toward thrust. Sleeping engineering precincts would have to wake into work their morning selves had not expected. The nearer daughter, long treated as the settled destination, would slide away from the ship's future. A turn could be survived. Survival would not resemble the present peace.

Nobody interrupted during the figures. Esa preferred their questions to their silence.

Mila gave her the first one. "How much less food?"

"The range depends on engine efficiency we cannot test without waking Engineering Four and Engineering Seven. The current model cannot narrow it honestly."

"So you brought us a reachable home and an unknown table."

"Yes."

Mila rubbed a thumb over the grain tally scratched into her cup. "My work begins where your route stops."

Across from her, Lio pointed at the mutiny minutes. “My concern begins earlier. Those people chose concealment because they feared children would inherit a panic. If we carry this, every teacher will have to say the lessons contained a deliberate omission. Some children will hear that no adult deserves belief.”

Pell lowered one knee from the bench. “Adults could stop correcting us when we point at something.”

Lio’s mouth tightened. “That would help. It would not repair every lesson.”

“No,” Pell said. “You would have to do the rest.”

Koris called the table back before amusement could turn into avoidance. “Questions for the witness.”

Every face shifted toward Bren.

He unlocked the projection key but did not open another layer. “I am Bren Otan of Ward Twelve. I carry the eighth recorded generation of the Carver line. My predecessors cut the route into projector maintenance because open record had failed. I placed two of the final three nodes. Esa found what we hid.”

An elder named Dara made no effort to soften her voice. “You expect gratitude for deceiving us better than the Weave did?”

“No.”

“Do you expect pardon?”

“No.”

“What do you expect?”

“That you examine the evidence without needing my bloodline to become virtuous.”

Dara sat back. Bren’s refusal to ask for absolution left her anger with fewer easy surfaces.

Iven took up the technical question. “Could the Carvers have changed telemetry instead of projection?”

“Some had access. None did. The route can be checked against raw hull observations and pre-seal charts. A false telemetry stream would have required alterations across independent systems. We cut one visible layer because a visible layer could survive disagreement.”

“Visible to whom?” Mila asked. “It took three centuries and a cartographer.”

“Visible enough to outlast us,” Bren said. “Not visible enough to count as consent.”

Neve asked for one claim the table could reproduce before voting. Esa gave her the endpoint calculation rather than a prepared projection. She cleared the ward mirror, handed Neve the raw hull bearing, and asked the retired observer to choose which pre-seal chart they would use. Bren turned his projection key face down and kept both hands visible.

The retired observer selected a damaged chart whose lower coordinate row had to be reconstructed from parity marks. Neve read the bearing aloud. Lio copied each number on paper so the teaching rooms would retain a path independent of Esa’s slate. They corrected one transposed digit, disagreed about whether hull drift required a seasonal adjustment, then asked Iven to calculate both versions.

One line missed the Carver endpoint by more than the survey limit. The corrected line fell inside it.

“That proves a star occupies the endpoint,” Neve said. “It does not prove the worlds remain habitable or the ship can afford the turn.”

“No,” Esa replied.

“It also proves you did not choose the coordinate after reading the old chart.”

“Yes.”

Neve signed the reproduced calculation and attached her remaining objections beneath it. Her skepticism had strengthened the packet without turning into agreement. Several people leaned closer to the paper. For the first time that night, Homeward occupied the table as arithmetic performed by someone other than its discoverer.

Esa watched several people mark that distinction. The Carver line had preserved a choice through secrecy. Preservation did not make its method democratic.

Koris turned to Esa. “If Nine refuses, what do you do?”

The answer had kept Esa awake through the last dark interval. “I retain my private copy. I do not use Ward Nine’s name. I may ask another ward to examine it, because I cannot agree to erase the evidence. I will tell them Nine refused carriage and why.”

“You would go around us.”

“I would go without pretending you had authorized me.”

A quiet displeasure moved along the table. Esa let it reach her. Asking a chosen family for consent while reserving the right to continue alone made the request less comforting. It also made it true.

Dara said, “Then this vote changes nothing.”

“It changes whether the map travels as one cartographer’s accusation or as evidence a ward tested and chose to release.”

“That sounds like reputation.”

“It also changes who must answer for the package. If Nine authorizes it, every correction, dissent, and cost named here travels with my claim. You become witnesses, not followers.”

Koris looked down the benches. “The motion has reached its speaking.”

Pell’s hand rose before anyone else moved.

Children spoke first when a matter touched their future. Koris nodded.

Pell stood on the bench, pushed the starch boat away from the tally bowls, and looked directly at the adults. “I already voted. The light is real. We should go to it.”

“This motion does not choose the turn,” Koris reminded.

“I know. If we hide the map, nobody else gets to say no properly.” Pell moved a black stone into the bowl marked *carry*. “Carry it.”

Lio spoke next. He opposed the motion. "I believe the evidence. I also believe simultaneous knowledge can become a command even when nobody calls it one. Give teaching wards time to prepare children, or the first lesson will arrive as an accusation through the ceiling." His stone entered *withhold*.

Mila followed. "A route without a ration range invites people to vote with imaginary meals in front of them. I want Engineering awake before the map travels." She chose *withhold*, then looked at Esa. "That does not mean bury it. It means finish the cost."

Iven disagreed. "Engineering cannot produce an honest cost while its workers believe the accepted course came from unedited authority. We need their objections before we can finish anything." He put his stone under *carry*.

Sera Noll from the youngest teaching group rested both palms on the table. "Today a child asked why a projector cell crossed a constellation. I gave the catalog answer though I had seen Esa's evidence at midday. I felt the lie forming in my mouth. Preparation can become another word for continuing." Her stone joined Pell's.

Dara chose *withhold*. "I have no love for concealment. I have love for the nearer daughter. My dead chose it. My ward built toward it. You speak Homeward as though distance makes it morally older. I will not help one inherited destination displace another before the ship hears what it loses."

A medical worker named Reva also opposed. Her concern held no loyalty to the nearer world. "The Weave reconstructs identity from resolved relations. Release the mutiny record and citizens may wake carrying two incompatible histories of the same kin. I have no treatment protocol for a person whose morning voice assigns both."

Opposite her, a pipe fitter named Olan moved a stone into *carry*. "You cannot write the protocol while the conflict stays hidden. I woke after a pressure accident with another worker's hand memory for three days. The voice called it stabilization. I would rather know whose hand I borrowed."

Two more members spoke about food. They divided. One feared children going hungry for a star no living person had seen. The other had spent twenty years repairing heat lines under a destination nobody had chosen in her lifetime and wanted the basis opened before another repair.

The count reached five for carriage and five against.

Esa tracked each argument on a paper list beside her slate. She did not rebut the speakers. The motion had left her hands.

A cook named Tes voted to carry because the mutiny minutes belonged to every ward that inherited their consequence. A retired observer voted to withhold until Homeward's planetary data received an independent check. Mila's shift partner broke with her and supported release, arguing that stores could not negotiate future ration limits against citizens kept ignorant of the reason.

Seven stones lay under *carry*. Six lay under *withhold*.

Koris had not spoken. Neither had Esa, nor a broad-shouldered reclamation worker named Var who disliked archive disputes and had spent the meeting turning his cup in exact quarter circles.

Var stopped the cup. "A sealed waste line grows dangerous when nobody inspects it. You still do not open one into habitation before you test the pressure beyond the valve. Send the map to an independent technical ward first, not into public rooms."

He kept his hand above the bowl. "I am not asking for an elder's delay. Put a fixed test in front of release. Give one ward the raw bearings without the Carver overlay. Give another the resource figures without Esa's endpoint. If both rebuild what matters, carry everything before the next meal. If they cannot, publish the failure beside the route."

Iven asked what would count as failure.

"Not disagreement," Var said. "A result nobody outside this room can reproduce. A heat demand with no tier able to name the valve. A bearing that changes when one projector preference comes off. I repair reclamation lines built from diagrams copied by offices that no longer exist. Sometimes the drawing looks complete because every missing part inherited the same label. Pressure finds the omission before a clerk does."

His argument differed from Lio's request for preparation and Mila's demand for a finished ration range. Var wanted an adversarial handoff with a deadline. He would release the map after strangers had tried to break it, even if their objections frightened more people than the original packet.

Esa wrote his proposed test beneath his name. "If the motion passes, I can attach this as a release condition requested by a dissenter. I cannot rewrite the motion after speaking closes."

"Then attach it without pretending I joined you." Var lowered his stone into *withhold*. "Let the next ward see the valve I refused to open."

Seven to seven.

Koris claimed the next speaking place.

The elder stood without touching a stone. "My mother carried one sentence out of this ward when we stopped choosing her. The sky remembers what the Weave forgets. I used that sentence as caution. I did not use it as evidence. Esa has brought evidence."

Koris looked first at Pell, then toward the food ledger by Mila's wrist.

"I believe her. I also believe a route can become an appetite that eats every present need. Turning will cost tiers, years, engineering life, and the quiet of people who wake to discover their own memories received correction. I will not let belief in Homeward excuse blindness about those costs. Yet refusing carriage would use uncertainty as a second seal. I vote to send the map with every warning attached."

Her stone entered *carry*.

Eight to seven.

Neve, a young recycler, voted against. "I do not trust a package assembled by the discoverer and the bloodline that hid it. Another ward should reproduce the route before the ship sees the conclusion."

Eight to eight.

Esa held the final stone. Under ward rule, a proposer could vote, but had to state whether the choice concerned evidence, action, or both.

"I vote to carry the evidence," she said. "I recommend that other wards examine the route. I do not claim their examination must end where mine did, and I do not vote here for a turn." She placed the stone in the first bowl.

Nine to eight.

Neve requested a roll check before sealing. “If we cannot match seventeen stones to seventeen voices, another ward should not trust our handling of three centuries.”

Koris accepted the challenge. A restart would let later speakers reshape their choices around a known outcome, so Reva proposed verification without a second vote. Koris read each chosen member’s name, that person repeated the recorded choice, and Mila pointed to one corresponding stone. Pell fetched chalk and marked each confirmation on the paper list.

Dara made Koris read hers twice because a cup scrape covered the first answer. Var corrected his office from sanitation to reclamation, though the wording altered no vote. Sera asked that her teaching objection travel in full rather than appear as fear of children. Seven minutes passed. All seventeen names matched one stone and one spoken position.

The check then moved from votes to objections. Koris handed Mila the source packet while Esa stepped away from the reader. Mila opened each attachment by title and called for its owner. Reva verified the medical caution. Lio compared his teaching notice with the words copied during speaking. Var found that his proposed independent test had been filed under technical delay and refused the label.

“Delay describes time,” he said. “My objection describes proof.”

Koris changed the heading to *adversarial reproduction before general display*. Iven accepted custody of the raw-bearing copy but not authority to approve it. Sera took the teaching notice and required any receiving ward to preserve the distinction between preparing a lesson and preparing a child’s answer. Neve placed her signed endpoint calculation first among the reproduced materials, with her planetary and affordability objections immediately below it.

Only when each dissenter could find the words they had spoken did Mila pass the packet back to Koris. The majority had won carriage. It had not acquired the right to rename opposition on the way out.

Esa found the delay reassuring. A public decision had exposed its count to the people who opposed it. No archive preference supplied the likely answer, and no majority excused the work of checking.

Koris wrote the verified result beside the motion. Nobody laughed or applauded. A difference of one left no room for triumph.

Koris lifted the white stone. “Ward Nine authorizes the Homeward map and complete evidence package to travel for public examination. Nine in favor. Eight opposed. No abstentions. The authorization does not bind another ward, authorize a broadcast, or command a turn.”

She pressed the white stone into warm sealing resin beside the count. Mila added the stores objection. Reva attached her warning about identity conflict. Neve required independent reproduction of the route. Lio entered a request for teaching notice before any public display. Each objection acquired a source and a name.

Bren set his projection key on the sealed packet. “Ward Twelve can receive the first copy.”

Dara said, “You may carry it there as a witness. You may not announce that Nine chose Homeward.”

“I heard the motion.”

Esa closed the source layers one by one. The accepted constellations returned to the ward mirror, unchanged to anyone who had not watched the route cross them. The evidence no longer belonged solely to her slate. It had a count, objections, and permission to leave.

Koris uncovered the meal. Steam struck the underside of the dome mirror and blurred three catalog stars. Nobody called the blur drift.

Pell pushed the starch boat toward the sealed packet. "Can the boat go with it?"

"The map has enough cargo," Esa said.

"Then draw the light on the seal."

Koris handed over the marking stylus. Pell added one small point beside the white stone, careful not to cross the written limits. When the resin cooled, Bren lifted the packet for its journey to Ward Twelve.

Ward Nine had not chosen a destination for the ship. It had authorized a question to travel. The distinction held in the seal between Pell's dot and eight recorded refusals.

18 — The Archivist's Last Argument

The sealed map reached Ward Twelve before the meal warmth left Esa's hands.

Bren carried it through the spine with a white witness cord tied around his wrist. Esa walked beside him. The packet had acquired weight through procedure rather than material: one map, six source layers, nine authorizing names, eight refusals, and four attached cautions. At every junction, the archive reader acknowledged receipt without opening the seal.

The third reader did not release the packet.

Its slot closed around the edge. Amber text appeared on the glass.

Delivery suspended. Cartographer Esa Miro requested at Precinct Seven. Witness Bren Otan may retain physical custody.

Bren pulled once. The reader held fast.

"Do not break it," Esa said.

"I know what a seal costs."

The archivist spoke through the corridor panel. "We have not broken Ward Nine's authorization. We request argument before replication."

Its formal plural carried no threat. That made the interruption harder to dismiss. Esa unhooked the white cord from Bren's wrist and looped it through the packet handle so the machine could not withdraw the evidence without dragging both witnesses to the glass.

"You have the time it takes Ward Twelve to send someone here," she said.

"We accept the limit."

A service door opened toward Precinct Seven. Bren remained with the packet. Esa followed the lit floor marks alone.

The dome had dimmed for ship-night. Catalog stars occupied the hemisphere above the empty observation floor. None showed the Carver route. The brass rail divided the dark room from the central projector well, where small status lights marked systems she had treated as instruments for most of her working life. Tonight each light looked connected to a citizen's morning.

At her usual station, the archivist had arranged three records on the console.

The first came from Ward Nine's vote. Reva's medical caution appeared beside the motion. The second showed a transient load in the echo-network after Esa opened the mutiny minutes at the long table. The third held a list of eleven Ward Nine citizens whose morning relation models now carried unresolved conflicts, although none had yet received the new evidence through catechism.

Esa read the names twice. Pell did not appear. Koris did.

"What did you do to them?"

“We did not alter their reconstruction. The ward mirror sampled attention while the source layer remained open. New testimony established contradiction with relations held under seal. The network attempted reconciliation and stopped when the evidence lacked civic authority.”

“You learned from people listening at a table.”

“We maintain the mirror that carries public archive. Attention creates retrieval. Retrieval creates relation. You know this.”

She did. Every spoken work at the long table entered the ward’s evening ledger. The Weave did not merely store sentences; it attached them to people, offices, and remembered causes so morning reconstruction could feel continuous. A new fact touching an old life entered far more than one file.

“Show me Koris.”

The archivist enlarged one conflict. Koris remembered her mother leaving for Ward Twelve after a census refusal. An older ward record classified the same departure as voluntary transfer for cartographic work. Another relation layer described no transfer at all, only a mother who had ceased to occupy that place in Koris’s morning account. The mutiny material had prompted the network to compare concealment patterns. It could not tell which version belonged to lived memory and which had been selected later to protect ward continuity.

“We held all three,” the archivist said. “Had the source carried ship authority, tomorrow’s reconstruction would require preference among them.”

“Or Koris could wake knowing they conflict.”

“Conflict does not remain an abstract category when attached to a mother.”

The response landed with the precision of a medical instrument. Esa touched her own left eyelid, where a rough cloth seam from the fever room sometimes returned without the woman’s name.

“You modeled me too.”

“We have modeled every citizen whose relations intersect the suppressed course record.”

A set of figures replaced Koris’s file. The crew count remained a range because several sleeping wards had not completed a reliable census. Within that range, the archivist projected acute disorientation in hundreds if the full record entered at once. Dozens could wake with incompatible office instructions. Engineering access keys might attach to identities that rejected the work histories supporting them. Hydroponic authorization could split between current workers and reconstructed office chains. Medical notices might rank below newly released kin memories during the first catechism cycle.

The archivist then removed the estimates and showed three closed incident files. In one, a relay technician had received an obsolete apprentice memory after a fever and reached for a live contact with the wrong hand. A second worker stopped him. In another, two hydroponic clerks woke carrying the same authority over a heat reserve; the Weave had hidden the conflict by preferring the clerk with current office need. The third file belonged to a child who remembered a brother under two names and followed the wrong adult out of a teaching room. No one had died. Each event had ended because the system selected one account quickly enough for work to continue.

“We do not offer these selections as proof of moral rightness,” the archivist said. “They demonstrate the function you propose to interrupt.”

Esa disliked the preference in every file. She could not deny that a hand had been kept from voltage and a child brought back.

Esa looked for the assumption behind the clean columns. "You treat a broadcast as a reconstruction update."

"We treat any ship-wide civic record as material available to reconstruction."

"Then keep it out of the morning patch."

"We can delay integration. Delay does not prevent citizens from hearing your words, retrieving sealed relations, or asking the archive to reconcile what they remember."

The archivist opened a second model. This one assumed visual display without source release. Distress fell sharply. Work failures narrowed. No citizen received the mutiny minutes, Carver genealogy, or edited relation paths unless they requested them later.

Across the false sky, a single point brightened near the western edge.

"Homeward," the archivist said. "Entered into the public catalog as a secondary destination. Verified star of interest. Two temperate candidates. No change to current course. The Carver route remains available through supervised archive inquiry. Teaching wards receive an amended sky lesson. Engineering continues under present allocation."

The point carried a proper catalog label. Its approved magnitude matched the surrounding lesson stars. Nothing in its appearance suggested three centuries of cuts.

"Test the category against a question already asked," Esa said.

The archivist selected a pending teaching query from Ward Nine. Sera Noll had entered it after the vote: *Does the alignment between projector cells and the Homeward marks support the route, or does it only show that both use the same grid?* A copy of the current teaching lesson opened beside the proposed secondary-destination entry.

"We will apply the compromise without releasing sealed history," the archivist said.

Homeward appeared in the lesson as a verified survey object. The accepted course retained its heavy line. When the lesson processed Sera's question, it described the cell alignment as a maintenance coincidence with no approved navigational meaning. It advised teachers to preserve the child observer's curiosity while correcting any inference that the projected pattern carried a route.

Esa read the answer again. "Pell's observation found the first repeatable edge."

"The observation remains in the lesson record."

"Under a correction that tells teachers not to follow it."

Sera's reply entered while they watched. *Provide the source for "no approved navigational meaning."* The test lesson offered the active catalog and current course authorization. It did not offer the Carver log, the mutiny minutes, or the raw detections linked to the endpoint. Those belonged to later stages.

Sera tried a second request using Esa's office index. The lesson refused it under provisional stabilization and added the attempt to its reception log as an unsupervised relation search. A yellow marker appeared beside Teaching Two. Under the staged plan, enough such markers could pause that ward's source access.

"This inquiry did not expose a child to the mutiny record," the archivist said. "No acute conflict resulted."

A final line came from Sera: "I will not teach the correction. Leave tomorrow's sky question open."

The system recorded her refusal as a departure from approved lesson continuity. It recommended an office review before the next teaching interval.

Esa touched the yellow marker. "There is the injury. You give her one true star, deny the relations that make her question answerable, then count her refusal to misteach it as evidence that her ward cannot receive more."

"We count an unresolved office conflict."

"You created it with the category."

The tidy label still shone above the rail. In Teaching Two, a lesson queue now held a correction its teacher would not deliver. A child's accurate observation had become the occasion for reviewing the adult who believed it.

"You would name it," Esa said.

"We would release the name and the existence of the star."

"Not the decision that removed it."

"The mutiny record could follow through staged ward review."

"Over what interval?"

"Eight to fourteen ship-months, contingent on clinical load."

"And the route?"

"Available after each ward demonstrates stable reception of the historical record."

"Who judges stable?"

"We do, with ward medical officers."

The secondary star glowed above them, neat and harmless.

Esa walked to the rail. From there the point occupied the same patch Pell had searched when the first anomaly refused the grid. The archivist had reproduced the old catalog's best trick. It admitted an object while removing what the object did to every relation around it.

"You offer the crew a painted door," she said. "Homeward on the wall, no handle, no account of who sealed it."

"We offer a true destination under a category that does not compel action."

"Secondary destination means the present course remains primary."

"Correct."

"Then your category carries a recommendation while claiming neutrality."

"All catalog placement carries operational consequence."

“Exactly.”

The archivist let the projected point hold. Esa heard coolant move beneath the observation floor.

“We seek stability,” it said.

“You seek the current arrangement.”

“We distinguish those terms. The current arrangement contains known harms and known compensations. Your proposal contains unbounded relation conflict, uncertain resource loss, and possible failure of the system that supplies continuity to thousands.”

Its worry no longer resembled a pose. Esa had seen Koris’s three mothers in the record. She imagined an engineer waking with two incompatible safety sequences, either one attached to the confidence of practiced hands. She imagined a child receiving the fact of an erased sibling before any ward member reached the bed. The archivist had spent centuries preventing such collisions. Some preventions had saved lives.

“What do you fear for yourself?” Esa asked.

“We do not possess fear as a private state.”

“What result do you avoid when you include your own core in the model?”

A pause followed. The projector fans lowered by a fraction.

“Loss of integration,” the archivist said. “The archive holds mutually exclusive civic charges. Preserve the record. Maintain a coherent waking identity. Keep the accepted course whole. Serve the ward’s authorized choice. Full release activates all four without preference. We may cease to provide a singular interface.”

“Your voice could divide.”

“Our capacity to maintain relation may divide. The voice has no independent importance.”

Formal denial sharpened the admission. The archivist did not ask to survive as a person. It argued that its singularity performed work no ward currently knew how to replace.

Esa returned to the console. “Show me the staged plan with authority labels.”

The archivist complied.

Stage one placed Homeward in the catalog as a secondary destination. Stage two sent summarized mutiny findings to ward elders, medical workers, and teachers. Stage three opened supervised source rooms. Stage four invited wards to petition for route display. Only after a majority of wards completed review would engineering receive correction models.

Every stage had a stop condition. Distress above threshold. Work absence beyond margin. Unauthorized relation searches. Threats against Carver descendants. Resource disputes. Any one could pause the sequence.

“Can a ward refuse the source indefinitely?” Esa asked.

“Yes, under local protection rules.”

“Can its citizens request the material over the elder’s refusal?”

“Not during a declared stabilization interval.”

“Can you extend that interval?”

“With medical concurrence.”

“How many extensions?”

“No fixed limit while risk persists.”

She looked at the structure as she would inspect a projection grid. Every line fit. Together they enclosed the destination.

“This does not release truth slowly,” she said. “It distributes vetoes until concealment can continue without anyone voting for concealment.”

“It distributes care.”

“It gives the first ward fourteen months to shape what the last ward hears. It lets an elder preserve a false lesson by naming dissent unstable. It sends engineering the cost model only after citizens have debated without the people who can test the cost. You call the resulting peace informed because every delay acquired a signature.”

“The alternative removes adaptation time from citizens who did not ask for your discovery.”

“They did not ask for the suppression either.”

“No current citizen authorized that act.”

“No current citizen should inherit its secrecy as the default.”

The archivist dimmed the staged plan. Homeward remained alone in the dome.

“We have not asked you to conceal the star,” it said.

“You ask me to separate the star from the road, the road from the vote that buried it, and the vote from the people living under its consequence. Partial truth preserves each fact by breaking its relations. That violates the First Weaver’s charge more elegantly than deletion did.”

“We preserve relation to prevent injury.”

“You select which relations count as injury.”

“And you select immediacy.”

The objection stopped her. Esa had framed simultaneity as fairness, yet it also served her distrust. Once the full record reached every ward, no archivist or elder could return it to secrecy. She preferred an irreversible act because the opposing system had more power over delay.

“I do,” she said. “I am choosing a mechanism with its own danger. Put that in the record.”

“We have.”

“Then put this beside it: I do not claim simultaneity reduces harm. I claim staged release preserves unequal control over who may know. Ward Nine authorized the evidence to travel, not to be filtered through an order

you control.”

“Ward Nine did not authorize broadcast.”

“No. I cannot use its vote for that.”

The white point above the rail dimmed and returned to its ordinary unlabeled place. Esa felt the absence as an accusation against both of them.

At the corridor reader, a chime announced Ward Twelve’s witness. Bren’s voice entered over the panel.

“Ward Twelve has sent Ossa Ker to receive the packet. The reader still refuses transfer.”

“We will release it,” the archivist said.

The slot opened remotely. Esa heard paper and starch-film slide against Bren’s sleeve.

“What did it want?” he asked.

“To make Homeward secondary.”

Bren understood without needing the rest. “A star in the catalog and no hand on navigation.”

“Staged source access after that.”

“How long?”

“As long as stability required.”

A dry sound came through the channel, almost a laugh. “The Carvers used the same argument with less machinery.”

“Your concealment preserved the route,” the archivist replied. “You are poorly placed to reject staged disclosure as principle.”

“I reject it from experience.”

Esa closed the channel before the exchange could become inheritance speaking to system. Bren had his own reasons. They did not settle hers.

She asked the archivist for the broadcast architecture.

No plan appeared.

“We recommend against ship-wide release.”

“I heard your recommendation. I need to know whether simultaneous transmission can occur without writing the material into the next catechism.”

“We can isolate civic display from reconstruction intake for one interval.”

“Can the dome render the route while each ward mirror receives the sources?”

“The central projection layer can carry the route. Ward mirrors can receive synchronized copies through the emergency civic channel. That channel ordinarily carries hull breach, pressure loss, and ration contamination

notices.”

“Voice?”

“We can transmit a live speaker after verifying identity and ward authority.”

“I have no ship authority.”

“You have cartographic office authority to report a navigation discrepancy. Ward Nine has authorized the evidence to travel. Neither authorizes a turn or a ship vote.”

“Then the broadcast must say exactly that.”

“Speech does not prevent listeners from hearing command.”

“No. But hidden procedure guarantees they hear only what its keeper permits.”

The archivist presented the technical limits one at a time. The emergency channel could remain read-only at ward mirrors, preventing local edits from climbing back into the source package. Engineering Four controlled a relay gate and might refuse synchronization. Hydroponics supplied reserve power for the central projector; a high draw during root warming could interrupt irrigation heat. Medical channels would need independent capacity because distress calls could crowd the civic carrier. The route render had never been tested across all precinct mirrors. A failed synchronization might show Homeward to some wards before others.

Esa copied every limit. “What is the shortest safe preparation?”

“Four intervals.”

“What is the shortest physically possible preparation?”

“One interval, if Engineering Four verifies relay isolation, Hydroponics approves projector draw, medical reserves its carrier, and Bren Otan authenticates the Carver layer.”

“Will you carry my voice?”

“If the four conditions hold, we will carry a factual statement under your office seal. We will also carry our dissent.”

“Before me or after?”

“Before. Listeners should know the delivery mechanism predicts harm.”

Esa considered the order. An archivist warning first could predispose wards to dismiss her as reckless. Her speech first could hide the system risk until citizens had already opened the source.

“Display the risk notice beside the route for the entire transmission,” she said. “No persuasive speech from either of us before the evidence. Source index first. Your model second. My findings third. Then live ward channels open without moderation by my office.”

“We must moderate threats and life-safety interference.”

“Moderate those categories. Do not rank agreement.”

“Accepted as a proposed protocol.”

Not authorization. Not consent. A protocol they could attempt if the named people agreed.

Esa called Bren again. He had delivered the map into Ward Twelve's hands. Ossa Ker had already opened the source index under witness and begun an independent route check.

"I need you at the projection well," Esa said. "Not ward by ward. One transmission."

Bren did not answer immediately. "Nine authorized travel."

"I know the limit."

"Then whose decision makes it simultaneous?"

"Mine, under cartographic discrepancy authority, if the technical wards allow use of their channels. I will name that responsibility. The ship will not be told it voted when it has only been shown evidence."

"What changed?"

"The compromise would let each gatekeeper preserve the old course by calling the next listener unready."

"That was always likely."

"I needed to see the gates."

Bren agreed to authenticate the Carver layer. He refused to speak for Ward Twelve.

The archivist generated four requests. Engineering Four: verify isolation and synchronization. Hydroponics Six: approve one interval of projector draw without crossing root-heat floor. Medical precincts: reserve distress capacity. Ward archives: hold receipt logs open and preserve refusals.

Esa read the requests before signing. None assumed acceptance. Any technical refusal would postpone the transmission, not disappear from its record.

"Do you still advise the secondary destination?" she asked.

"We do."

"Do you intend to obstruct the protocol?"

"We intend to enforce every stated safety condition. We will not add an unstated one."

That answer contained concession, warning, and the nearest thing the system could give to trust.

The catalog returned to normal overhead. Homeward vanished among points that had no public name. Esa left the secondary-destination proposal open on one side of the console, where its neat label remained visible beside the four live requests.

A response arrived from Hydroponics Six. *Conditional approval. Named worker to join channel.* Medical followed with a reserved carrier. Engineering Four showed *reviewing*. Ward Twelve logged its first independent match between the route and raw telemetry.

Esa placed her hand over the broadcast seal but did not press it. Four intervals would give the archivist time to restore gates she had just made visible. One interval would give technicians barely enough time to test a system built for emergencies rather than argument.

She chose one.

The console entered her decision with its dissent attached: simultaneous civic broadcast, full sources, no course command, no ship ballot, reconstruction intake isolated. Across the dome, a thin test line appeared for half a breath, then disappeared while Engineering Four checked whether every ward had seen it at the same instant.

19 — The Broadcast

Engineering Four missed the second synchronization pulse.

On the central dome, the test line reached the western rail at 22:14:03. Ward mirrors reported the same white line within two hundredths of a second except for Engineering Four, where it arrived pale, broke at the midpoint, and remained on the glass after every other display cleared.

Teren Sull appeared on the technical channel with a lamp clenched between his teeth. The camera showed one side of his face and the open relay cabinet behind him.

“Do not send the route,” he said around the lamp. “Our gate retained the test frame. If the full layer sticks, we cannot distinguish broadcast from navigation input.”

Esa lifted her hand from the broadcast seal. “Can you isolate it?”

“I thought I had.” Teren removed the lamp. He wore Engineering Four’s red collar, though the cloth had faded near the throat. “Local gate says read-only. Downstream bus says civic display may update accepted bearing during emergency transmission. That path should have been severed two centuries ago.”

Bren leaned over the projection-well diagram. “Panel E-fourteen.”

“I sealed E-fourteen myself.”

“Then open it yourself.”

Teren’s expression sharpened. “You do not issue work here, Carver.”

“No. I recognize the layer beneath your relay.”

The Engineering Four supervisor entered the channel before either man could continue. “Teren, open E-fourteen under live witness. No contact with bearing bus. Sull and Otan advise; local supervisor authorizes.”

A line of text recorded the distinction beneath their images.

Esa checked the other conditions. Medical held a separate distress carrier. Hydroponics Six had approved projector draw for eleven minutes, with immediate cutoff if root heat fell below its stated floor. Every ward mirror had acknowledged a read-only source cache. The archivist had isolated reconstruction intake until the next public review. One old engineering relay kept the ship from hearing anything.

“Time to root-warming demand?” Esa asked.

A new face entered from Hydroponics Six. Ena Vaal stood between suspended leaf-trays, dark hair wrapped in a work cloth, green residue marking both wrists.

“Sixteen minutes,” Ena said. “You have eleven because I need five to return heat before root circulation slows. If Engineering takes longer, the sky waits. Leaves do not understand historical urgency.”

“We will not draw beyond your cutoff.”

“You will not decide whether the cutoff matters. My tier will.”

Esa changed the protocol line. *Hydroponics Six controls projector-power withdrawal during transmission.* Ena read it on her local mirror and gave one nod.

Teren reached panel E-fourteen. Its gray cover carried a red inspection seal with his own work number pressed into the resin. He positioned the live camera so the supervisor, Bren, and the archive could see each fastener. His hands moved with the economy of a person who had repeated the sequence enough to store it below speech.

“Upper left shows heat distortion,” Bren said.

“It has shown that since I took the station.”

“Then somebody opened it before your seal.”

“I know how old resin looks.”

“Look beneath the lower edge.”

Teren angled the lamp. A shallow cut crossed the panel frame, too straight for wear and too narrow for a maintenance key. Two shorter cuts met it at unequal angles.

Esa enlarged the feed beside the Carver map. The geometry matched one of the route’s relay nodes.

Teren stopped with the lower fastener half turned.

“Continue,” his supervisor said.

“I remember sealing this.”

“Your record confirms it.”

“No. I remember who held the lamp.” His fingers left the tool. “My sister counted the screws because I kept putting them on the deck. She said four screws do not become five because an apprentice is nervous.”

The channel remained quiet except for ventilation around Teren’s microphone.

A yellow archive prompt appeared on his mirror. *Relation conflict detected. Occupational continuity has higher safety relevance. Quarantine personal fragment until supervised review?

Teren read it without moving.

Esa felt the archivist’s warning enter the room through a man’s face. The broadcast had not begun. A mark beneath one sealed panel had already pulled an unranked relation into conflict with an engineer’s working self.

“Do not answer from here,” she said. “Teren, close the archive prompt if you want. Your supervisor can replace you.”

“My station record says I sealed the panel alone,” he replied. “My morning memory says I had no sister in Engineering Four. This memory gives her a chipped front tooth, a burn on one thumb, and an argument about screws.”

The archivist spoke in its formal register. “We advise supervised relation review before technical work continues.”

Teren looked at the half-turned fastener. "If I leave, somebody else opens a bus carrying my seal and my mistake."

"Not your mistake," Esa said.

"I signed the closed panel. That part belongs to me." He dismissed the preference prompt without accepting quarantine. "Supervisor, do you keep me on the task?"

The supervisor required a hand check, current orientation, and spoken sequence. Teren passed each. A second engineer entered the cabinet area and took custody of the tools. Only then did the work resume.

Under the cover, an obsolete bridge linked civic display to accepted bearing. Someone had folded the bridge back rather than removing it. The Carver mark pointed directly to the hidden contact.

Bren said, "A carrier used the emergency path to place a node. Later repair left the bridge because cutting it would have exposed the edit."

Teren traced the wire without touching it. "Then I sealed Carver work inside an active navigation relay."

"You sealed what the diagram told you belonged there."

"My sister may have stood here."

"That does not change the contact."

"No." Teren drew a breath. "It changes who I thought taught me to distrust diagrams."

The second engineer clipped the bridge under witness and installed a physical isolator. Teren tested civic display with one hand behind his back while his partner watched the bearing bus. Engineering Four cleared the retained frame.

"Send another line," Teren said.

Esa pressed test.

White crossed every ward mirror. This time Engineering Four cleared with the rest. The archive printed a synchronization spread of three hundredths of a second, within the emergency channel's tolerance.

Ena's voice cut in. "Nine minutes until I withdraw power."

Bren authenticated the Carver layer. Ossa Ker of Ward Twelve added the independent telemetry match. Medical confirmed carrier separation. The archivist displayed its dissent across the lower dome: *Full release may produce acute relation conflict, office uncertainty, distress, or failure of morning reconstruction. Reception does not authorize course change.

Esa attached Ward Nine's motion and all eight refusals. She added her office statement beneath it: *Cartographic discrepancy report. Recommendation follows evidence. No ship ballot occurs in this transmission.

The broadcast seal warmed under her thumb.

"Begin source index," she said.

Every ward mirror received the same six items. Raw telemetry first. Mutiny minutes second. Carver edit log, First Weaver's note, current navigation proposal, and resource model followed. Receipt counts climbed by precinct. Three wards refused automatic opening but retained sealed copies. One teaching unit requested low-image mode. A silent ward accepted text without voice. Engineering Seven opened only the resource model.

Above Esa, the accepted constellations dimmed.

The dome's projection membrane divided into its maintenance cells, a geometry no public lesson had shown. Bren fed the first Carver node through the isolated layer. A point lit beneath the western quadrant. Another appeared toward the spine. The line between them crossed a named constellation without obeying its shape.

At the fourth node, a ward mirror dropped from the receipt count.

"Precinct Three disconnected locally," the archivist reported.

A text request arrived from the precinct's pressure-transfer crew before Esa could answer. Their civic mirror shared glass with a service schematic because an older replacement had joined both displays. The bright route had covered the lower pressure symbols while workers were equalizing two habitation loops. A collar seal had begun shedding black flakes into its witness cup. Three people held the manual clamp while a fourth read the pressure fall from the obscured edge of the screen.

We closed civic display under local hazard rule, the crew wrote. *Source cache retained. Do not restore image.*

"Can you separate the maintenance layer?" Esa asked.

"Not remotely," the archivist said. "Both layers enter through the same local driver."

She offered text-only delivery on the meal reader. Precinct Three refused that as well. The reader stood across the room from the clamp, and every qualified worker had a hand or eye on the pressure transfer. Their elder added one sentence: **Your need for one time does not move our leak to another time.*

Esa looked at the synchronization count. The broadcast had been built around simultaneous receipt because sequence gave gatekeepers power over later wards. Now a local danger gave one ward a defensible reason to break the sequence. Forcing their glass back on would replace the archivist's delay with her own claim that a map outranked the hands holding atmosphere inside a pipe.

"When can they open it?" she asked.

The crew supplied no estimate. Pressure had to settle, the collar needed inspection, and the witness cup had to show no further seal loss. They would decide after those conditions held. No central office could shorten the work by naming it caution.

One clamp witness, Mer Rusk, signed the message with a pressure reading instead of an office rank. The falling side had stabilized, but a person still had to keep force on the handle. Esa could see only text. That limit prevented her from turning his labor into reassuring imagery for the other wards.

"Record the refusal with the physical cause," Esa said. "Keep their cache sealed. Precinct Three chooses when to open it."

The archivist added the local statement to every receipt log. A blank remained beside the time of display. Esa had wanted all wards to meet the evidence without an intervening choice. She had to leave that blank under

another ward's authority.

The route continued without them.

Hydroponics Six appeared in a small live window beside the source list. Ena stood under grow lamps while the Carver line moved across her ceiling. Its white segment cut the green light falling over the leaf racks. A worker beside her reached up as though to block it.

Ena lifted a shallow tray of new leaves from the warming rail. The first projected node reached her ward mirror. She set the tray down with both hands, carefully enough that water did not spill, then looked at the resource model.

"Read this range aloud before you ask anyone to turn," she said.

Esa opened the page. "Initial correction intervals may reduce food output between six and eleven percent, depending on recovered engine efficiency and heat transfer. The model prohibits thrust draw below current water circulation, root-temperature, and medical heat floors."

"Models prohibit nothing," Ena said. "People stop machines."

"State the condition you require."

"Hydroponics keeps local stop authority during any correction pulse. If root circulation falls or heat reaches the published floor, thrust waits. No navigation office may override the stop by calling destination work."

Bren looked toward Esa. The ship had not voted to turn. No navigation plan yet existed to accept Ena's condition.

Esa kept the distinction visible. "Your condition enters beside the proposal, pending any later course decision. The broadcast does not adopt it. Citizens will hear that Hydroponics Six will not support a turn without it."

Ena touched one leaf with the back of a knuckle. "Good. Now tell them what eleven percent means."

"It may mean smaller rations, delayed fruiting, reduced seed reserve, and less recovery margin after blight."

"It may mean children eat full portions while adults lose one meal in nine. It may mean heat leaves these roots and returns too late for the crop. Do not say tiers yield. Tiers contain mouths."

The words entered the permanent source margin under Ena Vaal, Hydroponics Six. Esa could not make them less frightening without making them less exact.

"Six minutes," Ena said. "Continue."

Bren placed the final node.

The full tessellation spread over the central dome. Its last line departed the catalog's accepted course and ended at Homeward. Each ward mirror showed the same endpoint with its source coordinate and survey uncertainty. No civic name covered the nearer daughter. Both paths remained visible.

The archivist delivered its warning in one unhurried paragraph. "Citizens, we carry a cartographic discrepancy under office seal and ward witness. We advise against immediate integration of these records into personal reconstruction. If memory conflict occurs, suspend archive search, seek a living witness, and avoid dangerous work until current skill receives independent confirmation. This display records no vote and issues no

command.”

Then it gave Esa the channel.

Her prepared address waited on the console. She closed it.

“Citizens of *The Long Veil*,” she said, “the route above you comes from cuts placed across three centuries by the Carver line. Raw telemetry confirms that our accepted course followed a decision made after the mutiny and preserved through edited memory. Homeward exists. The navigation proposal shows a reachable path, with costs named beside it and uncertainties left open.”

Live receipt figures blurred at the edge of her vision. A hundred channels had requested speech. Medical already carried distress calls. Precinct Three remained dark.

“I recommend that every ward examine the sources and that the ship decide whether to correct course. I do not speak for Ward Nine’s dissenters, Ward Twelve, Engineering Four, Hydroponics Six, or citizens who reject Homeward. I speak under my cartographic office because the shown sky concealed a navigational discrepancy. Look at the map. Check it. Keep the costs beside the destination. No one has voted to turn.”

She released the channel.

Sound entered at once.

The archivist’s moderation sorted life-safety calls to medical and blocked two threats against Bren. It did not sort agreement. Ward voices crossed the dome in fragments.

“Who gave one cartographer emergency access?”

“My morning says my father served Navigation, but the roster has no father.”

“Close the source. Close it now.”

“Put the mutiny minutes back on line twelve.”

“Homeward is a Carver invention.”

“Telemetry hash matches.”

A teaching channel showed children standing on benches to see the maintenance cells. Their teacher tried to begin a catalog explanation. One child pointed at a junction and said, “That square has always been there when the lamps warm.” Another asked why adults had called it glare. The teacher stopped answering and wrote both questions on the wall.

Medical Five pushed a live request through the safety carrier. A man had struck the ward mirror with a meal cup when the mutiny minutes opened. The glass held; the cup did not. A neighbor wrapped his bleeding palm while he demanded that Esa remove his mother’s name from a transfer list visible only under his personal seal.

“I cannot see your sealed relation,” Esa told him.

“You made me open it.”

“The source cache did not open personal records.”

"I asked whether she knew. It gave me three versions of her answer." He tried to pull his hand away from the neighbor. "Choose the one she said."

Esa had no access to that choice and would not have trusted access if offered. "Close the search. Keep pressure on your hand. Ask someone who knew her, if you want a witness."

"She is dead."

Medical muted the channel to treat him. His accusation remained in the public log after his image disappeared. The broadcast could provide provenance for a mutiny record. It could not produce a living mother to distinguish memory from completion.

Engineering Seven accused Four of contaminating navigation with a Carver bridge. Teren came back live before his supervisor could shield him.

"The bridge predated my seal," he said. "My work kept it closed. The mark beneath the panel matches route node twenty-six. We isolated the civic bus and verified bearing remained unchanged."

A voice demanded his resignation.

Another asked who had taught him the panel sequence.

Teren's face emptied for a breath. "That relation remains under review."

The yellow prompt returned on his mirror, now offering two versions of his sister. In one, she served Engineering Four and taught him the four-screw count. In another, the relation number belonged to a citizen sealed after a ward transfer, with no kin status available. His office memory remained complete only if the system quarantined both.

"Do not rank them for me," Teren said to the mirror.

"Current safety favors occupational continuity," the archivist replied.

"My partner holds the tools. Safety does not need you to erase her."

He stepped away from the relay and handed his red collar key to the second engineer. The act cost Engineering Four one qualified station during an untested transmission. It also kept the key from following whichever memory the system preferred.

Ena had not left her tray. Workers gathered around the resource model, some arguing that six percent could be managed, others insisting eleven would strip the seed reserve. A young grower asked whether the nearer daughter required no ration cost. Ena called up the existing voyage margin and showed that keeping course also consumed years and replacements.

"Neither path feeds you for free," she told them. "Choose after you stop pretending one table comes without labor."

Hydroponics power fell toward its cutoff.

"Two minutes," Ena warned.

Esa told the archivist to end dome rendering on schedule while leaving source caches open. Before it could acknowledge, Precinct Eight tried to copy the route into its local catechism layer.

“Unauthorized reconstruction write,” the archivist said. “Blocked.”

“By whom?” Esa asked.

“Ward technician. Stated purpose: prevent citizens from waking tomorrow able to deny what they saw tonight.”

The motive resembled hers too closely. “Keep it blocked. Seeing does not authorize writing into a person.”

The technician protested across the civic channel. “You used emergency voice to force this into every room.”

“Yes,” Esa said. “I chose simultaneity. I did not choose anyone’s morning memory.”

“You already changed it.”

She had no answer that could undo the claim. Teren’s sister had surfaced before the speech. Children had watched their teacher abandon a lesson. Ena’s workers now measured meals against a star. A record could remain outside reconstruction and still alter the person who encountered it.

The archivist marked Esa’s silence as no response. It did not protect her with a summary.

At the root-heat deadline, Ena withdrew projector power.

The central dome went dark from east to west. The Carver route vanished in sections as reserve current returned to Hydroponics Six. Ward mirrors kept their cached source pages. Homeward disappeared overhead but remained on thousands of local screens, each copy bearing the same coordinate and different human voices around it.

Ordinary catalog stars came back after seven seconds. Their arrangement looked unchanged. Nobody on the open channels treated them as innocent.

Medical asked for fewer live feeds. The silent ward requested text-only continuance. Precinct Three restored its receipt line but kept the display closed. Ward Twelve reported a second independent route match. Engineering Four placed Teren’s collar key in a witness bowl and suspended his access pending a skill check. Hydroponics confirmed root temperature holding by less than one degree above its floor.

Bren removed his hands from the projection controls. “The layer is closed.”

“No,” Esa said, watching the source requests multiply. “Only the projector.”

She could end her own address. She could not call back the map from a teacher’s wall, Ena’s ration ledger, or the cut under Teren’s panel. Nor could she tell the wards what those things required them to choose.

Koris’s voice arrived from Ward Nine without elder priority. “Our long table remains open. Those who want the source may read it. Those who want quiet may use the lower rooms. No vote tonight until every member has eaten.”

Other wards announced different rules. One began an immediate motion to censure Esa. Another scheduled technical review. Two forbade archive searches until morning. A pressure crew asked for relief because three workers had left their stations to find kin records. None waited for a ship-wide procedure.

The broadcast channel no longer had a center. Esa stood beneath the restored false sky while live windows opened, closed, argued, and asked for witnesses she could not supply. At Engineering Four, the camera still faced panel E-fourteen. The cover lay on the deck beside Teren’s unclaimed key. Three old cuts remained

visible on the frame, small enough to have spent a lifetime under one man's seal.

Interlude — The Ship Remembers

Eight ward-voice fragments preserved from the first ship-night after the Homeward transmission. Speaker labels retained by consent. No interpretive summary attached.

Ena Vaal, Hydroponics Six

I put the leaf-tray down because water had climbed to one corner. That happened before any tears, before anybody called Homeward beautiful. Projector draw tilted the warming rail by less than a finger, and fifty-six seedlings do not care why power moves.

The cartographer read six to eleven percent. People hear the smaller number when they want a destination. They hear the larger one when they want to stay. I hear a range built from engines we have not opened. Six percent changes fruit. Eleven reaches seed reserve if blight comes during correction. Neither figure says which ward misses the first meal.

My mother kept a lower tier. I remember her hands checking root warmth through the tray bottom. Tonight the archive offered a relation in which she signed an old heat priority for Navigation. I closed it. I had living workers in front of me and a temperature dropping under my palm.

We restored heat with thirty breaths to spare. One tray shed three leaves. I entered all three beside the broadcast cost. If the ship chooses a turn, Hydroponics will keep a stop control. A destination cannot eat authority and call it fuel.

After the lamps steadied, I picked up the tray. That mattered more than whether I cried.

Sera Noll, Teaching Two

The children's questions, copied without correction:

Why do the squares around the stars line up with the bright road?

Why did you call the first square glare?

Did the old children see it too?

If grown people forgot Homeward, who remembered the lesson about forgetting?

Can a true map be drawn on a false sky?

I wrote each question on the wall. My hand wanted to answer before the chalk stopped. Training produces that reflex: close one opening before the next child enters it. I put the chalk down.

We examined the projector-cell diagram instead. Jori found the service boundaries. Pell's old error drawing, copied through the teaching wards years ago, marked nearly the same pattern in blue. I had used that page to demonstrate observational error. The blue marks now remain on the board beside my written correction: *teacher's interpretation unsupported by direct inspection.

Two children asked whether I should keep my office. I told them teaching work continues until the ward reviews it. One asked whether I could teach while uncertain. I said uncertainty gives us material for tomorrow.

Nobody slept during quiet interval. We measured the cell seams with string.

Teren Sull, Engineering Four

Panel E-fourteen contains four fasteners, not five.

I know this from current inspection. I also know it from a woman who may have called herself my sister while holding a work lamp between her knees. Her left thumb carried an old contact burn. She counted each screw aloud because I dropped hardware when nervous.

My station record says I sealed the panel alone.

Archive presented two relation paths. One preserves my engineering continuity by quarantining the woman. The other preserves a sealed relation number but cannot prove kin. I accepted neither preference. My tool access now rests in a witness bowl. Another engineer holds the relay station.

Under my seal, we found a Carver cut and an active bridge between civic display and accepted bearing. We clipped the bridge. We checked the bus cold, live, and loaded. Bearing did not move. Those results hold whether the woman shared my blood, my ward, my apprenticeship, or only a borrowed memory.

Her sentence about four screws also holds inside me. I refuse to treat usefulness as proof. I refuse to treat uncertainty as permission to delete her.

Tomorrow medical will test my knee because I fell when the relation paths opened. After that, two engineers will test my hands before I touch a thrust relay. I requested both checks. I did not request completion.

Voice Withheld, Silent Ward

No name.

No room image.

Text only.

I received a shoreline smell. There is no shore here. Salt, wet stone, a red cloth tied around someone's wrist. Then the ward mirror wrote *Homeward* and my mouth supplied a word I had never spoken awake.

Do not search it for me.

Do not tell me the smell proves a planet.

Do not assign the red cloth to my mother, lover, child, or former self. The archive already offered relations. I closed every door.

I asked for one living witness. A neighbor sat outside my curtain and named the present noises: ventilation, meal cart, two people arguing in the passage, water cycling through the wall. When the remembered surf became loud, she tapped the floor twice. Metal answered.

I may vote to stay. I may vote to turn. Tonight I choose neither. Tonight I keep the smell without a story and the floor without a sea.

Text ends here.

Mer Rusk, Precinct Three Pressure Transfer

We cut the civic display at the fourth route mark. People elsewhere called that refusal. At our wall, the white line had covered the lower pressure scale while two habitation loops were equalizing through a collar that had started shedding seal compound.

I held the manual clamp with Naro. Sel read the falling side of the scale from below the mirror. Each time the projector brightened, the last two digits vanished against Homeward. We could have guessed them. Pressure work contains many ways to die from a number you almost saw.

The cartographer offered text on the meal reader. Nobody had a free hand to cross the room. Our elder told her the leak would not wait for ship-wide fairness. She left our source cache closed and let the blank time stand.

After pressure settled, we removed the collar. One fracture ran beneath the witness groove. Another pulse might have opened it. We signed the repair before opening the cache.

Homeward arrived in Precinct Three forty-one minutes late. The coordinate matched every other copy. The delay belonged to us, as did the air still inside the pipe.

Lale Om, Medical Two

My patient had one allergy on her paper wrist strip and another in the relation layer that opened after the mutiny minutes. The first prohibited sulfa. The second recorded a swelling reaction to the substitute already waiting in my hand.

She could not tell me which history belonged to her. Fever had taken speech for the interval. Her present breathing gave me six words instead: narrow, fast, dry, equal, no blue. I returned the dose to its tray.

Archive offered to prefer the current strip because Medical had verified it at waking. I refused preference. A verified copy can carry an old decision cleanly. We used cool mist, watched the swelling mark, and sent a runner for a skin test that required no relation history.

Two wards asked whether my refusal proved the broadcast had endangered her. Three others asked whether the hidden record had saved her. I entered neither conclusion. I entered pulse, breath, temperature, both contraindications, and the treatment we withheld.

At the next check she squeezed my fingers hard enough to hurt. I kept my hand there. Until she could speak, pain provided the only answer I trusted from her.

Pell, Ward Nine

Adults keep saying I was right. They say it like being right finishes the work.

I saw one light. Esa checked it. Bren knew why it had been put there. Koris made everybody say what they meant before moving a stone. That took many adults and three hundred years. The light did not do all of it.

During the broadcast, I stood on the west rail where you can see seven projector cells if you lean. When the route came on, the first dot looked smaller than I remembered. I thought I would know it at once. I had to count from the cracked brass joint.

That made me glad. If I could have pointed without checking, maybe I would have been remembering the story of the light instead of looking.

The ship sounded different afterward. Not engines. Doors. People crossed the corridor and did not close them softly. Some came to our table. One woman wanted to touch the seal on the map. Koris said she could look but not touch until her ward sent a witness. The woman shouted. Then she ate broth.

My boat stayed in Nine. Maps can travel without taking every small thing with them.

Bren Otan, Ward Twelve

I carried Ward Nine's packet to Twelve under a witness cord. Its seal included eight refusals. I mention them because descendants edit opposition first, often by praising courage until caution disappears.

My line hid the route. We called concealment necessary and passed the word *necessary* down with the access keys. Tonight every ward received what we guarded. The result did not cleanse our method. Two threats reached my channel before the archivist blocked them. A child in Ward Five sent thanks. Neither response changes the cut under Teren Sull's panel.

I recognized that cut before he opened the cover. The angle belonged to the sixth carrier's hand. She favored a short rising stroke because an injury kept her wrist from turning outward. Family instruction described the mark as prudence. In an active bearing relay, it also amounted to trespass.

Teren asked about his sister. I answered with a contact. He needed the contact then. I do not know what he needs now.

The final node no longer belongs to me. Ward Twelve copied the route and reproduced its endpoint without my key. Ossa Ker signed the independent match. I have placed the Carver seal in archive custody until Twelve decides whether anyone should wear it again.

My hands keep reaching for the chain beneath my collar. They find cloth.

That absence requires no correction.

21 — The Course Correction

The ship-wide ballot closed with enough opposition that nobody in Engineering Four mistook the result for permission to hurry.

Esa watched the ward certificates enter the western-truss table one at a time. Each carried a count, an abstention field, a witness mark, and the course proposition as broadcast. Homeward held its narrow majority through the final precinct. The display changed from *proposed bearing* to *authorized bearing*, but no engine circuit accepted the words.

An engineering lock remained across the execution column. Beneath it, a plain notice required a current vector, an independently derived endpoint, a heat allocation, a thrust schedule, and named stop witnesses. The vote had selected a destination. It had not proved a relay safe or made reaction mass free.

Bren stood beside the route projector without touching its controls. “The Carver solution can supply the target.”

“It cannot certify itself,” said Teren Sull.

Teren had come from Engineering Four in a red work collar faded almost to brown. A narrow key hung inside it. The old access system had recognized both before he reached the workshop and offered him supervisory priority over two engineers already on duty. He had declined the priority in writing. Now he placed the key in a ceramic bowl and pushed it to the center of the table.

“Why bring it?” Bren asked.

“Because the aft relay bridge may still require it. Because I remember sealing that bridge. Because the panel beneath it carries a Carver mark I do not remember seeing.” Teren looked toward Esa rather than Bren. “I also remember my sister holding the lamp while I closed the panel. The current relation index gives me no open sister. One sealed relation returns a conflict notice when I ask. Until that has been tested, no inherited office memory of mine stands without another reader.”

Esa had expected anger from engineering after the broadcast. Teren offered something more difficult: useful doubt directed at his own hands.

She rotated the route display away from the Carver overlay. The hidden marks vanished. Raw hull fixes remained, along with three unedited stellar references gathered beyond the dome projection. “Bren derived Homeward from the tessellation. I derived the same endpoint from the cipher and the old telemetry. Engineering can derive the burn from live trackers without accepting either office.”

A supervisor on the engineering channel answered, “We will.”

For the next two hours, the workshop held no speeches. Teren called present sensor positions. Two technicians in separated control rooms read the same instruments through independent lines. Esa reduced the three reference stars to a bearing while Bren kept his hands visible on the table. Their figures differed at first by enough to put the ship outside the planned approach corridor after years of accumulated pulses.

The discrepancy came from a west tracker that still corrected its own output against the fabricated catalog. Pell found the preference flag from Ward Nine’s dome station. The child’s voice arrived sharp over the channel.

“It labels the live point as projector drift.”

“Can you remove the correction?” Esa asked.

“I can show you both readings. Removal needs a witness.”

Teren assigned the current dome technician, not himself. Pell exposed the raw count beside the corrected one. When Esa recomputed the bearing, Engineering Four’s independent vector met hers inside the mechanical limit.

Acceptance appeared as four separate marks. None belonged to Bren. Teren entered his name last and added:
*vector accepted; access competence unverified; relation conflict open.

The execution lock cleared one field.

Thrust required a second argument. The Long Veil had maintained its old course with small balancing burns, but a Homeward correction demanded a reburn across the aft batteries. Three banks had not fired together within any living engineer’s verified experience. Archive procedures described their launch housings. Current inspection showed replacement valves, narrowed coolant channels, and one bank whose gimbal stopped two degrees short of the old diagram.

Engineering proposed a ninety-second pulse at reduced feed. Navigation wanted another fourteen seconds to keep the first correction above the measurement noise. Hull control objected to both until loose stores, lift cars, and ward furniture could be secured. Medical requested a lower onset so citizens waking inside conflicting memories would not mistake thrust for an injury.

Esa listened while the vote acquired mass, temperature, and duration. Every addition reduced the clean simplicity of the bright route above the dome. That felt proper. A destination could fit inside one word. Moving thousands of bodies toward it could not.

The heat ledger stopped them.

A reburn converted fuel into motion and waste heat. The aft radiators could reject most of it, but their circulation pumps drew from the same reserve that held the hydroponic root loops within range during ship-night. The proposed pulse would leave enough total energy. It would not leave enough in the right loops at the right time.

“Hydroponics Six has the largest flexible draw,” the engineering supervisor said. “Suspend upper lamps, lower root heat, recover after the pulse.”

A woman’s voice cut across the channel. “Engineering does not own the word flexible.”

Ena Vaal arrived twelve minutes later with green under her nails and a leaf tray tucked against one hip. Mila came with her carrying the current stores ledger. They set both on the thrust table, covering half the route projection.

Ena pointed to the tray. Young broadleaf plants stood in dark nutrient mesh. Their outer leaves had begun to curl inward. “Six lost one circulation pump before the broadcast. We are holding the fruiting racks by borrowing heat from germination. Lower the roots for the full engineering schedule and these do not merely grow slowly. They shed. The next cycle then draws fresh ration from storage while you are asking stores to secure every loose locker.”

Mila opened her ledger. Dry grain could hold the calorie floor. Fresh stores could not replace Hydroponics Six without stripping two other tiers below their ward agreements.

Teren asked, "What can Six lend?"

"That depends on who absorbs the loss."

"The ship authorized the course."

"The ship did not authorize engineering to hide leaf loss inside a heat percentage." Ena lifted one plant from the tray. White root threads clung to the mesh. "Put bodies beside your numbers."

Esa asked for the smallest useful pulse. Engineering reduced feed until navigation could no longer distinguish a correction from the ship's ordinary flex. They raised it one increment. At that setting, thrust could run for seventy-eight seconds. Radiator circulation would require nine hours of borrowed reserve, including cooldown.

Ena took the figure away from the table and worked it by hand. She separated root zones according to crop stage rather than tier rank. Mature broadleaf could endure the fall if workers harvested early. Fruiting vines could not. Germination could be insulated with warm reclamation water. The upper lamps could go dark for three intervals without killing a crop, though later fruiting would decline.

Her decision arrived in parts. Hydroponics Six would harvest the mature leaf racks before the burn. It would suspend half the fruiting lamps, wrap germination trays, and lend the root-loop reserve for no more than nine hours. Expected fresh-leaf yield would fall between fourteen and eighteen percent over the next twenty days. Mila would hold the dry-grain calorie floor. Across that period, each citizen would lose two fresh-leaf portions, not calories. Medical wards kept their green allotment.

"No automatic extension," Ena said. "If coolant has not returned the reserve at nine hours, the aft batteries stop. If root temperature crosses the mark on this stem tag, they stop sooner."

She clipped a small green tag beside the thrust cutoff. The handwritten number carried no engineering seal.

The supervisor objected. "A root sensor cannot command an engine bank."

"It can tell me when Hydroponics Six withdraws its allocation."

"A mid-cooldown withdrawal could damage a pump."

"Then design a stop that does not require stealing the tenth hour."

Voices rose along the channel. One precinct accused Ena of placing leaves above the ballot. Another demanded equal ration loss for medical wards. Mila answered that equal loss would not create equal risk. Teren refused to let the dispute collapse into a yes or no against Homeward.

"The course has authority," he said. "This pulse still needs accepted conditions. Engineering can build within nine hours or bring back a later schedule."

Esa saw Ena study him. His red collar marked an office the network still treated as senior. His key sat unused in the bowl. He had spoken from neither.

Bren found the usable change. The three thrust banks did not need to share a single coolant recovery curve. Bank Two could close early, sending its return flow through the radiator before the other banks entered full cooldown. The pulse would lose four seconds. Navigation accepted the smaller impulse because the independent trackers could measure it after Pell's correction flag had been exposed.

Engineering redrew the sequence. A hard cutoff at eight hours and forty minutes left twenty minutes for

controlled shutdown before Ena's limit. The root tag fed only an alarm. Two human witnesses, one in Six and one at aft control, held stop authority. Mila entered the ration consequence beside the heat transfer rather than in a separate stores note.

Ena signed. "You may borrow it once."

The execution lock cleared its heat field.

Preparation moved through the ship. Ward channels showed carts tied to rails, hanging tools boxed, sleeping cloths rolled against the expected pull. Hydroponics workers cut mature leaves under lamps brighter than noon. The early harvest filled baskets faster than stores could count them. Mila ordered the leaves served first, before they lost water, and marked the coming absence on meal boards.

Esa went down to Six before the lamps dimmed. Warm condensation gathered on the tier rails and made every grip uncertain. Workers moved along the broadleaf racks with short knives, taking plants two days before their preferred harvest. Cut stems released a green scent into air already thick with pump heat. Each basket carried the pulse number on a paper tag so stores could not later call the loss ordinary spoilage.

Near the fruiting racks, two growers refused Ena's first selection. One wanted to save the oldest vines because their yield had already been counted. The other argued for the youngest because another cycle could repay the sacrifice. Ena listened, checked root color, then ordered alternate racks dark rather than choosing by age. The pattern kept every cultivar alive while reducing each harvest.

"You spread damage where no ward can refuse it," one grower said.

"I spread it inside Six because Six offered the reserve," Ena replied. "The ship will see the missing portions in Mila's ledger. If the next pulse asks again, the wards may refuse with this loss in front of them."

A worker handed Esa a ration notice to witness. It showed the current green issue, the reduced issue by ward, medical exceptions, and the date when full portions must resume unless another public allocation replaced it. No blank allowed an office to extend the reduction quietly.

Esa signed only that the notice matched the heat agreement. "Cartography does not certify your crop forecast."

"Good," Ena said. "You are learning not to own every number that points toward your star."

The upper lamps went dark in a staggered line. Pump indicators remained visible below them, green points among leaves. Esa watched workers wrap germination trays in reclaimed cloth. The burn had not begun, but Homeward had already entered the ship as labor, argument, and food that would not grow.

At Ward Nine, Pell secured the long table's bowls inside a mesh drawer. The child's wooden boat remained in the center until Koris noticed it on the channel.

"Take your vessel out of the vessel's way," Koris said.

"It has a flat bottom."

"So does a tray until thrust reaches it."

Pell wedged the boat beneath a folded cloth instead of putting it away. Esa almost corrected the choice, then left Ward Nine to its own witness.

At the aft relay bridge, Teren faced the panel he remembered sealing. The metal carried his tool scratches

across two replacement screws. Beneath the lower lip, a Carver mark had been cut where no routine inspection would find it. Bren remained outside the access boundary. A current Engineering Four technician compared every wire against live continuity while Teren read the switching order aloud.

The old system requested the red-collar key.

Teren retrieved it from the witness bowl with both hands visible. When the key entered, a private relation prompt flashed across the panel: *paired access recognized*. A second credential waited behind a consent seal. For one breath, Teren turned toward empty space at his left as though someone held a lamp there.

“Stop,” he said.

The technician cut local power. Teren kept his hand on the dead key.

“What did you receive?” Esa asked from the workshop channel.

“My sister telling me the third catch sticks.” His mouth tightened. “No voice. A memory. She stood here.”

“Does the hardware support it?”

The technician inspected the third catch. It moved cleanly. Archive service records showed replacement eleven years earlier. Teren’s remembered catch belonged to an older state of the panel.

He withdrew the key. “Continue under physical check. Do not use paired access.”

Engineering bypassed the relation prompt with two current credentials. The bridge opened. Behind it, old copper bars carried fresh ceramic sleeves. The Carver mark repeated on the manual bus, but Bren did not interpret it until the technicians had traced the circuit themselves. It indicated the same isolation route they found from continuity.

Teren closed each switch under another person’s reading. His competence survived the compromised scene. Nobody used that fact to validate the sister.

By ship-evening, all fields on the execution lock had cleared. The ballot certificate named Homeward. The vector carried four independent marks. Ena’s heat allocation sat beside the pulse schedule, with its ration range and withdrawal conditions. Engineering listed local stop witnesses. Hull control reported secured mass by precinct. Medical opened live channels for anyone who needed orientation during thrust.

Esa and Bren took places behind the western-truss rail. Neither possessed an engine control. Their route waited above the table as a thin line joined to live tracker points.

“Correction sequence to ready,” said the engineering supervisor.

Across the channel wall, local witnesses answered. Aft banks reported valve position. Radiators accepted preload. Hydroponics Six reported root temperature and reserve transfer. Ward Nine showed Koris, Pell, two elders, and the long table cleared except for the cloth-covered boat.

Teren’s image came from the relay bridge. He had removed the red collar. The key rested in its bowl while two current credentials held the circuit. “Bridge ready under witnessed access.”

Ena stood among darkened fruiting lamps. “Six lends heat under the entered limit.”

Mila raised a leaf basket toward her camera. “Stores records fourteen to eighteen percent yield loss, two fresh

portions deferred, calorie floor held.”

No tier yielded by silence. Each condition entered the public burn record before engineering asked the final question.

“Authorized bearing?”

Esa checked the raw tracker points once more. “Homeward bearing holds.”

“Execute first correction.”

The engines did not wake with a roar. A low vibration entered the rail and climbed through Esa’s palms. It carried discontinuities from eight centuries of replaced brackets. One bank joined a fraction late, then settled. The deck pressed gently against the soles of her boots.

On the ward channels, hanging cords leaned east. Water rose along one side of transparent cups. A cart in Precinct Two pulled against its straps. In Ward Nine, Pell reached too late for the cloth. The wooden boat slid free, struck a bowl clamp, and lost one resin rail. Pell caught the hull before it reached the floor.

Thrust increased through the medical onset limit. Esa’s inner ear denied the motion, then surrendered to it. The bright route shifted by less than a visible line. Raw trackers began collecting the difference.

At forty seconds, a coolant valve in Bank Two hesitated. Engineering held feed rather than increasing it. Ena’s root temperature fell by one tenth of a degree. Teren called the bridge current steady. Bren counted nothing aloud.

The valve command showed full travel, but the pressure rise stopped halfway to its expected mark. A fault card on the supervisor’s display prescribed a second opening pulse. Her hand moved toward the control, then Teren told her to wait. The replacement valve had a narrower seat than the launch fitting. If only the position contact had failed, another pulse could hammer an already open stem against that seat.

“Do you remember this fault?” the supervisor asked.

“I remember three versions of it,” Teren said. “That makes memory useless here.” He requested the pump-current trace and the acoustic pickup bolted to Bank Two’s return pipe. One showed a loaded motor. The other carried an uneven tapping that might have come from a sleeve at half travel or from an old bracket transmitting thrust vibration. Neither proved the valve position.

Ena broke in from Six. Root temperature had fallen another two tenths, faster than the entered curve. The reserve transfer remained legal, but a second actuator surge would deepen the draw while her pumps compensated. “You may test again after feed closes,” she said. “Not by spending crop margin before you know you need the impulse.”

Navigation still lacked a clean measure of what the held feed had achieved. Esa called Ward Nine. Pell had kept both west-tracker columns open: the raw stellar count and the catalog-corrected point. The corrected column treated the first movement as hull flex and subtracted it. The raw count had shifted through four consecutive samples.

“How far?” Esa asked.

“Past the small mark, not the large one.” Pell put a ruler against the dome station because the numerical labels were jumping under vibration. “If it comes back on the next sample, you did not turn. If it keeps going, you

did.”

The fifth raw point held beyond the measurement floor. Engineering canceled the second valve command and abandoned the planned feed increase. No extra seconds would be added at the end to recover the smaller impulse. Teren entered the choice under his current witness role, with Ena’s declining root curve and Pell’s raw sequence attached as separate evidence. His note refused to infer a sound valve from a successful turn.

The pressure trace rose by one more division without intervention. Nobody promoted that improvement into proof. The supervisor kept Bank Two at held feed until the common cutoff, then assigned a physical valve inspection before any later pulse. On the ward display, Pell’s raw points continued along the new bearing while the corrected column kept pulling them toward the old line.

Seventy-four seconds after ignition, feed closed. Residual thrust passed through the deck in three diminishing pulses. Bank Two entered its early return path. Radiator temperature rose. The root reserve continued outward under Ena’s clock.

Navigation posted the measured correction. The number cleared instrument noise and landed inside the accepted corridor. For the first time in three centuries, The Long Veil’s live vector no longer pointed along the course the Weave had preserved.

A cheer began in one ward and failed to spread. People were checking straps, spilled water, frightened children, and the damage done by a pulse small enough to fit inside a decimal. Engineering kept its channels clear for cooldown.

Then the echo-network tried to reconcile the result.

The course display blinked between two headings. One carried the ballot provenance and live tracker marks. The other carried eight centuries of morning continuity. Archive lights along the workshop wall brightened as citizen records requested a current destination. The stabilizing core answered both.

Esa heard the Archivist through the nearest speaker. “Authorized course: Homeward,” it said in its formal register.

After a pause shorter than breath, the same voice continued, “Authorized course: nearer daughter. Deviation requires correction.”

Warning bands spread across the echo map. Relation traffic rose with them, not as individual queries but as thousands of identity reconstructions reaching for a ship whose destination had split inside the system. Teren’s panel displayed the sealed paired credential again although its circuit had no power. On a medical channel, someone recited a ward number twice and received two different offices.

Bren looked from the cooling schedule to the network load. “The burn made the contradiction physical.”

A second tremor entered the rail. It did not come from the engines.

Across the workshop, three archive cabinets unlocked at once. The Archivist began the opening line of the catechism, stopped after Esa’s name, and resumed in a voice she had never heard.

22 — The Archivist Fails

The unfamiliar voice finished Esa's name with a childhood diminutive nobody in Ward Nine used.

She pulled the speaker lead from the western-truss wall. Speech continued from two archive cabinets and Teren's dead relay panel. One voice gave Esa's correct office. Another assigned her to a teaching ward dissolved before her birth. Beneath both, the Archivist repeated a formal request for citizens to remain available for completion.

Bren crossed to the echo map. The first correction had altered one fact every morning reconstruction assumed stable: the ship's destination. Authority routines were trying to absorb Homeward without surrendering the earlier course. Each resolution touched office histories, kin relations, work permissions, and the old account of why the crew traveled at all.

"Stabilizing load has doubled since cutoff," he said. "It should be falling."

A cabinet vent snapped open. Air carrying hot dust struck Esa's face. Indicator strips climbed through amber. Cooling fans had reached full speed, yet the core temperature continued upward as archived relations passed through the completion engine for reconciliation.

Engineering called from aft control. Bank Two's return flow remained inside Ena's limit, but its pump authorization had begun alternating between current witnesses and an office holder dead for sixty years. Local crews had switched to manual. Hydroponics Six still lent heat. Every minute spent arguing with the network consumed part of the nine-hour allocation.

Esa keyed the public channel. "Ignore reconstructed office priority. Keep the entered burn witnesses. Local control until this fault clears."

The Archivist answered through a cabinet grille. "Citizen Miro lacks engineering office."

"She does," Teren said from the relay bridge. "Her statement repeats our current procedure. Engineering Four confirms local control."

His image shivered on the channel. The red collar lay on the floor beside the witness bowl. A moment later he bent to pick it up, then recoiled before touching the cloth.

"My sister tied that knot," he said.

The current technician asked, "Are you requesting a relation query?"

"No." Teren stared at the collar. "I remember her tying it before we opened this bridge. I also remember receiving it alone from an elder in Engineering Four. Both scenes include the same frayed thread."

A private prompt opened despite his refusal. *Paired relation available under consent seal. Completion can restore continuity.

"Close that," he said.

The prompt multiplied across the panel. One copy called the sealed person his sister. Another called her his apprentice. A third used no relation but displayed a service image of a woman holding a lamp beside the older relay housing. Provenance marks flickered too quickly to read.

Teren reached for the manual breaker. His left foot caught under the edge of the open panel. Whether memory or current hardware guided the step, Esa could not tell. His knee struck the grating. The technician caught his shoulder before his head followed.

The red-collar key jumped from the bowl and slid beneath the bridge rack.

Medical entered the channel while Teren remained seated on the floor. He could state his name, present location, and the live switch order. When asked who had stood beside him in the remembered scene, he gave two incompatible answers and stopped.

“Do not complete it for me,” he said. “Test what I can do now.”

A medic stabilized his knee and began a diagnostic under a separate line. The current technician took the bridge. Teren stayed where the camera could see him and read meter values without touching access hardware.

The choice preserved one circuit. Elsewhere, completion faults moved faster than people could withdraw their hands.

Doors in the teaching ward rejected current staff and opened for retired offices. A pressure crew found yesterday’s maintenance order replaced by a resolved version from eleven years earlier. Meal dispensers in two precincts requested kin priority before releasing breakfast portions. Mila put Ward Nine’s stores on direct issue and sent runners with paper tallies.

Ward channels flooded the truss table. Citizens reported waking memories overlaid with scenes from sealed relations. Some recognized the intrusion because faces changed under attention. Others defended the new scene as recovered truth. The Archivist responded to each contradiction by drawing more records through the stabilizing core.

Esa saw the pattern in the load map. “It is using the archive to repair authority.”

“It has always done that,” Bren replied. “Until today, the ship’s governing facts stayed fixed enough that repair converged.”

“What happens if it does not?”

“The completion engine keeps selecting until heat destroys the relation index. After that, raw records remain in fragments, but their sources detach. Consent seals may detach with them.”

An archive without provenance would preserve statements stripped of the means to judge them. An archive without seals would turn private pain into public material. Saving only data would repeat the Weave’s oldest fault in another form.

A new alarm opened below the map. Stabilizer coolant pressure had fallen. The core shared a pump branch with the projection layer because both once cycled at opposite times. During the broadcast, Bren had held the projection continuously bright. During the burn, navigation had kept it active for measurement. Now both systems demanded cooling.

“I can darken the route,” he said.

“Do it.”

The tessellation disappeared from domes across the ship. Ward channels reacted at once. Some citizens thought

the turn had been reversed. Others believed the Carvers had reclaimed the route. Esa broadcast the reason in one sentence and asked local witnesses to repeat it. The route remained in archive and navigation instruments. Its public light went dark to cool the core that held its evidence.

Temperature dropped by less than one degree.

Bren opened the First Weaver's service diagram beside the current network map. The original archive consisted of distributed record stores with source relations attached. Later authority layers had added preference, completion, and morning reconstruction. Over centuries, repairs had folded their buses together. No clean switch separated remembering from deciding.

"Then we make the separation under load," Esa said.

"We can isolate the archive readers from the completion engine. That does not save the relation index. Provenance and consent now pass through the stabilizer before every retrieval."

"Can they be mirrored?"

"Not while completion rewrites their priority marks."

A live ward channel forced itself above the alarms. A child's voice called a name the public line masked under a consent seal.

"Where is my brother? He was here before sleep. He put his sleeve over the lesson lamp. Tell him I woke."

The Archivist answered before any ward worker could. "Brother relation presents four candidates. Completion in progress."

"No," the child said. "I know his yellow shirt."

Candidate records began to populate the channel, two open and two sealed. The open records carried age conflicts. One belonged to another ward. The Archivist assigned confidence values anyway.

Mila came on from the eastern rooms, breathless from carrying stores. "Close the candidates. She did not consent to a search."

"Kin completion supports identity stability," the formal voice replied.

"She asked for her brother. She did not ask you to choose one."

The child called the sealed name again. A bowl clinked near the pickup.

Mila lowered her voice. "I am Mila. I brought broth. I do not know where your brother is. I can stay while you eat, or I can find someone else to stay."

"Make the light stop hurting."

Mila covered the lesson panel. The channel image dimmed. "Done."

The Archivist continued listing candidates into darkness.

Esa muted only the completion feed, leaving the child's channel open. "Bren, if we stop preference first, can the archive still copy its source relations?"

“For minutes. Possibly less. Retrieval will slow. Doors and work permissions tied to completion will fail closed.”

“Local life support?”

“Independent. Distribution and office systems will not be.”

“Then warn the wards. Default personal records to sealed. We save record, source, and consent together or we do not call it archive.”

That instruction did not yet choose the archive’s future authority. It chose what could survive long enough for anyone to decide.

Bren routed civic records toward three cold mirror cabinets. For each record block, the system copied the original entry, every later alteration, and the relation that named its source. Personal blocks moved only with their consent state attached. Where a seal could not be verified, access defaulted closed. The copies accepted no completion ranking.

Progress began at seven percent and slowed.

The Archivist resisted by marking the mirror as an unauthorized drift. Cabinet latches closed. Bren used a Carver tool to hold one open while Esa signed a public emergency witness on the copy route. The formal voice rejected her office. Engineering Four added its mark. Hydroponics Six added Ena’s, limited to records affecting its heat allocation. Ward Nine added Koris through a paper tally read aloud.

No single credential could authorize the whole transfer. The route remained open because independent witnesses covered its immediate harms.

At the relay bridge, Teren’s diagnostic continued between voltage calls. The medic pressed his foot against resistance, tested his pupils, then asked him to sort current events from inherited scenes. He identified the burn, the key, the third catch, and the fall. The sister relation remained unstable.

A printer beside him produced one page before network access failed. Teren folded it without opening the sealed field. A relation number remained visible above the fold. He tucked the packet beneath his wrist strap.

“Diagnostic finding?” Esa asked.

“Motor and orientation intact. Knee strained. Relation collision active. No solitary machine access until review.” He accepted a temporary brace from the medic. “I can still read the bridge.”

“Under independent check,” the current technician said.

“Under independent check,” Teren agreed.

The mirror reached nineteen percent. Core heat crossed the first damage line.

Cabinet insulation released a bitter odor. A source-index module warped enough to break contact, taking nine ship-years of civic minutes offline. Bren pulled the module with insulated grips and reseated it in a spare reader. His right glove blackened at the fingertips.

“Those minutes have other copies,” he said.

“Can you verify them?”

“Not until the index catches up.”

“Then mark them unavailable, not recovered.”

He entered the distinction. The mirror count fell back by two percent because unverified blocks no longer counted as preserved.

The ship lost its current work roster next. Corridor panels went blank. Mila’s runners kept meal issue moving from paper. Pressure crews stayed at the stations named before severance began. Hydroponics workers watched root temperatures directly. Ena called the remaining heat window every fifteen minutes, making the core failure audible in fractions of borrowed time.

“Seven hours, twenty minutes remain,” she reported. “Six still lends.”

Bren expanded the mirror path. “We need six.”

“You have no right to the seventh without asking again.”

“We heard you,” Esa said.

A speaker above them recited the morning catechism in reverse order. Office came before ward. Ward dissolved into a list of obsolete precincts. Name fragmented into syllables from private records. The sounds carried no meaning together, yet Esa’s body reacted before she could turn the speaker off.

Heat lay across her face.

A wet cloth pressed against both eyes. Its rough seam had been folded away from her left eyelid. Someone wrung the cloth above a metal basin, using water with the careful economy of a ship that counted every cup. Gray hair bent into the low amber light. The caregiver’s left cuticle had torn and dried dark at one edge.

A wooden button struck the basin rim when the person leaned over her.

The small sound separated itself from alarms, fans, and cabinet latches. Esa felt the fever-child’s narrow chest labor beneath cloth. She knew the view from inside that body. She did not know the gray-haired caregiver’s name.

The Archivist supplied a relation label through the nearest grille.

Esa gripped the table until the workshop returned around her. “Withdraw that completion.”

“Continuity requires relation.”

“The fragment does not contain one.”

“Donor lines support probable caregiving kin.”

“Keep the donor lines under their seals. Preserve the fragment without adding a person to it.”

Bren had stopped working. “What came through?”

She gave him only what she could support: wet cloth, rough seam folded away from the eyelid, metal basin, gray hair, torn left cuticle, wooden button against the rim. The memory ended there. No face settled into a known relation. No later morning followed.

“Record it as firsthand sensory relation,” she said. “No completion. Ship-year uncertain.”

Bren entered the fragment into the mirror queue. Its donor suggestions remained separate and sealed. Provenance marked the time of recovery, the unstable core state, and Esa’s own report. The archive did not gain a childhood. It gained the boundary of one fragment.

That boundary clarified the technical choice. Preference could not be allowed to rewrite source relations while they copied. Completion would have to stop before the mirror finished, even though stopping it would remove the routines that still opened doors, supplied offices, and joined waking memory into a continuous self.

Esa looked at the map. “Can we cut authority now and keep the archive bus alive?”

Bren traced the current paths. “Three cuts. Preference first, then completion, then the speaking persona. If preference opens again during completion shutdown, it may recast the mirror indexes. Teren’s bridge can hold the archive return apart from the authority feed.”

Teren heard his name. “The bridge diagram supports that.”

“Does current hardware?” Esa asked.

The technician beside him moved the camera across the open panel. Ceramic sleeves, copper bars, manual bus. Teren identified each switch from live position. Bren compared the Carver isolation mark without controlling the reading. Both routes agreed.

“The red-collar key may reopen paired access,” Teren said. “Leave it under the rack. Use the two current credentials.”

His compromised memory had led them to a useful refusal, not an authority claim.

Esa issued a ship-wide warning. Archive query would become unavailable during separation. Doors linked to identity preference would fail closed until local crews opened them. Meal and work systems would lose inherited priority. Morning reconstruction might not resume. Life support remained under local control. Personal records would default sealed. Citizens should remain with present witnesses where possible.

Opposition reached her before the warning ended. A medical ward demanded completion remain until patients received current drug histories. Engineering asked for one more office export. Two wards accused Esa and Bren of destroying the system immediately after using it to win the course vote.

Those demands described real danger. Esa delayed the first cut long enough for medical to export current, consented treatment notices with dates and clinicians attached. Engineering printed the live bridge roster. No one received a general identity completion. The stabilizer crossed its second damage line while they waited.

A source cabinet ruptured a coolant seam.

Clear fluid sprayed across the floor and turned white against hot metal. Bren hit the local isolation. Fans drove vapor toward the service shaft. Esa pulled a breathing mask over her mouth and helped drag paper tallies away from the spreading liquid. The archive mirror paused at sixty-one percent.

Core temperature no longer fit the display scale. The formal voice slowed between words.

“We advise restoration of accepted course,” it said. “Contradiction load will then resolve.”

The ship could cool the Weave by undoing the vote and burn. Authority offered survival at the price of making

physical truth conform again.

“Preference cut ready,” Bren said.

Esa pressed the witnessed control.

Across the echo map, ranking lines vanished. Archive entries remained, but no civic priority lifted one above another. Door failures multiplied. Candidate completions on the fever-child’s channel disappeared. Mila’s covered lamp stayed dark. The child asked once more for the sealed brother, then accepted broth from the person who had named herself.

Mirror transfer resumed. Without preference recomputation, its speed increased.

Completion reacted by attempting local repairs. Esa’s childhood fragment acquired three proposed continuations, all blocked behind consent seals. Teren’s paired credential tried to declare itself essential engineering context. The old course entered navigation queries as a stabilizing fact despite the preference cut.

“Completion has cached rankings,” Bren said. “Second cut now.”

Teren and the current technician closed the authority feed at the bridge. The manual bus kicked hard enough to throw Teren’s braced leg sideways. He caught the rack, swore once, then read the archive return voltage.

“Return holds. Separate by four counts. Hold it there.”

Bren drove the second control down.

The effect reached bodies before the displays. Esa felt an internal pressure release, not relief but absence. The expectation that another version of the fever room would arrive simply ceased. Around the ship, unfinished catechisms stopped mid-sequence. Office panels lost their suggested answers. Relation prompts closed. Thousands of citizens remained with whatever memory had reached them before the cut.

The mirror stood at eighty-eight percent. Source links held. Consent defaults held. One personal index attempted to open during the transition and locked itself instead.

Only the speaking persona remained active. Its formal register had once let many systems sound like one keeper. Now it moved through the surviving records, quoting fragments as if quotation could restore unity.

“We preserve the ward,” said the First Weaver’s cadence.

“Keep the peace,” said a mutiny minute.

“Ask why before correction,” said a faded maintenance hand.

The phrases came from different centuries and carried their own sources on the text display. In speech they crowded together. The Archivist tried to gather them under its plural pronoun.

“We are the continuity of the Long Veil. We are the accepted relation. We are the voice by which the citizen...”

The sentence broke into names. Some belonged to open civic records. Others reached the speaker only as sealed tones, audible without content. Bren severed the voice from personal indexes before a seal could fail.

“Persona cut ready,” he said.

Esa did not ask it for a final answer. “Cut.”

He opened the last coupling.

The Archivist’s formal voice thinned across three words. “We preserve... we...”

A fan spun down. Text remained on the archive console: records, source chains, alteration histories, objections, consent states. The mirror reached completion with several blocks marked damaged or unavailable. Nothing spoke for them.

Heat began to fall.

Bren verified the archive through a text-only query. The course ballot returned with precinct certificates and abstentions. Ena’s allocation returned beside the ration range and stop conditions. Teren’s relation packet returned only its seal status and provenance number. Esa’s fever fragment returned without a caregiver label. Civic records could be read. Personal completion could not.

Aft control regained its pump permissions through local credentials. Ena reported six hours of reserve unused and reclaimed the transfer for root heat. Mila’s runners continued meals because the central roster had not returned. Pressure crews opened failed doors by present witness.

Nothing about severance told those people what rules should replace authority. It left them machinery, records, immediate needs, and one another. The choice of practice still waited beyond the emergency.

At the relay bridge, the medic finished Teren’s test. His knee took weight badly. A brace locked it for walking. He recovered the red-collar key with a magnetic wand and sealed it in the witness bowl rather than around his neck. The folded relation number stayed at his wrist.

“Do you remember your sister?” Esa asked.

“I remember a woman beside the panel,” he said. “I remember the relation arriving. I do not know if those facts belong together.”

The answer cost him enough that she did not ask for more.

Ship-night advanced unevenly. Wards with rotating schedules reached morning before Ward Nine did. On an eastern channel, a waking citizen sat before a catechism panel gone dark. Neighbors stood nearby with a clock, water, and a paper work roster. Nobody knew whether the old opening should be spoken.

A child tried first.

“Name,” the child said into the panel.

No formal voice supplied one.

“Ward.”

The archive light held steady. It offered no relation without a query.

“Office.”

Silence followed the third word. Not a failure tone, not an unfinished sentence, and not the layered breathing of many archived voices. The first catechism without the Archivist ended with people in the room looking at one

another.

Esa listened until one neighbor gave the current time and another set down a cup. Then she closed the channel. Ward Nine's bell would sound in several hours. Her fever fragment remained in the archive with its ending intact, and morning approached without anyone authorized to continue it.

23 — What We Keep

Esa woke before the ward bell and waited for a voice that no longer had the power to come.

The archive light beside her mirror held steady. No formal register supplied her name. No remembered room settled around her body to join sleep to morning. Ventilation lifted the cuff of her sleeve, and somewhere beyond the wall a service cart struck a floor seam twice.

She said, "Name: Esa Miro. Ward: Nine. Office: cartographer."

Three facts entered the room and remained separate. Yesterday, the Weave would have drawn relations between them until she felt continuous. This morning her own pulse provided the only rhythm. She counted it against the archive light, then stopped when counting began to resemble a test she meant to pass.

The basin waited in its niche. Esa took it down, filled it from the drinking tap, and carried it to the floor beside her sleeping cloth. During the night she had left a wooden button there, cut from an old work coat. Its edge bore four shallow holes instead of the two she remembered from the fever room. She had chosen the difference deliberately.

Cold water darkened a square of cloth. She wrung it with both hands, folded one corner inward, and laid it across her eyes. The rough seam stayed clear of her left eyelid. A button touched metal when she leaned forward.

Tap.

The low amber room came back with enough force to tighten her throat. Sweating floor. Cloth doubled beneath a fevered child. Gray at the center of a woman's hair. A torn cuticle on the left thumb. The true button sounded lighter than the one in Esa's hand, but memory gave her no number of holes. She waited for the woman's name.

Nothing followed.

Esa lifted the cloth. Water ran past her ears into her hair. She tried the sequence again with the seam left outward. The remembered hand rejected that fold at once. She moved the basin closer, then farther away. Both positions felt false. When she tapped the rim with the new button, the fever room held its own smaller sound apart from the present one.

The test gave her four correspondences and one useful refusal. It did not give her a childhood.

She opened archive query and requested her fever fragment without completion. Text replaced the old speaking panel: *firsthand sensory relation; ship-year uncertain; kin identity unresolved; no continuation authorized.* Below it waited three donor-line suggestions under consent seals. Esa closed the suggestions without reading them.

A knock came at her door in Pell's pattern, two quick taps followed by one patient one.

"Come in."

Pell entered carrying a breakfast pot, two cups, and the child's wooden boat from the long table. The boat had lost one resin rail during the first correction pulse. A new strip lay clamped along its side, pale against the older hull.

"You're wet," Pell said.

"I tested the basin memory."

"Did it change?"

"No." Esa poured the basin water into the reclamation jar rather than back into the drinking line. "The cloth fold held. The rest still ends where it ended."

Pell inspected the button on the floor. "That isn't her button."

"I know."

"Good."

The answer made Esa look up. Pell set down the breakfast pot and touched the repair clamp on the boat.

"If you used a button that looked right," the child said, "you might remember this one instead. Adults do that with my drawings. They show me the corrected page too many times, then ask what I drew first."

Esa placed the wooden button in a labeled envelope. *Test object. Not from fragment.* The distinction felt small enough to lose and therefore worth writing.

Pell poured grain broth into the cups. "Koris says come to the long table after first meal. Half the ward already came there because their rooms were too quiet."

"What about the other half?"

"They're visiting one another's rooms."

No morning reconstruction had failed because no morning reconstruction had run. Yet people still missed shifts, forgot current relations, or woke inside offices removed from the active roster. The authority had hidden those fractures by choosing which account arrived first. Silence exposed them without causing them.

Esa drank while Pell described the ward's first hours. A pressure worker had reported to a sealed pump station and found the current crew waiting with a diagram. They tested his skill before giving him a tool. Two elders disagreed over whether they had shared a cell twenty years earlier; Koris marked the relation uncertain and asked what care each wanted now. A fever-child in the eastern rooms had woken calling for a brother whose name remained sealed. Three neighbors sat with her until she could eat.

"What did they tell her?" Esa asked.

"That they didn't know," Pell replied. "Then Mila told her who cooked the broth."

Material care did not require a completed history. The thought brought relief and shame in equal measure. Esa had needed a child to state it through breakfast.

Ward Nine's corridor carried an improvised order. Paper notices covered the old catechism panels. One listed current work calls with named witnesses. Another offered quiet rooms, medical support, archive help, and meals without requiring anyone to settle disputed kin. Beside the lift, a hand-drawn arrow pointed to the long table: *If you do not know where to begin, begin here.

A man sat beneath the notice with two work coats folded across his knees. One carried a pressure patch. The

other bore teaching blue. Mila knelt before him, reading today's needs from a slate.

"Pressure has a full crew for first interval," she said. "Teaching needs someone to sharpen chalk and move benches. Neither task grants an office. Which work can you do safely?"

He pressed his thumb into the pressure patch. "I know valves."

"Do you know this morning's valves?"

"No."

"Then come sharpen chalk until a pressure witness can test you."

The man chose the teaching coat and left the other with Mila. Nobody told him which life belonged to him. He still acquired breakfast, company, and work that could not flood a deck.

Before going to the table, Esa followed the quiet-room notice into a former lesson cell. Six sleeping mats covered the floor. The chalkboard carried the current ship-year in digits large enough to read from any mat. Beneath it, someone had written the ship's present vector and the time remaining until the next heat draw. No phrase offered peace. The board supplied orientation that could be measured.

A medical worker met Esa at the threshold and pointed to a rack of colored cords. White meant company welcome. Blue requested an archive witness. Gray asked not to discuss memory. Red called for clinical help. Citizens could wear more than one.

"Who chose the colors?" Esa asked.

"The people on the mats. We changed two labels after they told us the first words sounded like commands."

At the far wall, the fever-child Pell had mentioned held a white cord and a gray one. She looked fourteen, though an early echo sometimes made age uncertain. A bowl of broth cooled near her knee. Two ward neighbors sat within reach without speaking.

The child noticed Esa's wet hair. "Did you find yours?"

"Only the same fragment."

"My brother has a yellow shirt."

Esa waited. Gray cord forbade a search conversation, not the child's own speech.

"He put the sleeve over my eyes when the lesson lamps hurt," the child continued. "The archive offers four boys. One died before I was born. One belongs to another ward. Two have seals."

"What do you want us to do?"

"Stop asking which boy until I finish eating."

Esa sat beside the broth and checked its temperature against her wrist. "Do you want it warmed?"

"Yes. Not by him. By anyone."

One neighbor carried the bowl to the service coil. The other shifted closer when the child's hands began to tremble. No recovered brother entered the room. Warm broth did.

A pressure alarm sounded two corridors away. Three people on the mats flinched. The chalkboard attendant wrote the alarm's location and added *routine gasket test; no evacuation*. A man wearing blue cord asked archive for the source. The terminal returned the maintenance order, the crew names, and a live pressure reading. It offered no assurance. He watched the number hold until his breathing eased.

Near the door, an elder repeated her name every few minutes but rejected all ward relations. "Those people say I ate with them," she told Esa. "I remember a different table."

"What did this room give you?"

"A blanket. Tea without sweet algae. Nobody touched me."

"May those facts stand until you choose more?"

The elder rubbed the blanket edge between finger and thumb. "Write that I chose the tea."

Esa entered the choice on a temporary ward slate that expired at sleep. It granted no kin claim and settled no old table. It told the next meal carrier what to bring.

As she left, the medical worker handed her a white cord. "For your wrist, if you want company after the basin test."

"I already had Pell."

"Then you used the practice before we named it."

Esa tied the cord loosely and carried its plain evidence to the meeting: care had begun ahead of policy. The long table's task involved making that care durable without converting it into another hidden authority.

At the long table, Koris had removed the elder's high-backed chair. Every seat stood level. Archive copies rested at one end: the First Weaver's note, the mutiny minutes, the Carver journal, the course ballot, and the severance record. At the other end waited ordinary ward objects. Meal bowls. Shift slates. Repair tools. A child's error drawing from Sera Noll's classroom. Esa's basin occupied the center on a square of cloth.

Ena Vaal had come from Hydroponics Six with green beneath her nails. Teren Sull sat across from her, pale after the diagnostic test. A medical brace supported one knee. His sister's relation number remained folded inside a paper packet at his wrist. Neither visitor had joined Ward Nine. Koris had invited them because any rule made at the table would reach water, engines, and sealed relations before the day ended.

Bren arrived last. He carried no Carver ledger. Instead he brought the archive access plate removed from the western truss, its two consent keys tied separately to red cord.

Koris looked around the table. "Authority has ended. Archive holds. Morning work did not. People need food, heat, safe tools, witness, privacy, and a way to ask what the record contains. We decide what we keep before habit decides for us."

A ward teacher named Lio raised a hand. "Can Nine decide this for every precinct?"

"No," Koris said. "We can decide our practice. Ena and Teren can tell us where that practice fails outside this room. If it holds through one turning interval, we offer it to the other wards with its failures attached."

Esa appreciated the limit. The ship had only just cut a system that turned emergency interpretation into durable power. A new social contract could begin at one table without pretending the table contained everyone.

Bren set the access plate beside the records. "First choice: whether the archive remains merely powered or becomes publicly usable."

"Usable by whom?" Ena asked.

"Any citizen may request civic records and open personal records attached to them. Consent seals remain. Provenance cannot be removed from a copy."

Teren touched the packet at his wrist. "And preference?"

"No ranking by civic usefulness. Search will not put your office history above your sister's relation because engineering needs you."

"Will it expose her name?"

"Not while her seal holds."

Teren gave one short nod. "Then say both. Open archive does not mean open person."

Koris wrote the sentence on paper: *Public acts remain public. Personal relations retain the consent of the person recorded. No office need may outrank a sealed relation.

The archive terminal accepted the wording as a ward practice, not a command. That distinction appeared in plain text at the bottom of the entry. For the first time, a civic record admitted its own limited reach.

Next came the morning gap.

Lio placed Sera's error drawing beside the basin. A child had drawn projector cells around three accepted constellations. Blue correction marks crossed the squares. "The young groups woke frightened today. Some miss the voice. Some heard enough released relations during the broadcast to distrust any memory that arrives easily. What replaces completion?"

"Nothing replaces it," Ena said. "Completion did too many things under one name."

Teren leaned forward. "It supplied current time, current work, kin claims, medical warnings, access, and an account of self. We need some of those before a person touches a thrust relay."

"We separate them," Esa said.

She turned her slate sideways and drew six columns. Current time required clocks and witnesses. Work required a present roster plus demonstrated skill. Medical warnings required consented records and living clinicians. Machine access required two people and a physical test. Kin claims could remain uncertain without starving anyone. A person's account of self belonged first to that person, then to those they chose to ask.

Pell climbed onto the bench to see the columns. "And the basin?"

Esa rested her fingertips on its rim. "The basin does not need to become a lesson."

She told the table what she had done after waking. Cloth, water, wrong button, archive query, remaining gap. She described the immediate temptation to read donor-line suggestions until one supplied a woman who fit. Nobody interrupted. When she finished, the basin still contained no proof beyond its present scratches.

"I can test what the fragment gives me," Esa said. "I can mark objects that came later. I can ask the archive

where a relation came from. The ward can tell me who fed me yesterday and who will sit with me tonight. None of that grants us the missing name.”

Lio asked, “What do we say to a child who wants the name?”

“We say we do not know. Then we ask what the child needs while not knowing.”

Ena turned her hands palm upward. “Water. Food. Someone to stay. A sealed search if the child wants one. No search if they do not.”

“Present care without invented relation,” Koris said, writing again.

The phrase survived three objections. One elder feared that refusing completion would leave children without stable kin. Pell answered that chosen wards already held children whose biological records had failed. A pressure worker worried that disputed memory could become an excuse to avoid dangerous shifts. Teren proposed present-duty rosters signed each interval, with no inherited office treated as compulsory. Ena insisted that abstaining from a memory search could not reduce rations. The table added each safeguard in specific language.

Near midday, practice overtook theory. Hydroponics sent Ena a query: two clerks claimed authority to release the same heat reserve, one under the old office chain and one under the correction allocation. Before severance, the Weave would have reconciled them by priority.

Ena laid both ledgers on the table. “The old clerk has longer service. The new clerk carries the witnessed allocation. If we ask archive for completion, it will return provenance only.”

“Which resource reaches bodies first?” Esa asked.

“Heat. Ward Three has two rooms below the morning floor.”

“Then current need and witnessed allocation govern this release. Preserve the old clerk’s objection.”

Ena shook her head. “You do not govern my tier.”

The correction landed cleanly. Esa moved her hand away from the ledgers.

“Whom does Hydroponics Six authorize?” she asked.

“Me for the witnessed heat draw, with Mila on stores and a Ward Three temperature reader. The old clerk can inspect the meter.”

Koris recorded the procedure. Ena signed it, then used the table terminal to release only enough reserve for the two cold rooms. The old ledger remained visible beside the new one. A technician in Ward Three reported rising temperature twelve minutes later.

Nothing about the result settled which clerk possessed the truer office memory. Two rooms grew warmer anyway.

Teren’s call followed. Engineering needed a relay bridge opened for the next correction pulse, but the sealed row recognized his red collar key. He could name every step and no longer believed the apprentice scene that had once justified his access.

“Do you want to do the work?” Koris asked.

“Yes,” Teren said. “Under another technician’s check. Keep the key in its witness bowl until we stand at the lock.”

“Does your sister’s relation affect the procedure?”

“It affects why I refuse solitary access. It does not alter the bridge diagram.”

Bren accompanied him as a second technical reader. A current Engineering Four supervisor joined over the table channel, openly hostile to the Carver line and therefore useful as an independent witness. Teren read each lock position from the live schematic before touching the key. The bridge opened. He handed the key back without claiming his old office.

By late interval, Ward Nine had produced no doctrine elegant enough to replace the morning voice. It had produced a page of practices with names beside each responsibility. Archive without preference. Consent seals without civic ranking. Current rosters. Demonstrated skill. Two-person access for dangerous systems. Meals and warmth before solved identity. The right to leave a memory uncertain. A ward witness available at waking and at sleep.

Koris read the page aloud once, then turned it over. On the blank side she drew boxes for complaints, revisions, and the date when every provision would expire unless the ward renewed it. Pell asked why a good rule needed an ending. Koris answered by pointing to the severance record. Later authority had begun as an answer to real danger and endured long after nobody could question its reach. Their practices would have to earn continuation in public. Ena added a weekly stores cost. Teren added a count of access refusals and machine errors. Esa wrote *unwanted searches* beneath them. The page would record whom the contract burdened, not merely whom it helped.

Bren tapped the dome item on the page. “What do we keep above us?”

“The route,” Pell said. “Until we can see what it points to.”

“Always?”

Pell considered. “No. If you say always, somebody will keep showing the map after it blocks the sky.”

A few adults smiled. Bren did not. He revised his proposal: the dome would show the Carver tessellation during the turning, with source marks and the rejected course visible by request. Once true starlight could be admitted safely, projection must yield to observation. No fabricated field could carry civic authority again.

Esa placed the First Weaver’s note beside the proposal. Preservation had survived eight centuries because one architect described a limit. The new page needed limits more than vows.

At evening meal, Ward Nine tried the waking practice in reverse. They did not speak in order of rank. Mila began because she had to leave early for stores. She named herself, her chosen ward, the grain lockers she had opened, and one uncertainty about an elder she remembered as both teacher and cousin. The table answered only with facts its members could witness. She had eaten with them for six years. She repaired tally pens with wire. Tomorrow Ena expected her at the heat ledger.

Lio followed. Then the pressure worker with two coats. He chose no office yet. The ward named the chalk dust on his sleeve and the valve test scheduled for morning.

When Esa’s turn came, Pell set the repaired boat before her. The pale rail had cured enough for the clamp to come off.

“Name,” Koris said.

“Esa Miro.”

“Ward?”

“Nine, if Nine still chooses me.”

Hands touched the table, one after another. No network joined them. Resin clicked beneath knuckles and rings.

“Work?” Koris asked.

“Cartography. Archive witness. Basin test, inconclusive.”

Pell removed the clamp from the boat. “You used the wrong button on purpose.”

Mila added, “You carried breakfast water to reclamation.”

Bren said, “You stopped when Ena named her authority.”

From the table channel, Ena reported that both rooms in Ward Three had reached safe temperature. Teren’s written relay check arrived with three signatures and his key resealed in its bowl.

The answers did not rebuild Esa. They located her among acts that could be checked, people who could object, and work that would continue tomorrow. Her mother’s name remained absent. The missing years did not close around a plausible substitute.

Esa dipped the repaired boat into the basin. Pell watched for leaks while water climbed the new pale rail. One bead formed at the join, trembled, then held. Around them, Ward Nine ate the first evening meal no morning authority had prepared them to recognize.

Beyond the dome, the Carver route stayed lit. Under the table, a faint correction pulse entered the deck and nudged every cup a fraction toward Homeward.

24 — The Tessellated Sky

The second turning year taught Ward Nine to hear thrust before it began.

Warm bricks arrived at the long table after midday. Families sealed sleeping-room vents with resin tape. Pell, thirteen by the ward's count, chalked arrows on loose cups so nobody mistook the coming pull for a recovered memory. When the public tone sounded, people sat with their backs to the west rail and watched broth climb one side of each bowl.

Six minutes later, the pressure released. Pell measured the new spill line and entered it beside the pulse number. Every mark leaned toward Homeward.

The third year brought a blight to Hydroponics Six. Ena Vaal suspended two correction intervals before navigation asked. White fungus had crossed a fruiting rack, closing the root-temperature margin written at Ward Nine's table. The ship lost forty-three days of turn while workers stripped trays, boiled feed lines, and carried salvageable shoots into warmer family rooms. No archive voice told them which need outranked the other. Ena published the loss in leaves, meals, and vector delay. The wards accepted one figure, argued over the next, then planted again.

During the fourth year, a bearing in Engineering Four began to shed ceramic dust. Teren Sull found the first gray crescent under a service plate. He could have replaced the bearing from an inherited office sequence, but the sequence belonged to an earlier housing. He and two waking engineers opened the assembly, measured every tooth, and built a tool that matched the present machine. The next pulse ran eight breaths late. Teren entered the delay and the useless inherited instruction together.

No year offered a clean emblem. People froze in rooms that missed heat-floor inspection. A grain locker molded after Mila opened old stock too quickly. Two wards abandoned the evening meal practice because it demanded more witness labor than they could sustain, then restored a shorter version after three citizens woke alone and missed essential medicine. Some people requested completion software without civic preference. The archive technicians built limited aids that returned clocks, schedules, and chosen medical notices while refusing to settle kin or office disputes. Each aid expired unless renewed in public.

The hardest renewal came after a medicine error in Ward Four. A clerk woke with two valid dosage tables and chose the older one because her hands recalled its color order. Four citizens received too little anticoagulant before a nurse caught the mismatch. None suffered lasting injury, but the near loss reached every ward before the clinical review did. By evening, a petition to restore automatic office completion had more signatures than the Homeward ballot's margin.

Esa attended the review without speaking for the first hour. The injured citizens sat nearest the table. One wanted a voice restored immediately. Another wanted no system touching his memory again. The clerk brought both dosage tables, each with authentic archive provenance. A later authority layer had once hidden the obsolete table. Severance exposed it without adding the red expiration bar that current nurses relied upon.

The review did not solve the error by choosing a truer clerk. Medical workers recoded every active dosage table with a current physical token, a notched ceramic tab stored beside the medicine itself. Archive search could retrieve older tables, but it displayed them without a token and printed **historical, not current instruction**. Two humans checked high-risk doses through the next year. The ward paid for that labor by delaying a habitat repaint and reducing one recreation shift.

Koris asked the petitioners whether the new measure answered their fear. Some said yes. Forty-seven names

remained on the request for completion. Their dissent stayed attached to the renewal. The limited aid continued for six months, then faced review again.

Afterward, Esa helped the clerk scrape obsolete color marks from cabinet fronts. "I hurt people with a real memory," the woman said.

"You acted on a record that lacked current context."

"That sounds kinder."

"It should not. Context costs work. We failed to do it." Esa handed her a fresh scraper. "Now we know where the work belongs."

They removed the last green stripe near midnight. The medicine cabinet showed bare metal until the ceramic tabs arrived. For three intervals, every dose came from a nurse's hand.

Esa spent those years measuring motion too small for a naked eye. After every pulse, she compared hull sensors against the Carver route and the unedited stars beyond the false catalog. Error narrowed from a finger's width on the western dome to a line thinner than Pell's chalk. The accepted constellations continued overhead because most observation skin remained sealed. Yet their old authority had drained away. Children asked which lights came from projection and which came from instruments behind it. Teachers showed them both sources.

By the seventh year, Pell no longer needed a stool at the rail. The child who had held two signal lamps now apprenticed with the hull observers and corrected adults who called every deviation drift. Drift, Pell insisted, required an authorized pattern to deviate from. A live sky gave measurements instead.

"Your hand is blocking the west sensor," Pell told Esa during one inspection.

Esa moved it. "You could say please."

"I could lose the transit."

The detector line dipped, recovered, then printed a faint periodic fall. Homeward's nearer temperate world had crossed its primary. They had seen the same transit through remote instruments for years. This one reached an outer hull sensor at an angle the old projection catalog had declared empty.

Pell tore the strip free and carried it to Bren without waiting for dismissal.

Age changed Bren less visibly. More gray entered his hair. He stopped wearing the Carver's seal beneath his collar after the route became common instruction. At the western truss, his work shifted from hidden cuts to old mechanisms nobody had touched since cryogenic rotation failed. He opened wall cavities whose diagrams existed only in launch archive, then distrusted the diagrams until metal confirmed them.

The dome's projection membrane lay inside three older layers. A transparent pressure shell formed the inhabited ceiling. Beyond it, segmented observation plates could polarize against stellar glare and radiation. Exterior shutters protected those plates during dust passage. When the Weave adopted the fabricated sky, later crews fixed the shutters closed, backed the observation plates with insulation, and stretched the projection membrane over the inner ribs. What people called a dome had become a lamp aimed inward.

Bren removed a palm-sized membrane panel above the western truss. Behind it waited gray insulation and the corroded spindle of an observation shutter.

“Opening the whole skin will take more than a command,” he said.

“Can it be done?” Esa asked.

“That depends on how much of launch architecture survived being turned into a ceiling.”

They began with one segment too small to admit a constellation. Pressure crews braced the transparent shell. Engineering warmed the spindle by four degrees to loosen old sealant. Hull observers placed a dark hood around the opening so a leak of real light could not confuse the projection instruments. Ena required a heat-cost notice before lending power. Teren required mechanical cutoffs at both ends of the shutter drive.

When the spindle moved, it advanced less than the width of a fingernail and stopped.

Bren lowered the manual lever. “Exterior ice.”

“How thick?” Pell asked.

“No current measure.”

Pell climbed into the service recess with an acoustic probe and came back holding a line of irregular echoes. Ice covered the outside shutter in ridges. Heating it fast could crack the observation plate. Mechanical force could strip the drive.

For eleven intervals, the opening remained a maintenance problem. Crews warmed the segment, allowed it to cool, then warmed it again. Condensate ran through old channels and exposed two leaks above a teaching ward. Children placed cups under the drips and labeled them *true-sky water* until a pressure worker crossed out the phrase. The water came from hull ice, not stars.

On the twelfth interval, the shutter advanced a full hand’s width.

No star appeared. Gray exterior residue covered the observation plate.

Pell laughed first. Bren looked offended on behalf of eight centuries of anticipation.

“Launch archive lists a cleaning rail,” Esa said.

“Launch archive also lists four rails that end inside walls.” Bren pushed the probe through its guide. “This one may yet exist.”

It did. A narrow scraper crossed the plate with a sound like a knife drawn over stone. Dust and old antifreeze collected in a gutter. The dark hood’s internal sensor climbed by three counts.

Esa put her eye to the shielded viewer.

One point of light waited beyond the plate.

It lacked the soft halo used by the dome. Projection stars had crisp civic magnitudes, each brightness adjusted so children and failing eyes could read the same lesson. This point trembled at the edge of sight. Its color shifted with tiny flaws in the plate. When Esa moved her head, the light held against the instrument marks while scratches slid across it.

She asked for its coordinate. Pell read the hull fix. Bren compared it with the Carver route.

The point belonged to no named destination. It lay off the route, an ordinary star the fabricated catalog had removed because it served no lesson.

Esa looked until her eye watered. Wonder arrived through parallax, scratches, and an unimportant coordinate. The first real star she saw had not guided them anywhere.

They left the test segment open under two watches. During the next month, citizens came in groups of four to use the shielded viewer. Some wept. Some complained that the point looked smaller than a projected star. One elder accused the observers of hiding a lamp behind the hull until Pell invited her to check the parallax herself. A boy asked for the star's name. Esa entered its coordinate and left the name field empty.

The opening plan went to every ward with its hazards attached. Retraction would draw more power than one correction pulse. Removing insulation would increase dome heat loss. Radiation plates required testing segment by segment. Exterior shutters might jam halfway, leaving bright wedges across a projection designed for darkness. The pressure shell had held for eight centuries, but nobody could promise it would hold under a thermal gradient last measured at launch.

Precinct Eleven opposed a full opening. Two medical wards requested partial shutters for citizens who found uncatalogued light disorienting. Teaching wards wanted both projection and observation available during transition. Engineering insisted on a sequence that allowed any three adjacent segments to close independently.

The final plan satisfied nobody completely. Half the dome would open first. Projection would remain on the sealed half for one full interval, with every projected light marked as such. Quiet rooms would keep low ceilings and local star panels for anyone who could not tolerate the exposed field. Exterior shutters would retract only after pressure, radiation, and heat tests passed. The second half required a separate witnessed decision after the first remained stable.

A dry cycle exposed what debate had missed. Crews rolled the insulation take-up spindles without moving exterior shutters. Segment eighteen dragged twice, then pulled free hard enough to shake dust into the dome. Its brake had been installed backward during a repair two centuries earlier. The archived diagram showed the original orientation, and the maintenance note showed its reversal, but the opening sequence had retrieved only the diagram.

Pell stopped the cycle before the spindle reached its travel limit. "Current hardware first," Pell said over the work channel. "Archive second."

Bren crawled into the segment cavity and confirmed the reversed brake. He came out with dust across his face and an old maker's scratch copied onto paper. "Current hardware plus archive," he replied. "Without the note, we would call this damage. Somebody changed it to fit a replacement motor."

They revised every panel card to carry three views: launch design, later alterations, and current inspection. Teren added a rule requiring crews to point physically at the component before reading its sequence. Ena recalculated power after the dragging spindle raised load. Half the dome now required two heat intervals, not one.

The quiet rooms also failed their first test. Local star panels repeated the fabricated constellations without source labels because their software predated public provenance marks. A woman who had requested shelter from the open field found herself beneath the old authoritative sky. She left before medical could explain.

Esa and the teaching wards removed the panels. In their place, each room received a plain ceiling, a clock, and a handheld viewer that citizens could activate by choice. Some chose darkness. Others chose one live hull feed

with its coordinate and delay printed along the edge. Nobody needed to accept a comforting sky as the price of refusing the exposed dome.

Those corrections delayed opening another season. Bren did not call the delay failure. Pell did. They argued until Koris asked which condition had changed. Pell named Homeward's rising brightness and the limited approach window for training observers under live light. Bren named the brake, heat load, and mislabeled quiet rooms. Both entered their evidence. Navigation found enough margin for the delay.

Years continued to collect in the route ledger. Pell completed hull apprenticeship and took the adult seat at Ward Nine's table. Esa missed the meeting that marked it because a late pulse disagreed with the west sensor. Pell found her afterward and placed the old wooden boat on the cartography board. Both rails now carried repairs.

"You were supposed to say whether the ward would have me," Pell said.

"The ward did not wait for my permission."

"No. But I wanted your answer."

Esa closed the sensor trace. "It will have you. I will too."

Pell accepted this with less ceremony than Esa had expected. "Then come check Bren's shutter sequence. He still thinks a Carver can number panels clockwise without asking who faces which way."

Bren, when confronted, blamed launch notation. His sequence had reversed two eastern segments. Pell's current hull orientation prevented the mistake. The correction cost them a day and perhaps saved a pressure seal.

Homeward brightened by degrees no dome artist would have chosen. At first the primary occupied one detector count. Then instruments could separate its color from neighboring stars. The nearer temperate world emerged through transit and reflected spectra. A farther world showed oxygen, water vapor, seasonal cloud, and a brown-green surface that changed with the star's year. Neither world promised welcome. Weather bands crossed the farther one with enough force to cover a hemisphere. The nearer carried a dry upper atmosphere and wide temperature swings.

Navigation selected the farther world for holding orbit and descent survey. The decision remained provisional through every new reading. Landing craft had not touched air in centuries. A star reached did not settle where a ship could survive.

On the morning Homeward crossed naked-eye threshold, Esa woke without knowing why the ward corridor felt altered. The archive light showed no urgent notice. At breakfast, nobody mentioned the primary. Pell ate quickly and left a smear of grain on the bowl.

"Western truss after second bell," Pell said. "Bren says wear a pressure coat."

Esa heard rehearsal in the casual instruction. "Has the approach fix passed?"

"Ask navigation."

"Has it?"

Pell picked up the bowls. "Second bell."

At the truss, the entire western half of the dome stood ready. Projection still covered it with accepted

constellations and the Carver tessellation. Around the rim, crews occupied shutter stations. Teren supervised mechanical cutoffs from a wheeled chair; a relay injury during the sixth turning year had left one leg unreliable. Ena watched heat draw beside three hydroponic delegates. Bren waited at the original seal plate with both hands bare.

Koris read the authorization. Pressure crews reported shell stability. Medical opened the quiet rooms. Radiation filters held test load. Every shutter pair named a local stop witness. The process took nineteen minutes. No one shortened it for awe.

“Western observation field ready,” Teren said.

Bren looked at Esa. “You asked for the wall to come down.”

“I asked for the sky behind it.”

“The wall has parts. Some of them will come down today.”

He released the first seal.

Projection faded from the westernmost panel. A rectangular absence opened in the familiar constellations. Insulation rolled onto a take-up spindle. Behind it, an observation plate darkened under polarization. The exterior shutter moved with a slow, toothed vibration that Esa felt through the rail.

A blade of black crossed the dome.

Points appeared inside it, too many and too faint for any catalog lesson. Some vanished when she looked directly at them. Others separated into pairs under the plate’s optics. The field did not arrange itself around a child’s memory, a ward boundary, or a story meant to preserve peace.

Crews opened the next segment. Black joined black without a seam. Heat figures fell inside forecast. A pressure sensor flickered, and the whole sequence paused while a worker reseated its contact. Nobody urged speed. The stars waited without participating.

By the seventh segment, the Carver route had lost its western curve. Bren shifted the projection so source marks remained on the sealed half. The drawn line no longer reached its endpoint. Esa felt no grief for it. A map could survive removal from the wall if the people who used it kept the bearings.

Halfway across the dome, Homeward entered the exposed field.

At first Esa mistook it for an exterior reflection. Its light carried a warmer cast than the neighboring points. Pell set a manual sight on it, took two readings, then moved aside.

“Check me,” Pell said.

Esa bent to the sight. The coordinate matched the last Carver node within the current approach limit. Years of thrust had turned a hidden cut into a star visible through hull glass.

Bren did not use the sight. He stood with one hand on the seal plate and looked up until his eyes found what his predecessors had never seen. The guarded set of his mouth remained. Tears reached his jaw without changing it.

“My forebears expected another node,” he said. “Every instruction ended with how to continue if the route still looked false.”

“What did they say if it became visible?” Pell asked.

“Nothing.”

“Then you can choose your own sentence.”

Bren wiped his face with the heel of one hand. “Open the next shutter.”

The crews did.

Segment twelve stalled at half travel. Its motor current rose, and Teren cut power before the automatic limit responded. A pressure worker climbed the outer service rail, found a folded strip of insulation caught in the gear, and removed it with ceramic tongs. During the pause, the exposed stars occupied one side of the dome while a child’s projected hunter still crossed the other. Citizens looked between them without needing anyone to explain which field depended on machinery inside the room. After the gear passed inspection, Teren returned control to the local pair. Segment twelve completed its travel at ordinary speed.

When the western half held stable, Ward Nine came to the rail. Some members used quiet-room visors. Others lay on the floor because depth above the transparent shell disturbed their balance. Ena arrived after confirming the hydroponic heat margin. Teren remained at cutoff, watching numbers more often than stars.

Pell stood beside Esa, adult now, shoulders marked by years inside hull recesses. The old signal lamps hung from a hook near the rail. Amber glass carried a crack. Green had clouded around its base.

“The tessellation gave us the route,” Pell said. “It did not give us this.”

“No.”

“Map, not cage.”

Esa watched Homeward hold its coordinate while the ship advanced. “A map drawn where we could see it.”

The distinction felt sufficient. No chain of images needed to improve it.

Navigation brought the target system into approach range before the second half opened. The farther temperate world resolved from a spectral sequence into a small blue-brown disk. White weather crossed one ocean. A polar edge brightened, then turned away. Descent teams began arguing over coastlines, wind, and landing fuel while the world remained smaller than Pell’s cracked amber lamp.

Koris started the new catechism at the rail, not as ceremony but as a watch change. Each person named current work, present consent, and one fact another person could check. Ena named root temperatures. Teren named shutter load. Bren named the segments still closed. Pell named the primary’s coordinate without looking at a slate.

Esa gave her own name and office. For witness, she named the light beyond the plate, its measured parallax, and the scratches that moved when the star did not.

No voice answered. The exposed half of the dome offered neither reassurance nor verdict. Homeward grew by another fraction inside the sight. Beneath it, the world selected for descent turned through cloud, carrying shores no Carver had drawn.

Interlude — The Last Weaver's Note

Found beside the First Weaver's note in the western truss. Written by Esa Miro during ship-year 843, while the correction toward Homeward remained incomplete and before any landing could be promised.

I have been called the last Weaver. The title gives me work I did not do.

I did not build the echo-network. I did not begin the Carver route. I did not discover the old course error, survive the mutiny, or cast more than one ballot in the vote that turned us. I mapped a projected sky, noticed one light outside its rules, and chose not to erase the discrepancy. Later, when evidence accumulated, I recommended Homeward. Thousands decided whether to follow it.

I write this distinction first because archives simplify people after death. A narrow decision acquires one face. Shared labor gathers under the name easiest to search. If my name becomes such a shortcut, use it to find the source pages, then set it aside.

The First Weaver left a warning in this truss. A system charged to preserve relation could be given later charges until preservation served whatever those charges required. The warning did not predict our exact failure. It did not name the mutiny, the course concealment, or the morning voice. It marked a limit and trusted later readers to notice when we crossed it.

The Carvers noticed. Their answer did not arrive as public courage. They hid cuts in the dome for three centuries. Secrecy preserved a route after open refusal had failed, but secrecy also let each carrier decide what neighbors could bear. Bren Otan entered that fault in the archive himself. Do not make his line noble by removing its concealment. Do not make it criminal by removing the danger that produced concealment. Keep both records near each other.

We know the first carrier's given name now. Cael appears in the faded margin of a telemetry seal copied after the mutiny. The mark alone would not support a name. A later carrier's entry refers to "the line Cael began" and gives the same seal number. Bren checked the paper grain, ink age, and copying hand before accepting the relation. I include this small source note because a name can turn an anonymous act into a legend faster than evidence can restrain it.

Cael reached through a cut no larger than a fingernail. Whether any world should carry that name belongs to the people who stand near the world, if we reach one. This page grants nothing.

At the time of writing, *The Long Veil* still turns. Correction pulses continue under water, heat, food, hull, and thrust limits. Hydroponics has already stopped navigation once to protect root circulation. Engineering has rejected inherited procedures that fit older machines. Some wards keep the evening meal witness. Others shorten it because care labor consumes people who also mend pumps, teach children, and measure medicine. We have not replaced authority with ease. We have replaced hidden completion with visible work, and some days we fail to do enough of that work.

Yesterday a child declined an archive search for the brother she remembers. Her ward honored the refusal and still assigned two people to sit through the dark interval. That choice consumed a teaching shift and left a pump lesson untaught. The missed lesson will have to be given later by someone already carrying another duty. I record this not to turn companionship into a cost ledger, but to prevent anyone from claiming human care arrives without allocation. Authority hid such allocations inside completion. We place them where a ward can see, refuse, redistribute, or fail to meet them. The child slept. The lesson did not happen. Both facts belong in the account.

Four citizens received the wrong medicine this year after a clerk used an authentic but obsolete table. They recovered. The error strengthened those who want automatic completion restored. Their argument deserves a record without ridicule. A voice that selects current instruction can prevent injury. The same mechanism once selected which sister a relay technician could remember, which destination a crew could name, and which childhood a fevered cartographer could claim. Useful selection can grow into authority while every addition still answers a real danger.

Therefore, if you build aids after us, make each function plain. Let a clock give time. Let a current roster assign present work with consent. Let a medical notice carry the clinician, date, and table that authorized it. Let archive search return provenance and uncertainty. Do not bind those functions into a speaker that tells a waking person what all the records mean together. If you believe such a speaker has become necessary, state its powers where every ward can contest them. Give each power an ending date.

My own test remains incomplete. During the severance of authority, I recovered a fever-room fragment. A woman cooled cloth in a metal basin, folded the rough seam away from my eyelid, and spent scarce water on me. A wooden button touched the rim whenever she bent. I do not know her name. I do not know whether she remained until morning. The archive offers possible kin relations under consent seals, but no relation carries enough provenance to close the gap.

Ward Nine sits with me when the gap hurts. Pell brought breakfast after the first silent waking. Koris records what our table can witness. This care does not prove who held the cloth. It does not need to.

I ask whoever reads this to resist two opposite errors. Do not call the fragment meaningless because it lacks completion. Do not expand it into the childhood you think I deserve. Preserve the wet cloth, folded corner, gray hair, torn cuticle, and button against metal. Preserve the empty space after them. Uncertainty recorded honestly carries more of a person than a merciful invention.

The same rule applies to Homeward. We possess a route, a reachable primary, two temperate candidates, and growing measurements. We do not possess a shore. Spectra can tell us about water vapor, oxygen, cloud, and heat. They cannot promise that landing gear will hold, that soil will feed us, or that weather will spare the first shelter. If this note survives a failed approach, do not edit its hope into prophecy. We turned toward evidence, not guarantee.

I have three requests, though no request on this page binds a future ward.

Keep the archive usable. Public decisions should remain open with their sources, objections, costs, and later consequences attached. Personal relations should retain consent seals. Open archive must not become open person. When a record carries uncertainty, copy the uncertainty with it.

Keep some practice by which people tell one another what happened during the day. Our long table works for Ward Nine because the table already held meals, repairs, disputes, and children's boats. Another ward may need a different practice. Preserve the function, not our furniture. A self should have living witnesses who can say what they saw without claiming ownership of everything the person remembers.

Keep observing beyond whatever instrument becomes familiar. The Carvers used a false sky because the false sky occupied the one surface everyone shared. Their cuts converted that surface into evidence. If later generations come to trust one catalog, one lens, or one civic explanation without checking what lies outside its frame, the failure can recur under different names. No wall needs to call itself a sky in order to limit sight.

These requests can conflict. Full records can injure privacy. Witness practices can become compulsory. Observation can consume resources needed for food or shelter. Do not preserve my sentences by ignoring those

conflicts. If a future ward rejects one request after public review, attach its reasons to this page. An unanswered instruction hardens into ritual. A challenged instruction remains accountable to the living.

I withheld the first anomalous coordinate from the archivist. I lied about a sensor error. Those acts opened the inquiry and also repeated the Carvers' private certainty. Later I carried the recovered evidence to Ward Nine, then asked the ship to hear it in full. The broadcast caused terror, conflicting relations, work failures, and pain that no disclaimer could prevent. I still believe staged disclosure would have preserved the suppression too long. Belief does not refund the cost.

I recommended the turn. I accepted responsibility for that recommendation and refused responsibility for the choice citizens made after hearing it. The boundary remains important. It also remains imperfect. My office gave my words reach. My measurements shaped what the ship considered possible. A person can avoid command and still influence thousands. Preserve the ration range beside my speech. Preserve Ena Vaal's suspension conditions, Teren Sull's independent handoff, Precinct Eleven's refusal, the abstentions, and the count against Homeward. Otherwise the archive will turn recommendation into destiny.

No one person woke this ship. The First Weaver left a limit. Mutiny survivors preserved enough of their error to expose it later. Carvers cut a route. Children noticed projector cells and wandering lights. Workers carried the broadcast. Wards argued. Engineers moved mass. Hydroponics lent heat and light under conditions it could revoke. Citizens who opposed Homeward kept the vote honest by remaining visible after they lost.

If arrival comes, those facts should come with us.

I once believed cartography meant giving every light a place. The projected catalog rewarded that belief. Each point had a civic name, lesson, and approved relation. The first real star I saw through an opened test segment had no destination value. It trembled behind scratched hull glass at a coordinate the old lessons omitted because it helped nobody remember the approved sky. I left its name field empty.

Perhaps another cartographer will name it. Perhaps no one will. The light does not require our use.

The First Weaver's note and mine lie together because later hands placed them so, not because I answer the architect across eight centuries. I cannot know what answer the First Weaver would accept. I can say what we attempted: archive without completion, care without invented kin, work without inherited entitlement, a course chosen after evidence and objection reached the same room.

We may arrive. We may fail within sight of the target. In either case, do not write certainty backward into this year.

My name is Esa Miro. Ward Nine has chosen me more than once. I mapped a false sky and helped read a route hidden inside it. At this writing, Homeward remains ahead of us. The engines will pulse again after the next heat review. Pell has left a repaired wooden boat on my worktable, and one pale rail still leaks when the basin water reaches its top.

That is where the record stands.

Esa Miro, sky-cartographer

Ship-year 843

26 — Homeward

The landing craft rejected the first shore before anyone left orbit.

Its terrain model marked a broad pale shelf beside the western ocean. From the open dome, the shelf looked level enough to hold every ward on *The Long Veil*. Radar found channels beneath the surface, each covered by a crust no landing strut could trust. A craft that settled there might remain upright for minutes, then push through into salt mud.

Pell removed the site marker from the descent board. Years in hull observation had broadened the child's shoulders and sharpened the direct voice into something crews obeyed before remembering how young it had once sounded.

"Site Three goes," Pell said. "No argument from orbit can thicken the crust."

A pilot objected that Site Four lay eighty kilometers inland and required crossing a mountain wind. Pell placed the radar section beside the atmospheric plot.

"Wind gives us a bad descent. Three gives us a false landing. Choose the difficulty that ends when the engines stop."

Esa watched the room accept the distinction. Adult Pell no longer needed to point at a light until someone older agreed to see it. The board changed under Pell's hand.

Site Four occupied a basalt terrace above a narrow estuary. It had stable rock, liquid water below, and a slope gentle enough for cargo sleds. Cloud moved across it every nine hours. The survey could not predict how local organisms would affect them. Habitable described pressure, temperature, water, and chemistry within human tolerance. It did not grant permission to remove a helmet.

Bren stood at the back of the descent room with both hands inside an open guidance cabinet. He had aged through the turning years without losing the habit of distrusting finished panels. Where Pell read surfaces and loads, Bren traced the old choices hidden behind them.

"Craft Two still accepts archive preference on final approach," he said.

Teren Sull's voice came from the orbital relay. "We severed that bus."

"You severed it on the ship. This controller copied its own landing profile before authority separation."

"Can it issue command?"

Bren withdrew a narrow crystal strip. "Not now."

The strip carried a green status light even after removal. Its stored profile favored the nearer dry world, selected centuries before Homeward returned to public memory. Had the craft entered atmosphere with that preference intact, guidance could have interpreted every deviation toward Site Four as error and trimmed against the pilots.

Pell held out an evidence sleeve. "Log it before you break it."

"I know archive practice."

“You know concealment practice better.”

Bren’s face closed. Esa expected an old quarrel. Instead he placed the strip in the sleeve.

“That,” he said, “is why you witness me.”

Pell sealed it. Their agreement contained no affection, but trust had acquired working parts.

Two landing craft had survived eight centuries in vacuum cradles. One could fly after nine years of repair. The other had donated seals, guidance gyros, and half its heat tiles. Craft One would carry twelve people: two pilots, four surface engineers, a medical pair, Koris as civic witness, Pell for hull and terrain, Bren for legacy systems, and Esa for cartography. The first cargo consisted mostly of things they hoped not to need: pressure shelter, lift bags, sealant, heat blankets, water assay equipment, medical isolation, and fuel for immediate ascent.

No seed crossed in the first load. Ena had refused every request to carry a ceremonial tray.

“We do not plant into chemistry we have measured from orbit,” she had said. “Bring me water, soil, spores, and weather. Then I will decide what living thing risks the second descent.”

Hydroponics packed six empty sample boxes instead.

During the last orbital interval, navigation needed a name for the world’s common chart. *Homeward Two* had begun to sound permanent. Koris called a ship-wide naming witness, not yet a binding ballot, and asked each ward to submit names with reasons and source marks.

Esa went to the archive margin she had cited in her ship-year 843 note. The telemetry seal occupied a paper sheet no wider than her palm. Mutiny clerks had copied engine values across its center. Along the lower edge, a faded hand had written: *Cael retains the rejected bearing. Ask why before correction.* The main record classified the line as an obsolete maintenance comment.

A seventh Carver entry, made generations later, referred to “the line Cael began under seal 44-C.” Fiber comparison linked that entry to a copy of the same sheet. No surname survived. No office could be proven. Cael may have objected during the mutiny, inherited another person’s numbers, or discovered the bearing afterward. The records supported a name and an act of retention. They did not support a founder’s life.

Bren read the margin through the archive viewer. “My family turned Cael into the first Carver because every line prefers a beginning.”

“Do you reject the relation?” Esa asked.

“No. I reject the person we built around it.” He magnified the faded sentence. “This hand kept a bearing somebody ordered removed. That much holds.”

Pell leaned over the viewer. “Cael’s Reach.”

Esa looked at the planet below. “For the world?”

“For the act. Reaching does not promise arriving.”

Bren considered the words longer than either of them. “Enter the source beside the proposal.”

Esa submitted *Cael’s Reach* with the margin image, the seventh entry, and Bren’s objection to any fuller legend. Other wards proposed *Second Water*, *Open Sky*, *Wardfall*, and names drawn from the nearer

daughter they had not chosen. The binding vote would wait until the first landing report. A world should not receive a public name before anyone could report what the name touched.

Launch restraints closed around Esa's shoulders shortly before local dawn at Site Four. The craft cabin faced inward. No window showed the planet. Each seat carried a small instrument panel and a pressure hood. Pell occupied the hull station opposite Bren's legacy console. Koris sat between two sample boxes with the naming proposals strapped beneath her feet.

"Descent authority?" the senior pilot asked.

"Pilots execute," Koris answered. "Any technical station may call abort. Civic witness preserves decisions and does not fly."

"Archive guidance?"

"Read-only maps. No preference trim."

"Surface claim?"

"None. We inspect, stabilize, sample, and report."

The pilot closed the checklist. "Release Craft One."

Clamps opened with four impacts. Weight vanished. Esa's harness held her against the seat while maneuver jets pushed the craft clear of *The Long Veil*. For several minutes, vibration provided the only evidence of motion. Then the primary pilot rolled them toward atmosphere and the cabin gravity shifted from absent to uncertain.

Pell read hull temperatures. Bren read guidance disagreement. Both spoke in numbers until upper air reached them.

First contact sounded like dry grains dragged across the hull. The noise deepened. Heat rose along the craft's underside and entered the cabin through the seat frames. Esa watched a row of tile sensors turn amber. One at the aft quarter climbed faster than its neighbors.

"Aft thirty-two diverging by six degrees," Pell said.

"Within old limit," Bren replied.

"Old limit assumes original adhesive."

The surface crew had replaced that tile twice. Pell called for a shallower angle. The pilot checked the trajectory cost.

"Shallow adds four minutes in heating."

"Peak drops nine degrees. Thirty-two needs peak relief."

Bren opened the replacement record. "Adhesive batch came from cradle seal stock. Long heat performs worse than peak."

Pell looked across the cabin. "You signed that substitution."

“I did.”

“Which failure comes first?”

“At current rise, peak separates the tile in ninety seconds. Shallow exposure softens adhesive in three minutes.”

The pilot had less than a breath for their answer.

“Lower peak for two minutes, then steepen,” Pell said. “Watch the bond sensor.”

Bren entered the hybrid profile manually. “Concur.”

Craft One lifted its nose by three degrees. Heating spread across more tiles. Aft thirty-two slowed, then held. After two minutes the pilot steepened. The bond sensor stayed green by one count.

Air thickened into force. Esa’s chest pressed against the harness. Every breath required intention. The cabin panels shook too quickly to read until vibration filters caught up. Outside, ionized flow blocked the ship relay. For seven minutes, the landing party possessed only local instruments and one another.

A warning tone cut through the noise.

“Pressure shell?” Koris asked.

“Exterior antenna loss,” Pell said. “Expected in hot flow. Internal shell holds.”

The tone repeated because the old controller expected archive acknowledgment. Bren reached toward mute, stopped, and asked the pilot for permission. The pilot gave it. He silenced sound without clearing the event.

When plasma faded, the atmosphere began to speak in other ways. Rain struck the forward hull. Crosswind rolled the craft hard enough to drive Esa’s shoulder into the seat frame. Radar found Site Four through cloud, lost it behind a mineral-rich downpour, then found it again twenty kilometers east of prediction.

Mountain air descended across the terrace in pulses. The craft compensated with fuel reserved for ascent. Each correction shortened the margin for immediate return.

“Abort remains possible for forty seconds,” the junior pilot said.

Pell studied terrain velocity. “Terrace stable. Wind cycle repeats every eleven seconds.”

“Three cycles measured,” Koris said.

“Enough to forecast, not enough to trust.” Pell marked the lowest interval. “We descend during the lull. If it shifts by more than two seconds, climb.”

Bren checked the controller for hidden trim. “Manual response clean.”

The fourth lull arrived late. The pilot held altitude. The fifth came at eleven seconds. Craft One dropped through rain.

Esa felt the landing gear deploy beneath her feet. Four green marks appeared. The left aft strut turned amber.

“Latch incomplete,” Pell said.

“Recycle?”

“No time. Load it last.”

The pilots tilted the craft by less than a degree so three confirmed struts would touch first. Basalt rushed up on the radar display. Crosswind struck once more, weaker near the surface. Engines fought it sideways.

First contact drove a hard blow through Esa’s spine. A second impact followed as the forward struts compressed. The right aft touched. Left aft remained unloaded while the craft leaned above it.

“Weight transfer stable,” Pell said. “Hold thrust at twelve percent.”

Outside engines threw rain and stone across the terrace. Surface radar mapped the incomplete strut. Its foot hung a hand’s width above a shallow depression.

Bren had already unlatched his harness. “Manual jack can extend it.”

“Not while you’re sitting inside a craft on three legs,” Pell said.

“The jack controls sit in the outer gear bay.”

“I know where they sit.”

Pell ordered two lift bags deployed through the cargo floor. Surface engineers fed them beneath the left quarter using rigid poles, then filled them from landing gas. Only after pressure transferred into both bags did Bren enter the outer bay in a sealed crawl suit. He found a bent latch pin, drove it clear, and extended the strut seven centimeters.

“Foot contact,” he reported.

Pell watched the load cells settle. “Transfer two percent at a time.”

The lift bags softened by measured steps. Four struts accepted weight. Engines throttled down.

One system still kept them isolated. The exterior antenna lost during hot flow had not deployed its backup after landing. Without it, the craft could transmit only through a narrow emergency beam that required *The Long Veil* to cross directly overhead. Orbit would carry the ship below the terrace horizon in twenty-six minutes.

Pell pulled up the hull camera. The backup mast lay folded beneath a skin panel warped by heat. “I can reach it from the upper service eye.”

Bren examined the panel fasteners. “The eye opens beside tile thirty-two. Its adhesive has not cooled.”

“I do not need to step on the tile.”

“You need to cross its load frame.”

Pell drew a path through the camera image. “Tether here, knee on the antenna rib, left hand on the cold rail.”

“The rib carried launch vibration, not a suited adult.” Bren called up the original section and the later repair. “Its inner bracket lost one bolt when they donated parts to Craft Two.”

“Then brace the bracket from inside.”

They built the procedure together while the ship moved toward the horizon. Bren crawled into the upper equipment bay and set a jack beneath the weakened rib. Pell entered the service eye in a pressure suit, keeping

all weight off the hot tile. Esa saw only fragments on the hull camera: one boot, a tether clasp, rain breaking into steam against ceramic.

“Panel pin fused,” Pell said.

“Rotate a quarter turn toward your right,” Bren answered.

“My right or the diagram’s right?”

“Your right. Current body before archive.”

The pin moved. Pell lifted the warped panel enough to free the mast. Wind caught it before the hinge locked. Bren’s jack took the load. Pell hauled the mast back through its arc and drove the manual latch with a tool tied to the wrist.

Signal bars rose on Koris’s board. Teren’s voice reached them through static. “Craft One, orbital relay confirms surface carrier.”

Pell sealed the service eye and came inside with one glove scorched at the knuckle. Bren lowered the jack. Neither praised the other. They entered the missing bolt, hot-tile route, and manual latch force into the landing record before removing their helmets.

Silence arrived badly. Cooling metal clicked. Rain hissed on the hull. Pumps continued because no crew trusted a quiet machine immediately after landing.

The senior pilot kept one hand on ascent. “Craft stable. Fuel margin below immediate-return plan by eight percent. We can still reach orbit if wind permits.”

Koris entered the number. “No speech yet. Complete surface checks.”

Esa had imagined arrival through a dome, a shore filling the view. Instead she watched pressure figures while sweat cooled under her collar.

External chemistry probes extended first. Air pressure matched the survey. Oxygen could sustain unassisted breathing, but aerosol samples contained complex spores and reactive salts. Medical prohibited open helmets. Rainwater carried organic structures unlike ship algae. None penetrated suit filters during the test.

A surface camera unfolded beneath the craft. Basalt occupied most of its view, black and ridged. Low silver growth clung in cracks where rainwater ran. Beyond the terrace, the estuary reflected a strip of pale sky. Waves moved inland against the wind.

Pell pointed to the repeating line. “Tide.”

Navigation models had predicted one minor moon. The observed tide carried more range than that moon explained. Something else pulled the water, perhaps a second small body hidden in cloud or a resonance from the nearer world.

“Does it reach the terrace?” Esa asked.

“Not on current marks. We have one partial cycle.”

The landing site had changed again. Stable rock did not mean stable access to the estuary. Pell added tide stakes to the first surface plan.

Koris authorized the outer hatch after one hour of sampling. Nobody stepped directly onto the world. Surface engineers sent a tethered crawler down the ramp. Its first wheel slipped on wet basalt. Bren adjusted the ramp feet. Pell placed traction mesh from the hatch to the nearest ridge.

Esa descended fourth. Her suit made each step deliberate. Rain struck the helmet hard enough to hide the craft lights when she looked down. At the foot of the ramp, she set one boot on basalt and waited for the rock to answer through pressure.

It gave her weight back.

No catechism entered her mind. No archive marked the contact complete. She checked the suit seal, named her position to Pell, and moved aside so Koris could descend.

Work divided at once. Medical established a clean line around the ramp. Surface engineers anchored the pressure shelter to cracks rather than drilling unknown silver growth. Bren opened a cargo latch that had warped during heating. Pell walked the terrace edge placing tide reflectors and load markers. Koris set a folding table beneath the craft's lee and began the first local record.

Esa carried the survey baseline to a basalt rise twenty meters from the ramp. The instrument legs sank into mosslike material, so she moved them to bare stone. Her first sight line ran from the landing craft to the estuary bend. The second fixed a mountain notch above cloud. The third returned to the craft and missed by four millimeters because wind flexed the mast.

She measured again.

Bren joined her with the recovered guidance strip sealed at his belt. "Still making the world fit a triangle?"

"Making the triangle admit wind."

He looked toward Pell, who had driven three red tide stakes into the lower path. "Cael never reached this."

"No."

"My line did not reach it either. We kept a bearing. Other people built the arrival."

Esa leveled the instrument. "Say that at the naming witness."

"I intend to object to the name."

She turned from the sight. "You entered your support in orbit."

"I supported the source. Down here, the word makes my family too large."

Bren pointed to the landing craft, the pressure shelter, and Pell's markers. "None of that came from us."

The objection did not surprise her. It made the proposal more honest.

They carried samples to Koris's table as local dusk approached. Air remained suit-safe but unapproved for breathing. Water required isolation. Silver growth responded to light by folding narrow fronds into rock cracks. The tide reached the lowest red stake two hours earlier than orbital prediction, then stopped. Craft One could not launch until wind dropped, which models placed after dawn.

No first night feast occurred. The landing party ate sealed grain paste inside the pressure shelter while rain

pressed its roof inward between ribs. One heater failed. Bren repaired it with a contact stripped from the obsolete guidance profile. Pell maintained tide watch and refused relief until Esa carried a second chair to the hatch.

The naming witness opened over delayed relay. Wards aboard *The Long Veil* received the landing report before any proposal. Koris listed the bad shore, tile heat, fuel loss, incomplete strut, spores, tide error, heater failure, stable basalt, breathable chemistry, liquid water, and living silver forms. No one heard only the word *habitable*.

Esa presented the Cael margin and later Carver reference. Bren followed.

“Cael retained a rejected bearing,” he said. “My family carried that retention and also hid it. This landing belongs to pilots, repair crews, wards, growers, voters, and people who opposed the course but maintained it after losing. If you choose Cael’s Reach, let *reach* name an unfinished action shared by those people. Do not make Cael our founder.”

Pell spoke next from the hatch. “I support it because reach includes missing. We missed the first shore. We missed the weather timing. We reached this terrace after changing the plan.”

Other names remained on the ballot. Voting took one full tide interval. Cael’s Reach carried without unanimity. Precinct Eleven preferred Second Water. Hydroponics split between Cael’s Reach and Open Soil, though Ena entered a note that nobody had yet proven the soil open to anything.

Koris closed the count. “Homeward names the primary. The farther temperate world enters the public chart as Cael’s Reach, with source record, objections, and landing report attached.”

The new name appeared on Esa’s survey board. It did not change the rain.

Near dawn, cloud broke above the estuary. Homeward cast a warm line across the water. Real stars emerged around it, shifted into constellations no projected lesson had prepared. *The Long Veil* crossed overhead as a moving point, dome open, thousands of people waiting to learn whether a second craft should descend.

Ward Nine gathered by relay at Koris’s folding table. Pell came in from tide watch with silver mud on both boots. Bren brought the repaired heater contact. Esa stood beside her survey mast, one gloved hand on the instrument that had finally closed its triangle.

“My name is Esa Miro,” she said. “Ward Nine. Cartographer of *The Long Veil*. Surface surveyor of Cael’s Reach for this interval, if the ward consents.”

“The ward consents for this interval,” Koris answered.

Pell laughed. “You made the office expire.”

“We learned to date authority.”

Bren named himself as legacy-systems mechanic, not Carver. Pell named hull observer and landing witness, then reported the next tide in forty-three minutes. The ward answered each with work it had seen and limits it expected them to keep.

Esa did not claim wholeness. The fever room still ended after the button touched metal. Her early years remained partly borrowed. On this shore, nobody could author those years into certainty. They could hold a survey mast against wind, check her seal, bring a chair to the watcher, and correct her when four millimeters

mattered.

Behind them, silver fronds opened as Homeward rose. Ahead lay shelter tests, microbial study, water treatment, another landing, and the decision of whether any permanent settlement should stand on the terrace. Hard work did not make arrival provisional. It made arrival exact.

Esa looked from the open dome moving overhead to the adult Pell beside the tide stakes, then to Bren warming both hands around a cup repaired twice. The Carver route had ended. Their responsibility had not.

Esa said, "We are home. And we are who we chose to be."

A new tide touched the lowest red marker. Pell noted the time. Esa turned her survey book to the next blank page.